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# Alcohol Marketing on Social Media During the COVID-19 Pandemic: Historical Perspectives, Modern Evidence, and Future Regulation

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# Alcohol Marketing on Social Media During the COVID-19 Pandemic: Historical Perspectives, Modern Evidence, and Future Regulation

**by**

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## Dedication

For the giants whose shoulders I stand on, Dyana and Patricia.

## Acknowledgements

The practice of science consumes a tremendous amount of time. The time I spent on this work came at the expense of precious time with my family during our formative years. I acknowledge Kirsti and Meridian. Know that these sacrificed hours were spent gazing optimistically towards the future. One where society is a little bit more sober and digital environments are a little bit safer.

Success in science depends greatly on mentorship. The path to success of this work was long and came with adversity. I acknowledge Keryn, whose unwavering patience and consistent encouragement were foundational.

A man’s pursuits - in one way or another - trace back to his father, and to his father before him, in an unbroken chain. I acknowledge this lineage of fathers, each life shaping the conditions that made my own path possible.

# Abstract

## Alcohol Marketing on Social Media During the COVID-19 Pandemic: Historical Perspectives, Modern Evidence, and Future Regulation

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This dissertation examines alcohol industry marketing on social media during the COVID-19 pandemic through three complementary lenses: historical perspectives, modern evidence, and future regulation. These historical perspectives illustrate the alcohol industry's propensity to leverage new communication technologies, capitalize on vulnerable people, and promote lasting media portrayals. Previous empirical research establishes links between alcohol marketing exposure and increased consumption, particularly among vulnerable populations such as young adults. Additionally, emerging studies during the pandemic identified specific alcohol marketing themes including isolation drinking, delivery services, and corporate social responsibility messaging on social media platforms. Building on this foundation, this dissertation investigates how the alcohol industry's pandemic-era strategies may represent a continuation of longstanding patterns of technological adaptation and targeting of vulnerable populations, now manifested through digital platforms during a period of heightened societal stress.

For this dissertation, two original studies analyzed nearly 6,000 tweets from major alcohol brands and over 486,000 tweets from twitter users before, during, and after initial COVID-19 lockdowns (January 2019-July 2021). Study 1 employed topic modeling, sentiment analysis, and A.I.-based classification to identify five primary marketing themes, with three showing significant increases during lockdowns: Alcohol Delivery & Isolation Drinking (+91.2%), Restaurant Support (corporate social responsibility-themed) (+189.5%), and Social Media Promotions (+150.0%). Notably, two themes, Alcohol Delivery & Isolation Drinking and Social Media Promotions, remained elevated post-lockdown, suggesting companies continued with these types of messages. The pandemic-emphasized themes displayed stronger emotional dimensions, potentially exploiting heightened psychological vulnerabilities during the COVID-19 pandemic.

Study 2 used cross-lagged panel modeling to examine bidirectional influences between alcohol brand tweets and user-generated content. Findings revealed a significant unidirectional pathway: increased potential exposure to alcohol brand tweets promoting alcohol delivery and isolation drinking predicted subsequent increases in similar user-generated content, with no significant inverse association. This suggests that alcohol brand tweets influenced user-generated tweets during this time period rather than individual tweets influencing alcohol brand tweets.

The empirical findings align with the historical perspectives and highlight how the alcohol industry may have leveraged communication technology during a crisis to appeal to a vulnerable population through messaging that contained high levels of emotional language as identified through linguistic analysis. Based on these insights, the dissertation proposes a regulatory framework including platform-level interventions (content moderation, enhanced age verification, algorithmic transparency), and industry accountability measures (mandatory reporting, liability frameworks, corporate social responsibility standards, and crisis marketing codes).

This research contributes to understanding alcohol marketing dynamics across time, particularly during societal disruptions, and provides evidence-based recommendations for mitigating public health risks associated with digital alcohol marketing practices.

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# Chapter 1: Executive Summary

## INTRODUCTION

This dissertation explores the dynamics of alcohol marketing, particularly in the context of social media during the societal disruption caused by the COVID-19 pandemic. It argues that the alcohol industry's strategies during this crisis are not novel but reflect long-standing historical patterns, amplified by modern communication technologies, and targeted towards a population experiencing heightened vulnerability (Standage, 2006; Gately, 2008; Phillips, 2014, Richards, 2022). By integrating three distinct but complementary lenses - historical perspectives, modern evidence, and future regulation - this work advances our understanding of alcohol marketing across time, in particularly on social media during times of societal disruption. Potential regulatory actions to mitigate public health risks associated with digital alcohol marketing are proposed.

## LENS 1: HISTORICAL PERSPECTIVES

Informed by historical perspectives (i.e., examining the past to understand present phenomena; Lawrence, 1984) *three key points* are suggested as contributing to the alcohol industry's enduring success and societal impact:

- Key Point 1: the alcohol industry’s success is linked to advances in communication technology;
- Key Point 2: the alcohol industry has a long history of distributing its products to, and exploiting, vulnerable people*;*
- Key Point 3: the media portrayals of the alcohol industry are lasting.

These historical perspectives provide a lens for investigating the alcohol industry's actions during the COVID-19 pandemic, a period marked by widespread psychological distress (Pfefferbaum et al., 2020; Panchal et al, 2023) and a surge in digital media consumption (Auxier et al., 2021; Bendau et al., 2021; Nabi et al., 2022; Price et al., 2022).

### LENS 2: MODERN EVIDENCE

The global burden of alcohol-related disease is staggering, with 5.1% of annual years lost prematurely to death or disability bring attributed to alcohol use (World Health Organization, 2019). Evidence demonstrates a clear link between exposure to alcohol marketing and increased consumption (Anderson et al., 2009; Smith et al., 2009; Jernigan et al. 2017), with candidate mediators suggested as marketing's effect on shaping favorable beliefs about alcohol and normalizing its use (Blume et al. 2003; Sudhinaraset et al., 2016; Collins et al, 2017; Martino, et al, 2016). The COVID-19 pandemic led to documented increases in stress, isolation, alcohol consumption and related mortality (Grossman et al., 2020; Koob et al., 2020; Pollard et al., 2020; Foster et al., 2021; White et al., 2022) alongside unprecedented reliance on social media for information exchange (Auxier et al., 2021; Bendau et al., 2021; Nabi et al., 2022; Price et al., 2022). The convergence of increased alcohol consumption, the rise of social media marketing, and the societal disruptions caused by the COVID-19 pandemic presents a unique opportunity to study the marketing practices of the alcohol industry at the nexus of the *three key points*. This dissertation conducted two original studies analyzing alcohol industry marketing and user-generated content on Twitter before, during, and after the initial COVID-19 lockdowns (total span, Jan 2019 - July 2021; initial lockdowns conceptualized as March-May 2020), focused on identifying the themes embedded in marketing content, the sentiment through which themes were conveyed, and how ideas about the themes spread across Twitter.

**Study 1: Marketing Themes, Sentiment, and Prevalence**

Using topic modeling (Grootendorst, 2022), sentiment analysis (Boyd et al., 2022), and A.I.-based classification methods (OpenAI, 2024) on nearly 6,000 tweets from 11 major alcohol brands, this study identified potential shifts in marketing strategy during the pandemic.

- **Key Findings:** Five primary themes emerged, with three showing significant increases during the initial lockdown period (March-May 2020) compared to pre-pandemic levels:
    - *Alcohol Delivery & Isolation Drinking* ($\Delta$ = +91.2%)
    - *Restaurant Support (CSR-themed)* ($\Delta$ = +189.5%)
    - *Social Media Promotions* ($\Delta$ = +150.0%)
    - *Sports* (stable across time)
    - *Gaming* (stable across time)
- **Lasting Shifts:** Alcohol Delivery & Isolation Drinking ($\Delta$ = +30.8%) and Social Media Promotions ($\Delta$ = +60.6%) remained significantly elevated post-lockdown compared to pre-pandemic levels, suggesting durable adaptation.
- **Sentiment:** The themes most emphasized during the lockdown (Alcohol Delivery and Isolation Drinking, Restaurant Support, Social Media Promotions) scored higher on emotional dimensions, potentially exploiting heightened psychological vulnerabilities (consistent with the Differential

Susceptibility to Media Effects Model; Valkenburg et al., 2013). Restaurant Support and Social Media Promotions also demonstrated stronger excitative characteristics through calls to action, while Sports and Gaming themes engaged more cognitive dimensions. While Restaurant Support messaging appeared socially beneficial as corporate social responsibility (CSR), it may have leveraged public empathy to promote product consumption through emotionally driven content, a common CSR tactic (Yoon et al., 2013; Mialon et al., 2018).

**Study 2: Alcohol Marketing Influence on User-Generated Content**

This study investigated the potential bidirectional influence between alcohol brand tweets (potential exposure) and user-generated twitter content related to Alcohol Delivery and Isolation Drinking during early lockdowns (March-May 2020). It analyzed tweets from 11 major alcohol brands (n=704) and 66,303 twitter users (n=486,786; Dashtian et al., 2021), and aimed to inform the origin and spread of this theme.

- **Key Findings:** Cross-lagged panel modeling revealed a significant *unidirectional* pathway:
    - Increased potential exposure to alcohol brand tweets promoting alcohol delivery and isolation drinking significantly predicted a subsequent increase in user-generated content on the same theme. (Specifically, a 1 standard deviation increase in potential exposure was associated with approximately 5% higher odds of related user tweet generation in the following period).

    - No significant pathway was found from user-generated content to potential exposure.
- **Implications:** The findings suggest an industry-driven dynamic where brand messaging influenced user-generated content around pandemic-related drinking, rather than simply reflecting existing behaviors. The high stability of both potential exposure and user-generated content over time suggests that once these variables are established, they are durable.

Study 1 and Study 2 synthesize to reaffirm the historical perspectives. There is evidence that during the COVID-19 pandemic, the alcohol industry leveraged advanced communication technology to appeal to a population made vulnerable by the pandemic. It employed specific themes, such as alcohol delivery and isolation drinking, conveyed through emotionally resonant sentiment, which transmitted to the broader community and appears durable across time.

### LENS 3: FUTURE REGULATION

The reaffirmation of historical perspectives via empirical findings calls for regulatory response to mitigate the risks of digital alcohol marketing, especially during crises. Lessons from effective public health interventions (e.g., comprehensive alcohol ad bans in Norway (Rossow, 2021), France's Loi Évin (Cogordan et al, 2014), and U.S. tobacco control (U.S. Congress, 1970B; National Association of Attorneys General, 1998; Federal Trade Commission, 2023)) demonstrate the potential impact of regulation. Based on this dissertation's findings, a multi-pronged approach is proposed:

- **Platform-Level Interventions:**
    - **Content Moderation:** Implement systems to identify and restrict marketing content that exploits crisis-related vulnerabilities (e.g., promoting alcohol as a coping mechanism for isolation/stress). Legal precedent exists for restricting harmful commercial speech (Redish, 1970; McKeon, 1981; Post, 2000; U.S. Congress, 2021; European Parliament and Council of the European Union, 2022).
    - **Enhanced Age Verification:** Mandate robust, potentially device-level, age verification to prevent underage exposure (Federal Trade Commission, n.d.; Blumenthal et al., 2023; Texas Legislature, 2023), crucial given increased youth social media use and pandemic-era rises in young adult alcohol mortality (White et al., 2022).
    - **Algorithmic Transparency:** Require platforms to disclose how alcohol marketing content is targeted and amplified, enabling oversight (Matz, 2025).
- **Industry Accountability:**
    - **Mandatory Reporting:** Require alcohol companies to report digital marketing expenditures and strategies.
    - **Liability Frameworks:** Develop enforceable penalties for known harms (e.g., National Association of Attorneys General, 1998; World Health Organization, 2021)

- **CSR Standards:** Develop clear standards to distinguish genuine corporate social responsibility from marketing disguised as public good (like potentially exploitative "Restaurant Support" themes).
- **Crisis Marketing Codes:** Establish enforceable codes of conduct for marketing during public health emergencies, with independent monitoring.

## CONCLUSION

By examining alcohol marketing through historical, empirical, and regulatory lenses, this dissertation suggests that the alcohol industry adapted its social media marketing strategies during the COVID-19 pandemic, reflecting historical precedents of leveraging technology and vulnerability. The findings underscore the significance of the industry-to-public pathway and highlight the need for updated regulatory frameworks governing digital alcohol marketing. Implementing the proposed measures could help mitigate public health harms and foster a healthier information environment, particularly during inevitable future societal disruptions.

# Chapter 2: Literature Review

## Part A: Historical Perspectives: Alcohol Use, The Alcohol industry, and Alcohol Media through the Ages

### Introduction

Chapter 2, Part A is a literature review that tells the narrative of how alcohol fits into human culture. Many histories of alcohol have been written, and several of them were invaluable to writing this review (Standage, 2006; Gately, 2008; Phillips, 2014). However, no previous work has been compiled in the form of the present review. Specifically, Chapter 2, Part A is a historical perspective focused on the evolution of alcohol marketing practices and how these practices coincide with cultural norms of alcohol use in society. In this review, the historical perspective acknowledges the combined impact of societal disruptions triggered by the COVID-19 pandemic and the rapid expansion of social media as a communications medium during that time. It aims to retrospectively document similar instances of societal shifts and changes in the media landscape throughout history, and explore how alcohol use and the alcohol industry may have been affected. Unlike historical research methods that focus on understanding the past, historical perspectives aim to deepen our comprehension of contemporary issues (Lawrence, 1984). For instance, an analysis using primary documents to systematically study the timeline of alcohol marketing in a specific era would be considered historical research. In contrast, examining historical perspectives on alcohol marketing helps us understand current phenomena by looking back at past events.

This review highlights five forms of media (i.e., the intervening methods of communication that are used to transmit messages and shape public opinion) used by the alcohol industry throughout history: *graphic media, culture as media* (i.e., aspects of culture such as language, beliefs, gestures, rituals, or other symbols serving as the intervening method of communication to transmit messages such as is the case with cultural meme)*, broadcast media, digital media,* and *social media*. The substantive portion of Chapter 2, Part A is organized into sections following these five forms of media. Simultaneously, the review also presents the narrative of how alcohol fits into human culture linearly, beginning in 7000 B.C.E. and extending to the early 2020's C.E. By reviewing the over 9000-year history of alcohol-related media, the reader will foster an understanding of *Three Key Points* related to the marketing practices of the alcohol industry, which are summarized in final section of Chapter 2, Part A.

### 2.A.1: The Intoxicating Medium: Relating to the definition of alcohol

Alcohol use, a.k.a. drinking, is the consumption of the intoxicating organic compound ethanol (Friedmann, 2013). Ethanol is created when certain bacteria metabolize sugar, a process chemically known as fermentation (National Center for Biotechnology Information, 2023). Drinking acutely acts as a depressant of the central nervous system and interrupts the communication between neurons; and chronically leads to neural atrophy, even when consumed in small doses consistently over time (Daviet et al., 2022). Alcohol is an important part of human history with historical records of alcohol use being associated with the earliest of human activities (e.g., cave paintings) and archeological evidence (e.g., fragments of vessels with traces of alcohol) indicating that that alcohol consumption precedes written history (Dietler, 2020). The earliest definitive evidence of the preparation of alcohol for consumption comes from analysis of residues found inside pottery near present-day northern China dating back to approximately 7000-6600 B.C.E. (McGovern et al., 2004).

Evidence suggests that early humans used alcohol mostly in ritualistic and/or spiritual contexts. For example, pagan cultures in neolithic Europe consumed alcohol during elaborate ceremonies in honor of the changing of seasons (i.e., the turning of the wheel of the year) (Lipscomb, 2020). Many hearths within settlements near Stonehenge are filled with large quantities of animal bones and broken pottery vessels commonly associated with fermented beverages, which suggests feasts and abundant drinking were common during the period (Albarella et al., 2011). The Bible even suggests consuming alcohol ritualistically to harbor a peaceful death or to cope with pain and adversity (Proverbs 31:6-7) (Carrol et al., 2008).

### 2.A.2: Graphic Media: Relating to drawing, engraving, or printing, and alcohol

As modern civilizations emerged, concentrations of resources, agricultural knowledge, and experimental curiosity kindled the development of many kinds of alcohol, including beers (fermented from cereal grains) (Oliver et al., 2011), and wines (fermented from fruit, typically grape) (Robinson et al., 2015). Archeological records indicate that citizens of Antient Greece often consumed wine with meals and mixed wine with their drinking water for fortification (Garnier et al., 2016). Heroic-age literary works - the Iliad and Odyssey (first written by Homer in the 8$^{th}$ century B.C.E.) - depict a society that had a sensual and religious affinity for alcohol, often referencing wine's powers to bring courage and strength to the meek or suggesting that the Gods would not pay mind to any ritual or prayers if drinking was not involved. This can be seen in Book 1 of The Iliad when Agamemnon organizes an elaborate feast, complete with libations of wine, in an attempt to gain favor from the gods and inspire his army towards victory in the Trojan War (Fagles et al., 1998).

Records from Antient Rome indicate that wine was a staple in the diet of nearly all Roman citizens and a part of the ration for legionaries (Davies, 1971). Romans began to standardize the cultivation of grapes and formulas for wine (as they did for much in Roman culture: roads, weapons, sewers, building codes, senate rules, and indeed the very concept of Republicanism), culminating in the first known treatise on producing alcohol, contained in *De Agri Cultura* by Marcus Porcius Cato (~160 B.C.E.) (Cato et al., 1934). *De Agri Cultura* quickly circulated among Roman landowners. Thus began the emergence of industrial alcohol production which, according to *De Agri Cultura,* required land with south-facing slopes and good drainage to establish a vineyard, substantial investment in slaves, building infrastructure, and establishing elaborate shipping/distribution practices. With abundant supply (driven by industrial-scale production by Roman landowners), drinking behaviors during classical antiquity shifted from predominantly ritualistic to the everyday mundane, and sometimes problematic (Laes et al, 2013). In fact, while drinking was ubiquitous, drunkenness (i.e., drinking to the point of intoxication) was considered a public nuisance and thought to be a health risk to some philosophers. Prominent Roman stoic Seneca even wrote that habitual drunkenness weakens the mind and that its consequences are felt long after the drinking has stopped (Motto et al, 1990).

The Medieval Period of Europe (September 4, 476 - May 29, 1453) was marked by a shift from centralized Roman imperial governance to a distributed feudal system of peasants and lords (Wood, 2013). Life in this system was difficult for the commoner, living often on the brink of hunger or despair. Ale (fermented from cereal grains) became an important source of nutrition for the population at large (Dyer, 1988; Carlin et al., 1998), and drunkenness became widespread among all sections of society (Glatt, 1977). During this time, many towns expanded, and the alcohol industry flourished among them with many breweries and drinking taverns. In 1389, King Richard II of England enacted a law stipulating that places selling alcohol must hang a sign to indicate that fact, which led to competition between establishments to have the most distinguished and attractive looking signs (Richards, 2022). It is important to note that suggesting alcohol merchants to declare themselves as such was not an idea original to King Richard II (or England, for that matter), but rather, some historians suggest, appropriated from antient Chinese lore that tells of villagers hoisting their wine banner when they had fresh wine to trade (Liao, 1959). Nevertheless, King Richard II's Medieval law likely contributed to a culture of extravagant marketing competition within the alcohol industry that echoes to this day, and English Pubs particularly continue the tavern-sign tradition (see Illustration 1, Illustration 2). Also in the late 1300's, a brewery named Den Hoorn was established in Belgium and its shop sign featured an elaborate golden horn (Kelly-Holmes, 2016). Den Hoorn was purchased in 1717 by Sebastian Artois, who renamed the brewery boasting his namesake: Stela Artois (Persyn et al., 2011). Interestingly, Stela Artois remains a globally popular brand today, and its logo still sports a golden horn reminiscent of Den Hoorn's, which makes the horn the oldest active marketing campaign (see Illustration 3). Brewery signs were not the only traces of the alcohol industry one might encounter in Medieval towns. Town criers could be heard all over Europe shouting local news, royal decree, new laws, or even announcements promoting deals at local alcohol establishments. So-called "wine criers" living in France during the rule of King Philip Augustus (1180 to 1223) were permitted to shout outside of taverns and subsequently charge the owner a fee for bringing in customers (Presbrey, 1929; Richards, 2022).

BRIDGE-FOOT, SOUTHWARK.
(*Showing the Bear Inn in* 1616.)

Illustration 1. Southwark a neighborhood in London near the southern end of the London Bridge, historically know for drinking establishments. Image depicts a tavern - Bear Inn, 1616. Tavern sign is highlighted. From Shelley, H. C. (1909).


Illustration 2. Present-day tavern sigh of The George Inn (also in Southwark, London), one of the oldest drinking establishments in London still operating according to the National Heritage List for England (n.d.).


Illustration 3. Stela Artois (n.d.) logo sporting the golden horn.

The Medieval period spawned The Renaissance (beginning in the 14th century and extending into the 17th century, with precise dates varying depending on the region), in which the cultural movement looked to Christian philosophy to invoke excessive drinking as contributing to the decay of society (combined with general poor conditions and widespread disease) (Monfasani, 2016). Early temperance movements during The Renaissance preached that God created alcohol for consumption in moderation, for pleasure, enjoyment, and health; however, drunkenness was a sin (Austin, 1985). Perhaps in contradiction to the Christian doctrine concerning alcohol, advances in science during The Renaissance led to the proliferation of distilled spirits, which are alcoholic beverages containing high concentrations of ethanol. To distill alcohol, a fermented beverage is heated to between 78.37°C (the boiling point of alcohol) and 100°C (the boiling point of water) for some time. Because alcohol boils at a lower temperature than water, the vapor emitted from fermented beverages boiling below 100°C is a concentrated form of alcohol, which can be cooled and collected into a liquid spirit (Sachaschke, 2014).

Distillation and the production of spirits were not invented by Europeans, but rather 15th century German apothecaries appropriated methods from sacred centuries-old Islamic alchemy (Tschanz, 2003; Edriss et al., 2017) into a potent product marketed as a health tonic, called *Gebrant wein* (burnt wine, a.k.a., brandy) (Gately, 2008). Although brandy carried a reputation of being deadly if consumed in excess, it quickly became popular, and in 1500 German doctor Hieronymus Brunschwig published *Liber de arte distillandi de simplicibus* (Book on the art of distillation out of simple ingredients) which achieved widespread circulation (Bruschwig et al, 1500). Reminiscent of how Cato's *De Agri Cultura* shifted the paradigm of wine consumption across the Roman Empire 1660 years before, *Liber de arte distillandi de simplicibus* enabled the development of a global spirit industry. Historically, excessively drinking fermented alcoholic beverages came to be associated with intoxication, health risk, and social nuisance; and it quickly became evident that widespread consumption of distilled alcoholic beverages brought even more intoxication, health risk, and nuisance to society.

### 2.A.3: Culture as Media (Part 1): Relating to the emergence of American identity and alcohol

By the 17th century, distilled spirits were available throughout the western world (Phillips, 2014). The emerging triangular trade routes linking Europe, Africa, and the Americas exploited the natural sugar resources of the Caribbean and served as a catalyst that furthered the proliferation of spirits worldwide (Higman, 2000). In the triangular trade, slave traders from Colonial America brought rum to Europe, and using the proceeds, they would purchase enslaved Africans. The enslaved Africans were brought to the Caribbean and sold to sugarcane plantations to harvest the sugar, which was then either used to manufacture rum in distilleries located in the Caribbean (such at Mount Gay Rum, established 1703 and still globally popular today) or brought from the Caribbean to the American colonies and sold to rum producers and distributors (See Illustration 4, Illustration 5). Triangular trading of sugar and alcohol during the age of colonialism facilitated massive economic expansion of the alcohol industry in America and played a significant role in tensions leading up to the American revolution (McCusker, 1970).

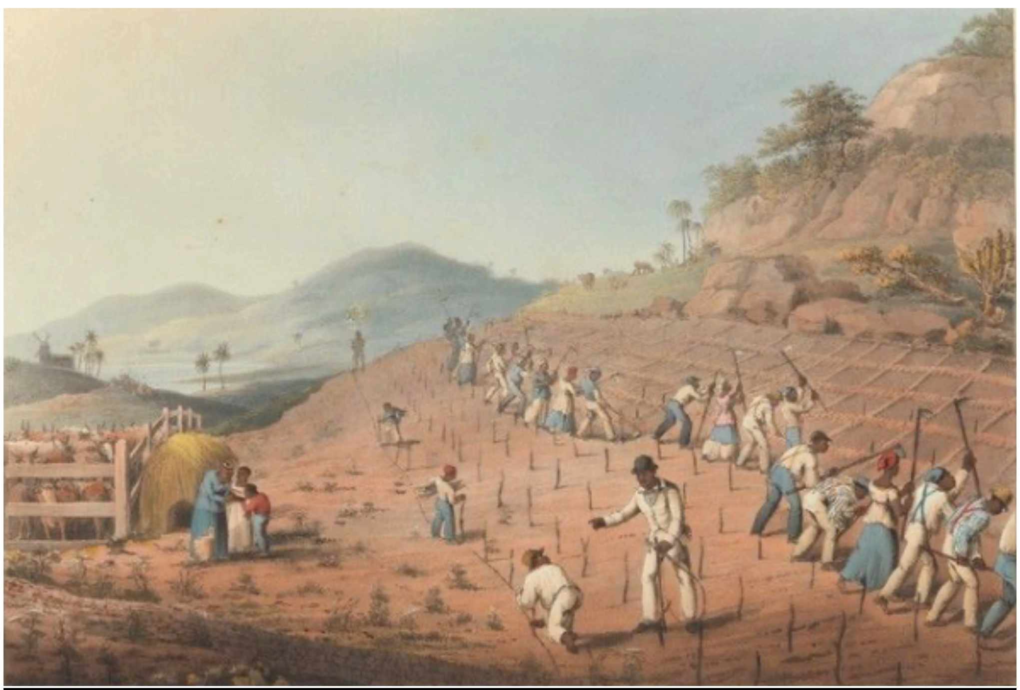


Illustration 4. Caribbean sugarcane plantations tended to by enslaved Africans, typical of transatlantic triangular trade of the time. From *Ten views in the island of Antigua: in which are represented the process of sugar making, and the employment of the Negroes, in the field, boiling-house and distillery / from drawings made by William Clark* in the Yale Center for British Art (n.d.).

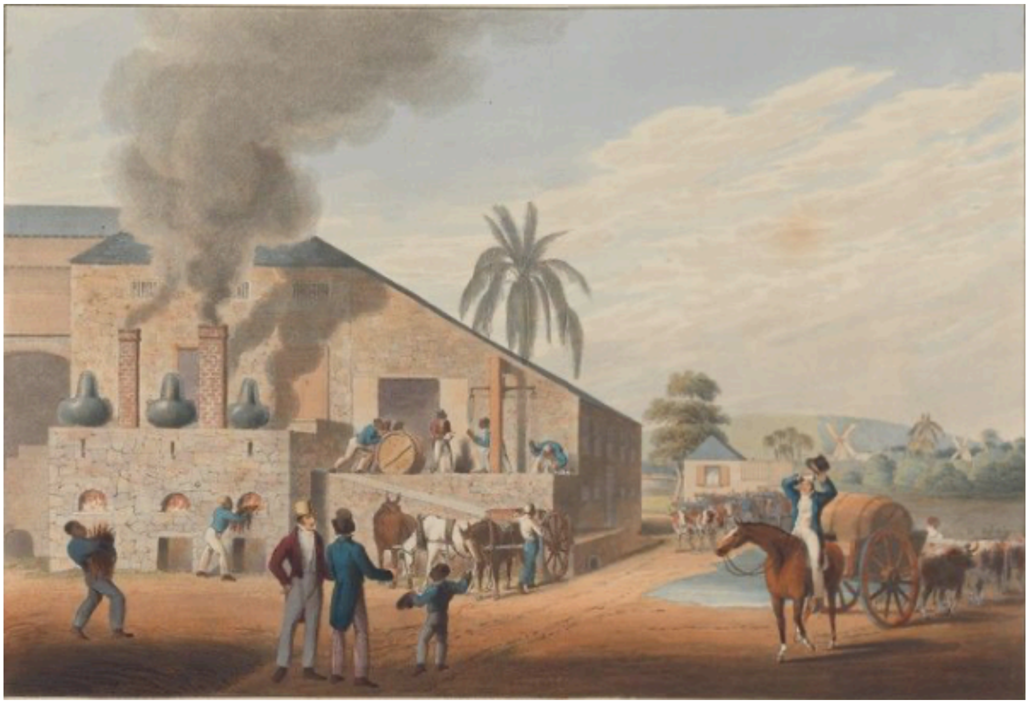
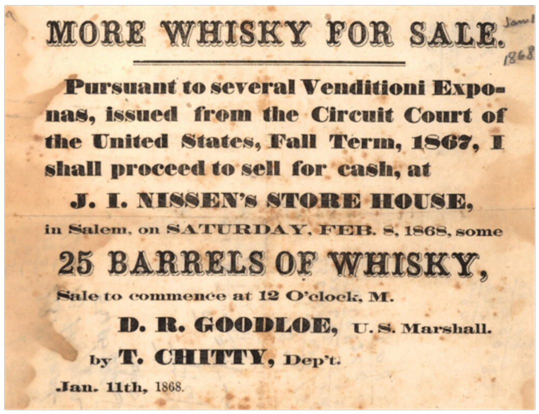


Illustration 5. Caribbean distillery tended to by enslaved Africans, typical of transatlantic triangular trade of the time. From Ten views in the island of Antigua: in which are represented the process of sugar making, and the employment of the Negroes, in the field, boiling-house and distillery / from drawings made by William Clark in the Yale Center for British Art (n.d.).

During the founding of America, many of the founding fathers had historical associations with the alcohol industry. For example, George Washington owned a distillery at Mount Vernon tended to by enslaved distillers (Breen, 2004). Washington's distillery was one of the largest of its time, was very successful, and contributed to his wealth. Washington was also known to have used alcohol as a political tool, as he tried to win over voters with large amounts of free rum, wine, and beer during his campaign for the Virginia House of Burgesses in 1758 (Chernow, 2011). Thomas Jefferson owned a vineyard at Monticello, although Jefferson was not nearly as successful in the alcohol industry as was Washington (Hailman, 2006). Nevertheless, Jefferson was a well-known enthusiast and importer of European wine (leaving a legendary collection in his estate's cellar), and he left his mark on the alcohol industry as leading one of the first attempts at cultivating fine wine in America (Meacham, 2012). Washington and Jefferson's interest in alcohol and business dealings in the alcohol industry are a reflection of the cultural norms of the time, as drinking was a very common social activity and economic force in early America. As such, the founding of America is linked to the burgeoning alcohol industry.

The industrial revolution and American capitalism advanced the alcohol industry in numerous ways during the $18^{th}$ and $19^{th}$ centuries. In 1775, the oldest American distillery still in operation was founded, called Buffalo Trace, which produced whiskey (Pacult, 2021). Early American whiskeys were commonly made by distilling fermented corn mash (as opposed to old-world whiskeys which are derived from barley, rye, and wheat mash). Corn was first domesticated by indigenous peoples in southern Mexico and is considered a sacred crop to many American Indian cultures (Huff, 2006), which suggests that American alcohol distillers appropriated corn in ways reminiscent of German distillers appropriating Islamic alchemy 200 years earlier. Many modern American whiskey producers have roots in the industrial revolution such as when Robert Samuels began making the whiskey that would go on to become Maker's Mark Bourbon, or when Jacob Beam came to Kentucky and sold whiskey that would later be named Jim Beam (Veach, 2013). The industrial revolution also ignited the advertising industry. The first commercial billboards (known as broadsides which were often glued to walls) appeared in the United States in 1835 (see Illustration 6, Illustration 7), direct mail advertising was firmly established in the 1840's (see Illustration 8), newspaper advertisements took off in the 1860's after color newspapers began circulating (see Illustration 9), promotional items such as branded paper fans and burlap sacks became popular in the 1880's, and the 1890's brought widespread color advertising in magazines (Richards, 2022). Just as tavern signs in medieval Europe pushed the envelope of advertising at the time, the alcohol industry in the United States was also on the frontier of each of these emerging media technologies during the industrial revolution.



MORE WHISKY FOR SALE.

Pursuant to several Venditioni Exponas, issued from the Circuit Court of the United States, Fall Term, 1867, I shall proceed to sell for cash, at

J. I. NISSEN'S STORE HOUSE,

in Salem, on SATURDAY, FEB. 8, 1868, some

25 BARRELS OF WHISKY,

Sale to commence at 12 O'clock, M.

D. R. GOODLOE, U. S. Marshall.

by T. CHITTY, Dep't.

Jan. 11th, 1868.

Illustration 6. Early Broadside Advertising Alcohol, From Duke University Libraries (1868).


Illustration 7. Early Billboard Advertising Alcohol, From Duke University Libraries (1916).



Illustration 8. Early Alcohol Direct Mail, From Duke University Libraries (1910)


Illustration 9. Early Color Newspaper Advertising Alcohol, From Duke University Libraries (1912)

The alcohol industry in the United States during the time of emancipation and Jim Crow (late 19th century to mid-20th century) was heavily racially segregated and regulated. The sale and consumption of alcohol was often prohibited or restricted for Black Americans, both legally through discriminatory laws and socially through racial discrimination in bars and other establishments (Guffey, 2012). Despite these restrictions, some Black Americans were able to establish their own successful businesses in the industry. One example of a Black-owned alcohol business in the early 1900's is the A. Smith Bowman Distillery, which was founded in 1934 (shortly after prohibition ended) by Abram Smith Bowman and his three sons in Fairfax County, Virginia (Sanfield, 2005). The distillery was one of the few Black-owned distilleries in the United States at the time, and it produced a variety of spirits, including bourbon, gin, and brandy. However, the

Bowman Distillery and other Black-owned businesses of the time often faced significant financial and legal challenges due to discrimination and lack of access to resources, which stifled their long-term success. For example, many Black-owned bars and nightclubs were targeted for closure by law enforcement, government officials, and white supremacist groups who used various tactics such as nuisance laws, zoning regulations, and hostile raids to shut them down (Godsil, 2006). White supremacists often said this was done in order to “maintain public safety" or "moral order," but in reality it was a way to suppress Black businesses and limit the social mobility of Black Americans (Trounstine, 2018). Jim Crow laws and culture, combined with the rapid growth of the alcohol industry at the time, interacted to concentrate wealth and alcohol industry leadership into nearly-exclusively white capitalists in America, driven by rum as-a-product-of-slavery and corn whiskey as-a-product-of-colonial-genocide-of-indigenous-people. While the A. Smith Bowman Distillery remained successful and stayed in the Bowman family for several generations, white capitalists representing a large New Orleans conglomerate - The Sazerac Company - eventually forced the Bowman’s out in a series of mergers and acquisitions helmed by billionaire William A. Goldring in the late 20$^{th}$ century that cemented Sazerac as a global spirits leader (Cunningham, 2003).

The alcohol industry in the late 19th century and early-20th century in America was also defined by the temperance movement and prohibition laws (Aaron et al., 1981). From 1920 to 1933, the United States implemented a nationwide ban on the sale, manufacture, and transportation of alcohol as a result of the 18th Amendment to the Constitution (National Archives, 2022). The temperance movement, which aimed at reducing or eliminating alcohol consumption, had been gaining momentum in the late 19th and early 20th centuries, primarily led by religious groups and social reformers who believed that alcohol was the source of many societal problems such as poverty, crime, and domestic violence (philosophically, the temperance movement of this period mirrored the earlier temperance movement of The Renaissance) (Andersen, 2013). The 18th Amendment had a significant impact on the alcohol industry, including producers, distributors, and retailers. Alcohol businesses were forced to close, adapt to the new laws, or turn to illegal means to continue producing and selling alcohol (Drexler, 2020). This led to the rise of illicit production and sale of alcohol, known as bootlegging, and many established industrialists participated in business that enabled bootlegging during prohibition. Anheuser-Busch Brewing Company, which is currently one of the largest beer manufacturers in the world, is one example of a well-known company that continued to operate during prohibition. During prohibition, Anheuser-Busch modified its production line to create a product they called “near beer,” which had a low alcohol content (see Illustration 10). However, historians have suggested that the company also continued to advertise and sell malt syrup and yeast to organizations intent on producing and distributing alcohol illegally through a network of underground speakeasies and other illegal channels (Klein, 2019). Another example is the Joseph Schlitz Brewing Company, which was one of the largest beer producers in the United States before prohibition. During prohibition, Schlitz also pivoted to producing near beer and other non-alcoholic products (see Illustration 11), but they maintained a significant operation producing and distributing malt extract that enabled illegal production of full-proof beer (Okrent, 2010). Many other alcohol manufacturers participated in the practices of selling near beer and ingredients needed to illegally produce alcoholic beverages (See Illustration 12). Organized crime syndicates assisted with distribution or even took control of much of the bootlegging operations, leading to an increase in violence (Mappen, 2013). Prohibition brought about several negative consequences, including an increase in crime and corruption, rise in consumption of dangerous forms of alcohol, and a decrease in government revenue from alcohol taxes (Thornton, 2014). As the public began to turn against prohibition, the 21st Amendment was ratified in 1933, which repealed the 18th Amendment and ended prohibition (National Archives, 2022). After the repeal, the alcohol industry quickly recovered, and legitimate businesses resumed their operations. However, the criminal element that had grown during prohibition persisted, and many organized crime groups continued to influence the alcohol industry.

BAVARIAN PREMIER KILLED AT MUNICH

GREAT EXCITEMENT PREVAILS AMONG THE PEOPLE IN THE STREETS OF MUNICH.

DEPUTIES ARE PANIC STRICKEN

IMPORTANT NEWS THE WORLD OVER

IMPORTANT HAPPENINGS OF THIS AND OTHER NATIONS FOR SEVEN DAYS GIVEN

THE NEWS OF THE SOUTH

PREMIER CLEMENCEAU NOW CONSIDERED OUT OF DANGER

LAGGING COMMITTEE WORK HASTENED BY CONFERENCE

TWENTY THOUSAND SOLDIERS ON THEIR WAY FROM FRANCE

MUNICH POLICE STATION STORMED BY SPARTACANS

CERTIFICATES HAVE BEEN OVERSUBSCRIBED

The Government has lifted the ban on cereals and relinquished its lease on the new Bevo building. Our plant, voluntarily tendered the Government, is now ready to resume full capacity production of

Bevo

THE BEVERAGE

America's Cereal Beverage

Like all Americans, we have made our sacrifice to help win the war. Now we are ready to renew our full duty as a great National industrial institution.

Anheuser-Busch St. Louis

Elgin and Waltham Watches

Watch, Clock and Jewelry Repairing

Z. T. HADLEY

Jeweler and Optician

GRAHAM, N. C.

Your Public Utilities.

THE GOVERNMENT is asking us to use our advertising space to ask you to economize in every way in the consumption of Electric Current.

PIEDMONT POWER & LIGHT CO.

Burlington, Graham, Gibsonville, Elon College, Haw River, Mebane.

Illustration 10. Prohibition-era Newspaper Ad for Bevo Near Beer in The Alamance Gleaner, From the North Carolina Digital Heritage Center (1919, February 27)


Illustration 11. Prohibition-era Broadside advertising Schlitz FAMO Near Beer, From the Smithsonian Institution. (n.d.)


Illustration 12. Prohibition-era Broadside on vehicle advertising 'Malt Syrup' which was a necessary ingredient to illegally product alcoholic malt beers at the time, From Klein (2019).

### 2.A.4: Culture as Media (Part 2): Relating to war culture/patriotism, trauma, and alcohol

During the early 20th century, the temperance-prohibition era coincided with World War I (WWI), the first large-scale war among fully industrialized nations. WWI was characterized by horrific trench warfare conditions, with soldiers living in small, cramped spaces under constant threat of enemy attacks and disease. This resulted in severe psychological trauma, including what was then termed "shell shock," a condition now recognized as post-traumatic stress disorder (Shephard, 2001). Of the 4.7 million American soldiers who served, many who experienced trench warfare returned home with profound psychological trauma after the Armistice in November 1918. In the teeth of profound trauma, alcohol was likely a prevalent coping mechanism used by soldiers during WWI due to its easy availability (The National WWI Museum and Memorial, 2023) (see Illustration 13). Veterans' and popular accounts suggest that alcohol was consumed at high levels, although definitive academic research on the topic is elusive. Some reports even suggest that military officers provided it to bolster morale or inspire a fighting spirit (Kamieński, 2019), a practice reminiscent of Agamemnon's effort to inspire his army during the Trojan war.


Image 13. U.S. Signal Corps photograph captioned: *"American soldiers in a captured German trench drinking beer out of steins and smoking cigars."* From the National WWI Museum and Memorial (2023).

After WWI, veterans faced challenges with alcoholism (i.e., addiction to alcohol). This was not a new phenomenon - throughout history mental health issues and alcohol abuse among soldiers has been well documented (Birmes, 2003; Schumm, 2012; Andreas, 2019) - but the scale of the problem was unprecedented (Shephard 2001). Exacerbating the problem, WWI veterans returned to an America entering prohibition, which meant that they would have to resort to illegal channels should they wish to continue to drink. Advertisers for most consumer goods, including within the alcohol industry, were quick to capitalize on demand for their products among veterans, often using themes related to nationalism or the war in order to sell their products (Pope, 1980). For example, Anheuser-Busch launched legal marketing campaigns for near beer specifically targeting veterans, using patriotic imagery to appeal to their sense of duty and sacrifice (see Illustration 10, Illustration 14). These marketing campaigns normalized beer as omnipresent in society (despite prohibition) and firmly established consuming beer as iconic among American war veterans.


Illustration 14. Anheuser - Busch Advertisement for Bevo Near Beer using Patriotic Themes, From Anheuser-Busch (n.d.).

Following the repeal of prohibition in 1933, the alcohol industry resumed its pre-prohibition practices of producing and marketing full-proof products. However, the industry had to navigate a more regulated landscape, with stricter rules and guidelines on how they could advertise their products (Pennock et al., 2005). The 21st Amendment gave states the authority to regulate and tax the alcohol industry, and while the laws did vary a bit from state-to-state, a major change from pre-prohibition practices across the country was that marketers had to be more truthful and accurate in their advertisements (Jurkiewicz et al., 2007; Congressional Research Services, Library of Congress, 2023). Whereas pre-prohibition, the alcohol industry often made exaggerated claims about the health benefits of drinking. After Prohibition ended, people were hopeful that new rules and regulation could help limit the negative effects of alcohol and encourage folks to drink responsibly; but it turned out to be more complicated than that (Jurkiewicz et al., 2007). It became clear that alcohol was a public health issue, leading to alcohol-related disease, and societal issues like addiction, underage drinking, and drunk driving (Katcher, 1993; National Safety Council, 2004).

The Great Depression, a global economic crisis that took place between 1929 and 1939, had a significant impact on the United States, both in terms of financial stability and psychological well-being (Bernanke, 2009). The Great Depression resulted in widespread poverty, with millions of Americans losing their jobs, homes, and savings (Benmelech et al. 2019). This caused a significant increase in psychological distress, leading to hopelessness and despair among the population. Historians suggest that many individuals abused alcohol to cope with the hardships of the Great Depression (Elder, 2018).

During the overall economic downturn, the alcohol industry developed adaptive marketing strategies, including messaging about thrift, patriotism, and attempts at counteracting social stigma about the effects of the great depression (e.g., humiliation about losing one's job) (Lefebvre et al., 2020). For example, advertisements for beer and spirits were targeted toward men and featured characters of hypermasculine roles such as cowboys of frontier explorers (see Illustration 15, Illustration 16). Historians have suggested that the hypermasculine marketing strategies of the time were intended to counteract American males' feelings of emasculation caused by their incapacity to financially provide for their families (Armengol, 2014), in effect promoting alcohol as a means of mitigating the psychological effects of unemployment and boosting confidence. Simultaneously, advertisements for wine were marketed toward women (see Illustration 17), presenting drinking wine as a sophisticated behavior to engage in (perhaps juxtaposed with the common lived experience of poverty). Marketing campaigns of the time commonly used references to beauty and aesthetics to appeal to women (Parkin, 2007; Golia, 2016) who commanded a growing share of household purchasing power (Lavin, 1995). Interestingly, these gendered social norms regarding beer/hard liquor (appealing to men) and wine (appealing to women) persist well into the 21st century (Carrotte, 2016; Auter, 2016). The combination of economic stress, psychological trauma, and opportunistic marketing by the alcohol industry contributed to an increase in alcoholism during the Great Depression (Elder, 2018). This was a cause of concern for public health officials, leading to the implementation of laws and regulations aimed at reducing alcohol abuse, such as restrictions on advertising and the hours of operation for bars and liquor stores (Fosdick et al., 2011). President Franklin D. Roosevelt's New Deal policies helped to ease the impact of the Great Depression and eventually led to economic recovery (Kennedy, 2009). However, the effects of the Great Depression persisted for many years afterward, highlighting its enduring impact on American society.


Illustration 15. Rodeo Lager Advertisement Depicting a Hypermasculine Cowboy, From Lehmann Printing and Lithographing Co. (1936).


Illustration 16. Lucky Strike Bourbon Advertisement Depicting a hypermasculine Frontier Explorer, From Lehmann Printing and Lithographing Co. (circa 1930s).


Illustration 17. California Belle Port Wine Advertisement Depicting a Woman With Idealistic Beauty Standards of the Time, From Lehmann Printing and Lithographing Co. (1935).

World War II (WWII; 1939-1945) was one of the most significant events in human history, with far-reaching consequences in almost every aspect of life. The technology of mid-century war and the brutal tactics employed by Adolf Hitler cast the German Blitzkrieg and Holocaust as the most violent and destructive campaigns in human history. More than 60 million souls were lost to direct warfare and genocide, and related famine, disease, and human displacement (Sturgeon, 2015). Fear and anxiety were high around the world (Bell, 2009; Shapira, 2013; Frounfelker, 2018). Accordingly, the alcohol industry was greatly impacted by WWII. During the war, the demand for alcohol increased and exceeded pre-prohibition levels, as soldiers, civilians, and war workers sought to relieve fear and anxiety by drinking (Lender et al., 1987; National Archives, 2015). Alcohol also became an important source of revenue for the United States government to fund their war efforts, using government-imposed taxes on alcohol production and sale (Ripy, 1999). However, WWII also had a negative impact on the alcohol industry. The resources required for alcohol production, such as grain, sugar, and fuel, were in short supply due to the demands of the war effort (Sturgeon, 2015; National World War II Museum, 2020). As a result, many countries eventually imposed restrictions on the production and sale of alcohol. In the United States, for example, the government imposed strict controls on the production of alcohol that used staple food grains and sugar, which were needed for food production. Despite waxing and waning production capability, alcohol marketing remained active throughout the war, often relying on emotional appeals to reach their customers, and using patriotic themes to connect their products with the war effort. For example, they encouraged Americans to be frugal and used images of war iconography to suggest that consuming their product was a way to support the war effort (See Illustration 18, Illustration 19) (Witkowski, 2003).


Illustration 18. Schenley Whiskey Advertisement circa 1943 promoting frugality and the priority of purchasing US. War Bonds. From American Century Shop (2024).


Illustration 19. Schenley Whiskey Advertisement using patriotic themes, From American Foreign Service Association. (1943).

Over 16 million Americans served in WWII and many of them experienced profound trauma from fighting on the frontlines. Historians have written extensively about drinking culture among American soldiers, including widespread celebrations with abundant French alcohol supply following the successful Normandy landings (see Illustration 20), U.S. infantry receiving daily rations of beer (See Illustration 21), and British naval soldiers receiving daily rations of rum (See Illustration 22) (Sturgeon, 2015; National World War II Museum, 2021). It is well documented that WWII veterans suffered disproportionately high levels of alcoholism (Herrmann et al., 1996; Shephard, 2001). Like the wave of alcohol-related problems that soldiers brought home from WWI, WWII accelerated the growing culture of alcohol abuse in America.

Illustration 20. American Soldiers Drinking on the Paris Streets following The Battle of Normandy, From Popperfoto (1944, July 1).

Illustration 21. American Soldiers receiving their beer ration on Sterling Island, 1944. From National World War II Museum (2021).


Illustration 22. British Navy Seaman receiving their Rum Rations circa World War II. From Central Press (1942, June 30).

The end of the war brought about a newfound sense of freedom and leisure time for many Americans, which led to increased consumption of alcohol per capita (a postwar trend that extended into the 1960's and accelerated in the 1970's) (Landberg, 2009). This period also saw the rise of a new postwar consumer culture, which normalized heavy drinking with a flurry of catchy and appealing marketing campaigns.

**2.A.5: Broadcast Media: Relating to television and alcohol**

The entire advertising industry in the 1950s and 1960's experienced tremendous growth, kicking off what many historians refer to as the Golden Age of Advertising (perhaps cemented in the public eye half-a-century later with the iconic fictious portrayal of Golden age advertising executive Don Draper in the cable television series Mad Men) (Cracknell, 2012). Following WWII, the American economy was booming, and businesses were ready to meet the increased demand for goods and services. Advertising agencies, which had played a crucial role in the war effort, turned their attention to promoting consumer products and mass consumption (Scott, 2009). Marketers sought to tap into Americans' desire for new things, promoting their products as essential for a modern post-war lifestyle. To attain maximum reach, advertisers developed a new strategy, television, which had emerged as a dominant force in American entertainment in the 1950s. Companies created television commercials that showcased their products in the best possible light, often featuring happy families, sleek new cars, the latest technology, and other aspirational vignettes (Samuel, 2001). Another feature of television advertisements was the use of catchy and memorable slogans and jingles. Advertisers began to position their products as integral parts of American life, using slogans like "Things go better with a Coke" (Coca-Cola), "Melts in Your Mouth, Not in Your Hands" (M&M's), and "When you say Budweiser, you’ve said it all” (Anheuser-Busch) to create lasting brand associations in consumers' minds (Richards, 2022).

In the mid-to-late 20th century, the television marketing space was shaped by a unique agreement within the alcohol industry. Spirit advertisements were voluntarily excluded, not by law but by a self-regulatory action within the industry (Hacker, 1998). This move was an industry-enforced caution rather than a mandate, and it paved the way for the alcohol industry to evade regulators in ensuing decades because they had a supposed track record of being responsible (although their “responsibility” is not supported by evidence). In a self-regulated system, the industry establishes its own advertising guidelines, and there are no repercussions for noncompliance. This is problematic because research has shown that despite self-regulation - for example, geared towards limiting youth exposure to alcohol advertisements - undesirable outcomes related to alcohol advertising are still common - for example, underage drinking (Anderson et al., 2009). It is worth noting that self-regulatory exclusion of spirits did not necessarily limit the number of alcohol ads appearing on television, but rather that vacuum was filled by prolific advertisements for beer (Miller, 2002) (See Illustration 23).

Illustration 23. Budweiser television commercial advertisements through the late 20th century. Left: ’Where there’s life, there’s Bud’ (1956) commercial depicting Budweiser beer alongside beach imagery and grilling burgers, suggesting that their beer is a necessary complement to everyday leisure. Right: ‘Wassup’ (1999) commercial depicting Budweiser beer being consumed by a group of young adults talking to each other on cordless phones, suggesting that their beer is a necessary complement to modern socializing. From BetterBrandsSC (2013) and Barnett (2022, June 14), respectively.

The Golden Age of Advertising also coincided with a heightened societal awareness and growing concern around alcohol abuse. The United States was living through the aftershocks of three significant historical events: World War I, the Great Depression, and World War II. Each of these events, traumatic in their own right, created a syndemic of trauma on the collective American psyche (Jones et al., 2007; Hirschberger, 2018). These shared societal traumas formed a backdrop against which concerns about alcohol abuse took on heightened importance. In the post-WWII era, scholarly research initiatives such as the Yale Center of Alcohol Studies (Wiener, 2020) and grassroots movements such as Alcoholics Anonymous (Wilson, 2001) put forth a disease theory of alcoholism that meticulously described how alcoholism was a result of underlying biological, psychological, and social mechanisms that were treatable, rather than moral shortcoming’s that temperance movements had argued throughout history (Pennock et al., 2005). The government began recognizing the gravity of these issues, leading to increased efforts to tackle alcohol abuse. The focus shifted from merely controlling the supply side, as evident from the industry's self-regulation and government alcohol policy, to addressing the demand side through research, education, and treatment programs. The 1970s marked a significant milestone in the formal study and understanding of alcohol abuse and alcoholism. Under the Nixon administration, the Comprehensive Alcohol Abuse and Alcoholism Prevention, Treatment, and Rehabilitation Act of 1970 was signed into law on December 31, 1970 (U.S. Congress, 1970A). This Act established the National Institute on Alcohol Abuse and Alcoholism (NIAA), cementing alcoholism as a national concern warranting systematic research and intervention.

With vitalized national attention towards addiction and calls for the government to act, cigarette advertisements were removed from radio and television in 1971 (Warner, 1979); however, alcohol advertisements were allowed to persist. This discrepancy arguably reflects the societal norms of the time, where alcohol, despite being recognized as potentially harmful, was still deemed acceptable in moderation. Perhaps recognizing impending regulatory action on their products that would parallel action against the tobacco industry, the alcohol industry began evolving new advertising techniques that would infuse alcohol advertising into all aspects of life (Schuster, 1987).

### 2.A.6: Culture as Media (Part 3): Relating to product placement, sponsorship, and celebrity endorsement

Product placement is a form of advertising where branded goods are shown within a popular production, such as a film or television show. This advertising technique has been increasingly popular since the 1970s and became ubiquitous after the 1982 success of the film E.T. masterfully placing Reese’s Pieces candy in its pivotal moments (Babin, 1996). Product placement allows brands to incorporate their products into popular culture, influencing consumers in a more subtle and nuanced way (Newell et al. 2006). Alcohol product placement in the James Bond films during the late 20th century and early 21st century is particularly notable. Wilson et al. (2018) systematically analyzed six decades of James Bond films and found 109 alcohol product placement events (4.5 per film) that included a variety of alcohol brands including Smirnoff Vodka and Heineken Beer (See Illustration 24). The James Bond anthology is just one example of many alcohol product placement campaigns throughout films and television shows. Other noteworthy instances include Budweiser's omnipresent product placement across countless films and television series, such as "Top Gun" (Scott, 1986) and "Friends" (Crane et al., 1996).

Illustration 24. James Bond's long history of Smirnoff product placement. Left: Dr. No (1962). Right: No Time to Die (2021). From Young (1962) and Fukunaga (2021), respectively.

Sponsorship involves a company paying to be associated with a specific entity, such as an entertainment event, a sports team, or an artistic performer. This advertising technique tries to create a positive association between the brand and the sponsored entity, increasing visibility and brand awareness (Meenaghan, 2021). In other words, if someone is having a good time at a sporting event, and they see an alcohol advertisement, then they might associate the brand of the alcohol being advertised with the good feelings they are experiencing from the sporting event. From the 1970s onwards, alcohol brands have sought to associate themselves with many sports events, music festivals, and cultural gatherings (Crompton, 1993). For example, Miller Lite's long-term sponsorship of the Dallas Cowboys NFL football team is one of the most recognized partnerships in the sports industry. Starting in 1991, the Dallas Cowboys-Miller Lite collaboration allowed Miller Lite to tap into the global audience of Dallas Cowboy's fans and positioned the brand as a key component of the fan experience (Fischer, 2021) (See Illustration 25). Such enduring sponsorships of beloved entities creates life-long brand association and infuses entire generation's affinity for a sports team with alcohol brands, and research has demonstrated that alcohol sports sponsorship is associated with increased levels of drinking amongst youth and with hazardous drinking among adults (Brown, 2016).


Illustration 25. Advertisement depicting the Cowboys Lite House, sponsored by Miller Lite, which is an area directly outside of Cowboys Stadium in Arlington, TX that is specifically designed to drink Miller Lite before beer goes on sale in the stadium (pregame) and after beer sales are cut off in the stadium (postgame), effectively creating an all-day Miller Lite drinking environment. From Dallas Cowboys (n.d.).

Celebrity endorsement is an advertising technique where famous people, such as Hollywood stars, musicians, athletes, or prominent cultural figures, are used to promote a product. These endorsements can be incredibly influential as they offer the brand a chance to leverage the celebrity's fame, charisma, and appeal (Bergkvist et al., 2016). Beginning in the early 1900s and rapidly accelerating in the 1970s, alcohol brands have capitalized on the power of celebrity influence to elevate their brand's image and reach a wider audience (Atkin et al., 1983). Perhaps one of the most iconic examples of this is the partnership between rapper Sean 'Diddy' Combs and Ciroc Vodka, which experienced a meteoric rise in the mid-2000s and accelerated vodka consumption among young Black Americans, a demographic that was previously hard to reach by vodka brands (Siegel et al, 2012). Ciroc's penetration into hip-hop culture via celebrity endorsement speaks to the power of the advertising technique. Another popular example of alcohol celebrity endorsement is George Clooney and his tequila brand Casamigos. Launched in 2013, Casamigos quickly became a market success, largely thanks to Clooney's star power and influence. By aligning the brand with Clooney's persona and lifestyle, Casamigos was able to increase its market share and enhance its image as a premium, aspirational product (Kellershohn, 2022) (See Illustration 26).


Illustration 26. Advertisement for Casamigos Tequila featuring celebrity endorser George Clooney. From Domenic (n.d.).

Product placement, sponsorship, and celebrity endorsement all gained popularity within the alcohol industry in the late 20th century and are ubiquitous in present day. Research has shown that these forms of advertising are effective in promoting brand awareness and increasing alcohol use, particularly among youth and young adults (Jernigan et al., 2008; Bragg et al., 2018; Sargnet et al., 2020). Furthermore, the infusion into popular culture achieved by the alcohol industry using these forms of advertisements have solidified a culture of drinking in America. As reviewed here and in previous sections of this literature review, (1) the formative period of American identity (i.e., the era of the American Revolution and proliferation of American Capitalism) established dinking as iconic, (2) drinking has been sought by people in the teeth of trauma and adversity, and (3) drinking has become an integral part of all aspects of culture, in essence appearing in all forms of mass media, events, and popular proceptions. Near the end of the 20th century, the omnipresent alcohol industry was poised to further their cause with the advent of an entirely new class of technology.

**2.A.7: Digital Media: Relating to the early internet and alcohol**

In the mid-1990s, the emergence of the internet started to garner widespread public attention (Dutton, 2013), and the alcohol industry was quick to identify its potential for advertising. Some of the earliest forms of digital advertising were static online ads. These consisted primarily of banner ads placed on websites, similar to a digital billboard (Manchanda et al, 2006). These early static ads were simple in design, mainly text-based and with low-resolution images. Brands like Budweiser were among the pioneers of this new advertising technique, purchasing banner space on popular platforms to tap into their large user base (Davik, 2000). The static nature of these ads, however, had limited potential to fully engage consumers.

The growth of the internet and advancements in bandwidth and speed soon made room for more immersive forms of advertising, one of which was homepage website advertising. Major alcohol brands started creating their own dedicated websites, a digital home for their products (Gordon, 2010). These websites served as both a form of direct advertising and as platforms for consumers to learn about their products. For example, Jack Daniel's launched its dedicated website in the mid 1990s (and remains active to this day), filled with information about its whiskey-making process, cocktail recipes, and historical tidbits, subtly telling a compelling story centered on the brand (McCune, 1998) (See Illustration 27). Another form of website marketing is programmatic advertising, which emerged as a means to more specifically target potential consumers. In this technique, alcohol advertisers would pay to place their ads directly onto certain websites that were frequently visited by their target demographics. Programmatic advertising differs from banner ads because programmatic content is dynamic, interactive, and often links directly to the product’s homepage. This allowed for a more efficient use of advertising budgets and a higher likelihood of reaching consumers who would be interested in their products (Hammersley, 2003).

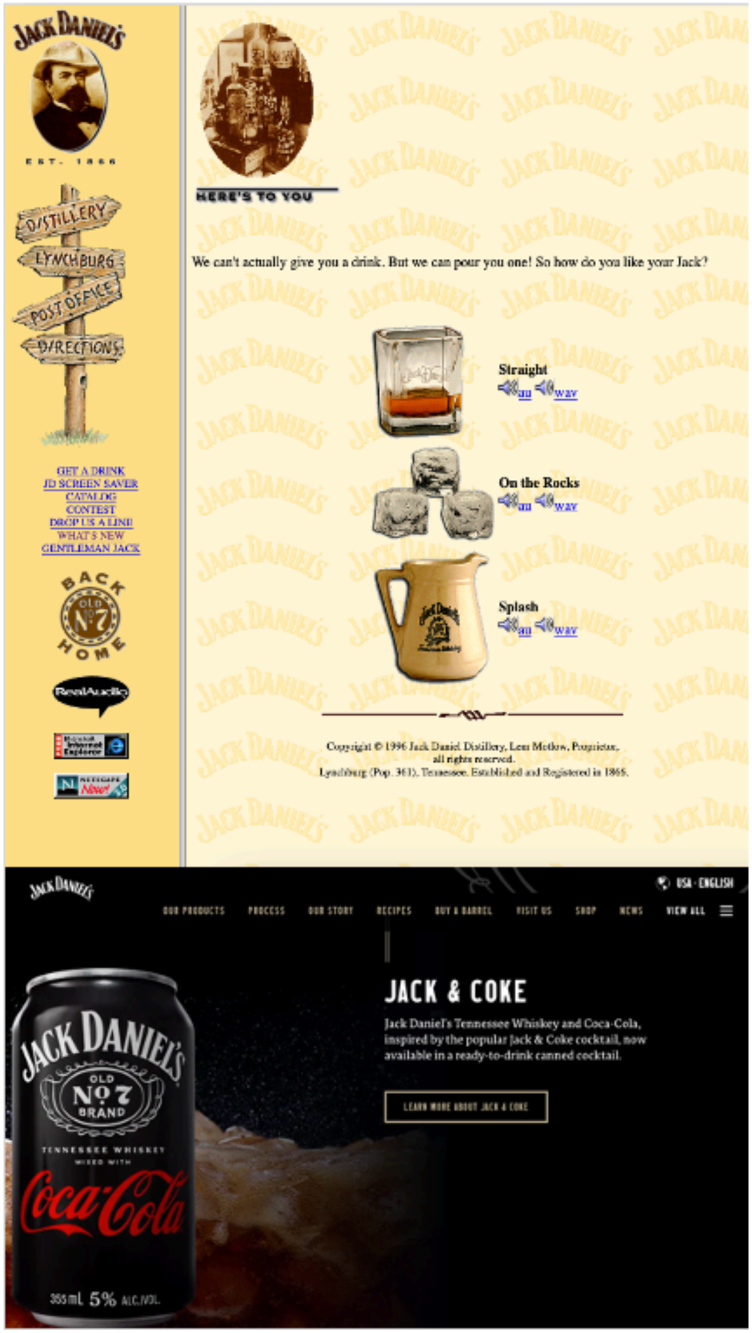



Illustration 27. Early (top, circa 1997) and modern (bottom, circa 2023) stills of Jack Daniels website. Both display various forms of consuming Jack Daniel's as well as hyperlinks to other information about their product with technology and design aesthetics typical of their time. From and Jack Daniels (1997) and Jack Daniel's (2023), respectively.

The use of email campaigns further personalized brands' connection with consumers. As the internet became a primary means for communication, alcohol companies started collecting email addresses of those who showed interest, often from their own websites or through online contests and promotions (Hastings, 2009). Email campaigns allow for brands to leverage specific information about the consumer, such as their name or geographic location, to tailor a message to each specific receiver, which helps foster a deeper connection with the brand (Kingsnorth, 2022). Thus, alcohol email campaigns offered a more personalized mode of advertising, allowing brands to connect directly to consumers in unprecedented ways.

Moving into digital mediums signaled a turning point in alcohol advertising. Through digital ads, alcohol brands were not only able to reach a wider audience but also customize their advertising content to appeal to specific demographics and individual consumers. As a result, the dawn of the digital age expanded the reach of alcohol advertising exponentially, setting the stage for the next chapter in the industry's long and storied history.

### 2.A.8: Social Media: Relating to the modern internet and alcohol

Social media marketing (SMM) is a digital advertising strategy that uses social media networks like Facebook, Twitter (now X), Instagram, and LinkedIn to promote products or services. It involves creating and delivering tailored ads that appear in the social media feeds of a specified audience. These ads can take different forms, such as still images, videos, carousel ads (i.e., compilation of multiple images or videos with unique links in one ad), and sponsored content that blends into the platform's standard content, making the ad appear alongside standard content and blurring the lines between content generated by users and by commercial entities. Influencer marketing is another form of SMM, similar to celebrity endorsements, but involving social media users with a large number of followers who may not necessarily be famous outside of their platforms. The primary goal of SMM is to connect with a specific audience segmented by demographics, interests, behavior, among other relevant factors. This precise targeting allows businesses to advertise their products or services to a group of potential customers effectively, potentially boosting brand awareness, customer interaction, and sales (Dwivedi et al., 2015; Jamil et al., 2022; Kingsnorth, 2022).

Since the late 2000s, the alcohol industry has been effectively using SMM (Hastings, 2009). Social media platforms have allowed alcohol brands to establish direct connections with their consumers by advertising their products, crafting their brand image, and directly engaging with their audience (Lobstein et al, 2016; Noel et al., 2020). A noteworthy example of this strategy involves Smirnoff Vodka's Facebook page. An analysis of internal documents from the alcohol industry suggests that Smirnoff's social media approach was intentionally designed to appear as if a friend, rather than a company, was posting content (See Illustration 28). This technique, along with the strategic encouragement of user comments, was aimed to amplify the brand's appeal. However, it was also revealed that Smirnoff insiders were aware that nearly 75% of their Facebook followers were under the legal age to purchase and consume alcohol (Hastings, 2009). Echoing earlier periods in history, Hasting's (2009) report raises important questions about the ability (or lack thereof) for the alcohol industry to self-regulate on social media, as research has shown that SMM of alcohol is associated with initiating, using, and abusing alcohol, especially among youth and young adults (Moreno et al., 2014; Jernigan et al, 2017, Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022).

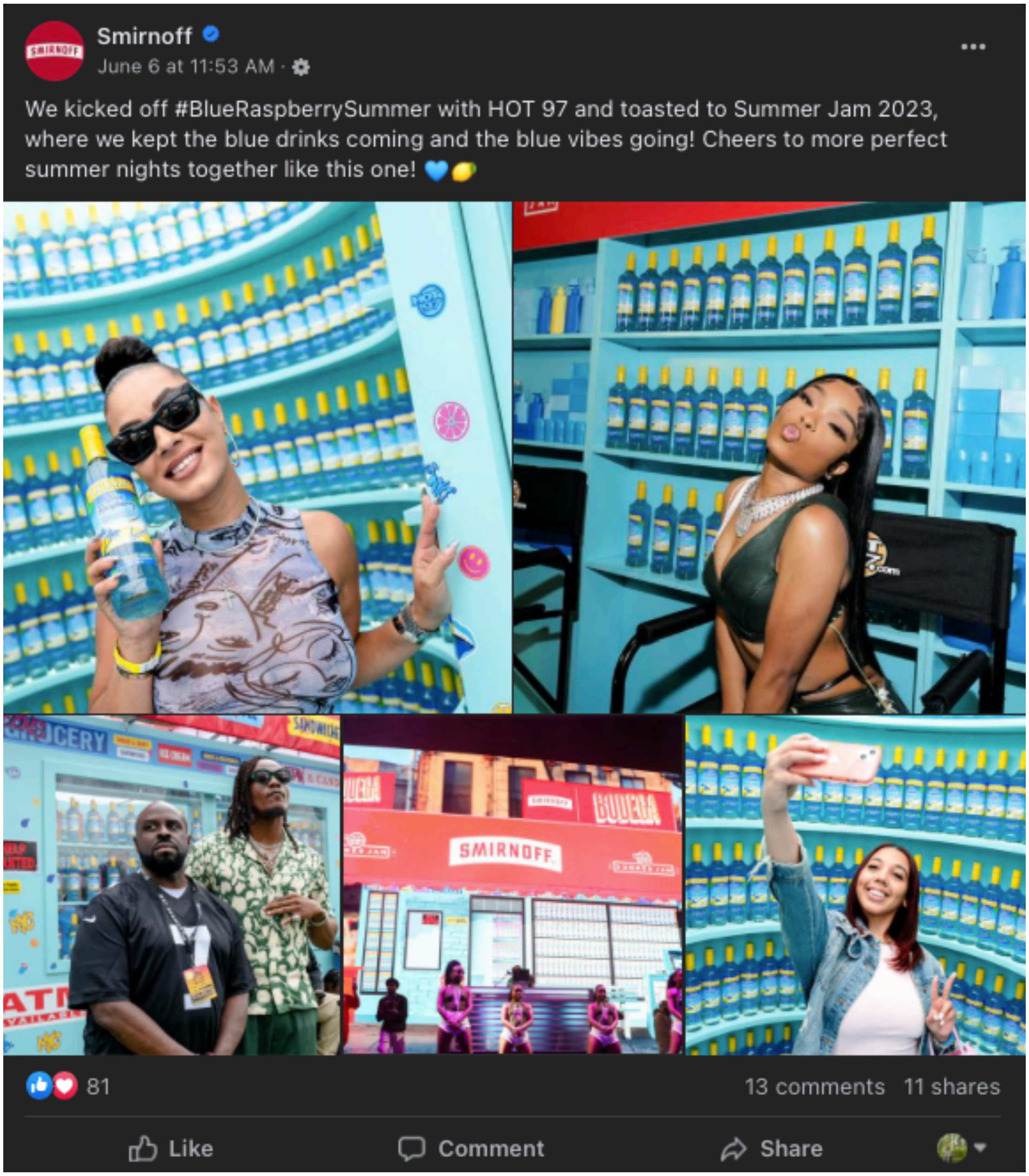



Illustration 28. Facebook post from Smirnoff Vodka, circa 2023. The post shows the insidious nature of SMM, with content appearing like user-generated content (i.e., selfies, hashtags, emojis, likes, comments, and shares), virtually indistinguishable in form. From Smirnoff US (2023).

The proliferation of SMM by the alcohol industry presents notable risks, especially considering alcohol companies' ability to deliver tailored messages to consumers combined with the knowledge that many social media users are underage (Auxier et al., 2021). The blurred lines between user-generated and commercial content raise serious questions about transparency, deception, and knowingly exposing underage youth to alcohol advertisements. Furthermore, in the face of a major societal disruption, such as a war, economic recession, or social upheaval, the potential perils associated with social media alcohol advertising could be amplified. The alcohol industry has a long history of leveraging the fear and anxieties of society in times of challenge (e.g., WWI, the Great Depression, WWII; reviewed in detail in previous sections), and it should be assumed that the alcohol industry will continue the exploitative practices associated with marketing amidst chaos. Looking ahead, the intersection of alcohol advertising and social media will need to be a focus of scrutiny and regulatory innovation. Researchers, policy makers, and society as a whole must grapple with these challenges to ensure that the alcohol industry does not abuse SMM.

### 2.A.9: Summary: Relating to the Three Key Points

Chapter 2, Part A has documented a historical narrative of alcohol and the alcohol industry. However, the prevalence of alcohol consumption alongside this narrative has not been fully examined. The historical prevalence of alcohol consumption is complex and nuanced, with definitive empirical evidence unavailable. Furthermore, while metrics such as historical alcohol consumption per capita exist, they often fail to capture the multifaceted drivers of alcohol use across different historical contexts. Before the Industrial Revolution, it is particularly challenging to disentangle alcohol consumption driven by essential needs - such as nutrient supplementation and water fortification - from consumption for recreational purposes or abuse. For example, estimates suggest that in medieval Europe (circa 1500), when alcohol fortification of water was essential for survival due to waterborne pathogens associated with increased population densities, alcohol consumption may have reached as high as 100 liters per person per year (Hanson, 2013). More systematic tracking of alcohol consumption begins in the late 19th century. For example, Holmes et al. (2017) tracks global alcohol consumption per capita from 1890 to 2014, indicating a general upward trend across the majority of countries. Specifically, in the U.S., per capita alcohol consumption increased from 5.3 liters in 1890 to 7 liters in 2014 (Holmes et al., 2017). These trends suggest that, at least in a modern context, alcohol consumption has risen alongside the proliferation of the alcohol industry.

The proliferation of the alcohol industry has a context deeply rooted human history. For thousands of years, alcohol was produced and consumed in small batches for rituals and celebrations (McGovern et al., 2004; Carrol et al., 2008; Albarella et al., 2011; Dietler, 2020; Lipscomb, 2020). The modern alcohol industry emerged in ancient Rome when the accumulation of power and knowledge of the Roman empire enabled wine to be produced at scale (Cato et al., 1934; Davies, 1971; Standage, 2006; Gately, 2008; Laes et al, 2013; Phillips, 2014). From the advent of the alcohol industry arises *Three Key Points*.

***Key Point 1: The alcohol industry's success is linked to advances in communication technology.***

The relationship between the alcohol industry and communication technology began with hand-written documents like *De Agri Cultura*, which were some of the first resources that discussed large-scale alcohol production (Cato et al., 1934). Additional hand-written or hand-illustrated communication technology that furthered the alcohol industry was pub signs, which make it clear where passers-by could stop to consume an alcoholic beverage (Richards, 2022).

The next significant advance in communication technology that served the alcohol industry was mass printing. Moving from hand-written to printed documents, *Liber de arte distillandi de simplicibus* made information about alcohol distillation widely available and helped spread this knowledge to a larger group of people (Bruschwig et al, 1500; Gately, 2008). Printed copies of *De Agri Cultura* and *Liber de arte distillandi de simplicibus* advanced the practice of alcohol making from small local batches for specific uses to one of the largest and most influential industries in history. Printing technology also allowed for information and advertisements about alcohol to start appearing in flyers, billboards, newspapers, direct mail, and magazines, making alcohol brands more widely known and popular (Richards, 2022). The alcohol industry was able to leverage printing to develop specific brand appeal and make it widely known to the public where to purchase their products.

Television made alcohol advertising even more impactful by using visual and sound effects (Miller, 2002; Cracknell, 2012). Alcohol ads could now show a more enticing lifestyle associated with drinking, making the messages even more appealing. Television also broadened the alcohol industries' reach as televisions began to appear in nearly every household and became a focal point of entertainment (Samuel, 2001).

Another impactful advancement in communication technology came with the internet and social media (Dutton, 2013). This expanded the reach of the alcohol industry to an even larger audience and allowed the industry to engage with customers in a more direct and personalized way with unlimited access from the comfort of one's computer or mobile device. Furthermore, with targeted advertising and interactive content, alcohol brands could connect with consumers on an individual level (Kingsnorth, 2022).

In short, each new technology has provided new ways for the alcohol industry to grow its reach and influence people's attitudes and behaviors. This relationship has been a key factor in the alcohol industry's success.

***Key Point 2: The alcohol industry has a long history of distributing its products to, and exploiting, vulnerable people.***

Our narrative of alcohol's association with vulnerable people begins in ancient Rome where wine rations were an important source of hydration for the legionaries as well as a tool for motivation and a form of control (Davies, 1971). Alcohol was used to influence these soldiers, setting the stage for an enduring trend of exploitation of soldiers, an echo from early literature such as *The Iliad* and a phenomenon that would repeat in various war contexts throughout history (Fagles et al., 1998; Shephard, 2001; Schumm, 2012; Sturgeon, 2015; Kamieński, 2019; National World War II Museum, 2021).

In medieval Europe, alcohol was often consumed in large quantities by peasants as a form of nutrition and an escape from harsh realities (Carlin et al., 1998). This exploitation was twofold: first, these peasants were essentially a captive market, as they had few other options for nutrition or entertainment. Second, the consumption of alcohol and proliferation of drunkenness often perpetuated a cycle of poverty and dependence, further exacerbating their vulnerability (Glatt, 1977).

In the United States, the powerful alcohol industry was built on a foundation of profound injustice. Slavery and the displacement and genocide of indigenous populations played a crucial role in the establishment of alcohol production, especially in rum and corn whiskey distilleries (McCusker, 1970; Higman, 2000; Huff, 2006). This represents one of the darkest aspects of the alcohol industry's history, and its impacts are still felt today in the form of racial inequities and disparate societal harm.

In more recent times, the industry has often preyed on those living on society's fringes, including those wrestling with psychological trauma, such as war veterans and people experiencing poverty (Pope, 1980; Armengol, 2014; Lefebvre et al., 2020). This predatory behavior further demonstrates an industry that has capitalized on the vulnerability of others.

In each of these instances, the alcohol industry has, intentionally or otherwise, profited from the vulnerability of certain groups. This aspect of the industry's history is a complex one, and it is crucial that we acknowledge and address these issues as we move forward.

***Key Point 3: The media portrayals of the alcohol industry are lasting.***

Media portrayals of the alcohol industry have had a lasting and powerful influence on societal attitudes and behaviors. The concept of war, soldiers, and patriotism has been frequently intertwined with alcohol, tracing back from the fictional ancient Trojan Wars and extending to modern war campaigns (Pope, 1980; Shephard, 2001; Schumm, 2012; Sturgeon, 2015; Kamieński, 2019; National World War II Museum, 2021). In many narratives, alcohol is seen as a companion to soldiers, underpinning their courage and camaraderie. Alcohol's role in coping with stress has also been strongly emphasized in media, from wartime traumas to the hardships of the Great Depression and the struggles of poverty (Elder, 2018). The use of alcohol as a coping mechanism against life's harsh realities has been a recurring theme, often with little acknowledgement of the potential for dependency and harm.

Additionally, alcohol media has played a significant role in reinforcing gender norms within the context of alcohol consumption. Men are commonly associated with beer and hard liquor, denoting strength and masculinity, while women are often depicted sipping wine, symbolizing refinement and femininity (Parkin, 2007; Armengol, 2014; Golia, 2016; Carrotte, 2016; Auter, 2016; Lefebvre et al., 2020). These enduring portrayals have shaped public perceptions, reflecting, and reinforcing societal gender norms and expectations.

The portrayals of the alcohol industry in media extend beyond war, stress, and gender norms. Alcohol, in many forms, is often tied to notions of sophistication, status, and celebration. From the martinis that James Bond sips in luxury settings to the suave George Clooney nursing a glass of tequila, these depictions subtly suggest that consumption of alcohol is a sign of success and affluence (Wilson et al., 2018; Kellershohn, 2022).

These portrayals, while sometimes subtle, have a lasting influence, shaping societal attitudes and behaviors towards alcohol, and reinforcing its central role in various aspects of life, both in times of joy and sorrow. However, they often overlook the potential negative consequences of alcohol.

***Limitations of these Historical Perspectives and the Three Key Points***

In Chapter 2, Part A, the *Three Key Points* have been presented as central to understanding the history of alcohol marketing. It is important to note that this historical perspective is not the only historical perspective that could be argued. John Lewis Gaddis

- one of the most the most accomplished historians at work today - writes in “The Landscape of history: How historians Map the Past” that History is not just about listing facts; it is about trying to understand and interpret series of events in meaningful ways (Gaddis, 2022). According to Gaddis, history is a balance of objectivity and context, where storytelling helps to make sense of the past in a way that is both accurate and accessible. Gaddis begins by drawing an analogy between historians and cartographers. Just as cartographers select certain elements to include in a map based on its intended use, historians select aspects of the past to construct narratives that are comprehensible and relevant. No map ever accounts for every feature of a landscape. Indeed, it would be impossible to. Similarly, no history ever accounts for every event in time, because, again, it would be impossible to do so. This analogy elegantly illustrates the selective nature of history and the responsibility that historical reviews bear in shaping how we understand the past.

Through the ages, the historical perspective that is presented in Chapter 2, Part A frame the persistent narratives that the alcohol industry is associated with - *leveraging advances in communication technology, exploiting vulnerable people, and creating lasting portrayals* - as consequential to the alcohol industry’s prosperity. This historical perspective suggests that these persistent narratives drove the alcohol industry forward though thousands of years. However, it is crucial to reiterate that this is not the only historical perspective. Future research can build upon the perspective presented here to further explore the intricate history of the alcohol industry, particularly in terms of how it utilizes media to advance its commercial practices.

### *Convergence of the Three Key Points: Alcohol, Media, and the COVID-19 Pandemic*

The ever-evolving landscape of alcohol media has been significantly influenced by technology, societal vulnerabilities, and lasting portrayals (i.e., the *Three Key Points*). Global events have also had an impact on how media develops and transforms. Beginning in 2019, an unprecedented global event was widely covered by all forms of media, the infectious virus - SARS-CoV-2 which causes COVID-19 - rapidly spread across the world, causing chaos, mass death, and widespread societal disruption (Centers for Disease Control and Prevention, 2023). The COVID-19 pandemic presented a unique crossroad for the *Three Key Points* of the evolution of alcohol media. The COVID-19 pandemic brought massive changes in the use and function of media (Bendau et al., 2021; Nabi et al., 2022; Price et al., 2022), profound psychological trauma (Pfefferbaum et al., 2020; Panchal et al, 2023), and the proliferation of certain media narratives (Cinelli et al, 2020; Evanega et al, 2020), producing a convergence of the *Three Key Points* in an unforeseen interaction.

First, COVID-19 intensified the public's relationship with the digital world. The lockdowns and social distancing measures, designed to control the spread of the virus, pushed people online in droves, including participating in online activities that were previously held in person such as education (Pham et al., 2022), work (Fernandez et al., 2020; Al-Habaibeh et al., 2021), family visits (Kelly et al., 2023) and leisure (Meier et al., 2021), which was associated with an unprecedented surge in social media usage (Pandey et al., 2020; Cho et al., 2023). This digital migration was not just a social phenomenon; it became a significant tool for governance and public health communication (Anwar et al., 2020). The President of the United States at the time, among other leaders, leveraged social media platforms as primary channels of communication. Simultaneously, key safety and health agencies used these platforms to disseminate crucial updates about the pandemic.

These phenomena positioned social media, particularly the social media platform Twitter, at the forefront of communication (Haman et al., 2020; Rufai et al., 2020). As suggested by history, this powerful development in communication technology was available to the alcohol industry as a tool for expanding their commercial activities.

Second, central to the convergence of the *Three Key Points* lay the human condition - one marked by fear, trauma, and vulnerability caused by the COVID-19 pandemic (Pfefferbaum et al., 2020; Panchal et al, 2023). These emotions, coupled with the physical limitations of lockdowns, compelled many individuals to seek solace by any means. Drawing from qualitative and quantitative research about coping, scholars have taxonomized methods that people used while seeking refuge from pandemic-related trauma, including “reaching up” (i.e., cultivating spiritual or religious practices such as meditation or prayer), “reaching down” (i.e., connecting with nature and the Earth through activities such as birdwatching or gardening), “reaching in” (i.e., tapping into internal resilience forged through other difficult experiences in one’s life), and “reaching around” (i.e., fostering social connection while maintaining physical distancing, particularly online) (Fraenkel et al., 2020; Garfin, 2020). Historically, such moments of mass societal vulnerability have been used by the alcohol industry (Pope, 1980; Shephard, 2001; Schumm, 2012; Armengol, 2014; Sturgeon, 2015; Kamieński, 2019; Lefebvre et al., 2020; National World War II Museum, 2021) and this instance was likely no exception.

Third, the media portrayed many narratives during the COVID-19 pandemic. From pro-health ‘flatten the curve’ messages to counterproductive vaccine conspiracies, the narratives that were pushed across many forms of media became imbedded into the minds of many people and shaped their attitudes and behaviors about COVID-19 (Cinelli et al, 2020; Evanega et al, 2020). Though, messages related directly to the virus were not the only thing proliferating. Media portrayals from the alcohol industry during this period also had potential for lasting impressions. From the new availability of delivery alcohol to the promotion of virtual happy hours, alcohol was often depicted as a necessary component of lockdown life, a way to cope with stress, or a means of maintaining social connections in an isolating period (Atkinson et al., 2021; Martino et al., 2021; Gerritsen et al., 2021; Pakdaman et al., 2021). These narratives, delivered through print, broadcast, and digital media, quickly became part of the COVID-19 cultural consciousness.

The confluence of high social media usage, heightened psychological vulnerability, and influential narratives around alcohol during the COVID-19 pandemic created an unprecedented environment within which the alcohol industry could expand their commercial activities. Armed with an in-depth understanding of the *Three Key Points*, and positioning them alongside the COVID-19 pandemic, the remainder of this dissertation reviews and investigates how the alcohol industry capitalized on the high prevalence of social media use (i.e., advances in communication technology - *Key Point 1*), to advertise alcohol to people experiencing pandemic-related trauma (i.e., distributing its products to, and exploiting, vulnerable people - *Key Point 2)* which coincided with new and persistent changes in drinking behavior (i.e., media portrayals of the alcohol industry are lasting - *Key Point 3)*.

## Part B: Modern Evidence: Epidemiology of Alcohol Use among Young Adults, Social Media, and the COVID-19 Pandemic

### Introduction

Alcohol use is a prevalent behavior among young adults in the United States, and it is associated with a range of health risks and negative outcomes (Klingemann et al., 2001; Rehm et al., 2003; Rehm et al., 2010; Friedmann, 2013; Park et al, 2015; McCabe et al., 2018; Krieger et al., 2018; Rehm et al., 2018; GBD 2016 Alcohol Collaborators 2018; World Health Organization, 2019; Castillo-Carniglia et al., 2019; Case et al., 2020; McCrady et al., 2021; Daviet et al., 2022). Young adulthood is a key developmental period in the human lifespan, during which many behaviors that are established become lifelong (Arnett, 2014). As such, attention on alcohol use during young adulthood is important. Therefore, the first section of Chapter 2, Part B provides an overview of the health risks, prevalence, and etiology of alcohol use among young adults, with particular focus on the relationship between alcohol marketing and alcohol use.

Alcohol marketing primarily targeting young adults has escalated alongside the growing popularity of social media platforms. This is to be expected considering the history of the alcohol industry, illustrated by *Key Point 1*. The second section of Chapter 2, Part B explores what is known about the relationship between alcohol marketing on social media and alcohol use among young adults.

Societal traumas, such as the COVID-19 pandemic, change how people behave (Center for Substance Abuse and Mental Health Services Administration, 2014; Hirschberger, 2018; Silver et al., 2021). For example, they might drink more alcohol and spend more time online. The third section of Chapter 2, Part B provides an overview of existing research about how COVID-19 affected drinking behaviors and related health issues, as well as online behavior and social media use among young adults.

The focus of Chapter 2, Part B is not just about young adults drinking more (Grossman et al., 2020; Koob et al., 2020; Pollard et al., 2020; Foster et al., 2021; Moon et al., 2021; Sharma et al., 2021; White et al., 2022) and using social media more during the COVID-19 pandemic (Ellis et al., 2020; Pandey et al, 2020; Pandya et al., 2021; Sikorska et al., 2021; Stuart et al., 2021; Lee et al., 2022). Moreso, it is about how alcohol companies might have used social media platforms to promote advertisements by using COVID-19 related themes to advance their commercial activities (Colbert et al., 2020; Foundation for Alcohol Research & Education, 2020; Huckle et al. 2021; Martino et al., 2021; Gerritsen et al., 2021; Kennedy et al., 2022). This phenomenon is reminiscent of countless times throughout history when the alcohol industry exploited vulnerable people for profit, illustrated by *Key Point 2*. So, the fourth section of Chapter 2, Part B reviews the literature related to how the alcohol industry used the COVID-19 pandemic as an opportunity to ramp up their marketing on social media.

The broader context of alcohol-related content on social media also changed during the COVID-19 Pandemic. The alcohol industry exists alongside an ever-changing social context, and it may be that society affects the alcohol industry just as the alcohol industry affects society - that this relationship is bidirectional. Considering the broader context of alcohol-related content on social media is important when examining alcohol marketing practices on social media so that we can begin to understand how messages spread between the alcohol industry and the broader social media community. Therefore, the fifth section of Chapter 2, Part B summarizes current literature about changes in alcohol-related content on social media during the COVID-19 pandemic.

The social media marketing practices of alcohol companies during the COVID-19 pandemic and their subsequent effects on drinking behaviors is incompletely understood. Thus, the sixth section of Chapter 2, Part B outlines the critical gaps in the research that the present dissertation helps fill. Advancing the understanding of how alcohol marketing on social media potentially affected drinking behaviors during the COVID-19 pandemic is crucial to forecasting how these behaviors may persist into the future - a notion that echoes *Key Point 3* - and critical to developing interventions that mitigate them. Included in this section are notable media effects theories that attempt to explain how exposure to media messages can influence the attitudes, beliefs, and behaviors of individuals. Understanding these theories is important for understanding the impact of alcohol marketing on young adults during the COVID-19 pandemic, and necessary to develop research that advances this area of study.

The information presented in Chapter 2, Part B, Sections 1-6 raises concerns about alcohol use among young adults, particularly during the COVID-19 pandemic, and how marketing on social media during the pandemic paralleled changes in alcohol use. Grounded by the *Three Key Points*, the conclusion of Chapter 2, Part B summarizes the public health significance of the present dissertation and highlights the need for public health research aimed at exploring the harmful effects of alcohol marketing on social media during the pandemic.

### 2.B.1: Health Risk, Prevalence, and Etiology of Alcohol Use Among Young Adults

#### *Health Risk of Alcohol Use*

Young adulthood is a critical developmental stage marked by the pursuit of independence, the exploration of identity, and the establishment of behavioral patterns that often endure for a lifetime (Arnett, 2014). During young adulthood, individuals encounter novel situations and are exposed to many potential influences that can shape their long-term health and well-being. Research into alcohol use during this stage is important because it can help identify factors contributing to alcohol use and lead to the development of prevention and intervention strategies, which can lead to better health outcomes and promote long-term healthier lifestyles (Merrill et al., 2016; Wood et al., 2018).

Alcohol use among young adults aged 18-25 is associated with numerous health risks, both acutely and chronically. Acutely, alcohol use impairs judgment, coordination, and reaction time, increasing the risk of accidents, falls, injuries from interpersonal violence, sexually transmitted infections, unplanned pregnancy, and HIV transmission (Rehm et al., 2003). Research has shown that young adults who engage in binge drinking (i.e., consuming 5 or more drinks in one drinking session for men or 4 or more drinks in one drinking session for women) are more likely to experience alcohol poisoning, and engage in risky behaviors such as drinking and driving, illicit drug use, social aggression, or unprotected sex (Krieger et al., 2018). Chronically, alcohol use can lead to liver damage, cardiovascular disease, and an increased risk of several types of cancer (Rehm et al., 2003). Alcohol use has also been linked to mental health problems such as depression and anxiety, which can have lasting effects on an individual's wellbeing (Rehm et al., 2010; Friedmann, 2013; Daviet et al., 2022).

Another long-term health risk associated with alcohol use among young adults is the development of alcohol use disorder (AUD; also known as alcoholism, a chronic and relapsing condition characterized by the compulsive and excessive consumption of alcohol despite negative consequences) (Alinsky et al., 2022). Young adulthood is a critical period for the development of AUD, and becoming a person with an AUD during this developmental stage can lead to long-term negative consequences that extend into later adulthood such as poor physical and mental health, dysfunctional relationships, and reduced occupational and social achievement (Castillo-Carniglia et al., 2019). Young adults who engage in binge drinking behavior are at an even higher risk of developing an AUD compared to their moderate drinking peers (McCabe et al., 2018). AUD - if left untreated to run its full course - can lead to liver failure, suicide, or alcohol poisoning, so-called deaths of despair (Rehm et al., 2018; Case et al., 2020).

It is not just the individual who consumes alcohol that is harmed. A 2015 systematic review demonstrated that children with alcoholic parents have lower self-esteem, lower academic and cognitive ability, and more difficulties with temperament (Park et al, 2015). Spouses of alcoholics have higher rates of depression, anxiety, and psychological distress (McCrady et al., 2021). Furthermore, the community as a whole is at elevated risk due to alcohol use with a notable proportion of motor vehicle fatalities, violent crimes, and property damage being attributable to alcohol (Klingemann et al., 2001). Given the determinantal effects that alcohol use has on self and others, it can be considered among the important urgent non-communicable health issues facing society. In fact, the global burden of disease studies have suggested that alcohol consumption accounts for 5.1% of the annual burden of disease worldwide (132,600,000 disability-adjusted life years), and alcohol abuse was the leading risk factor for morbidity and mortality among people aged 15-49 worldwide (GBD 2016 Alcohol Collaborators 2018; World Health Organization, 2019).

### *Prevalence of Alcohol Use Among Young Adults*

Alcohol use is prevalent among young adults aged 18-25, with many individuals engaging in binge drinking behavior. According to the 2021 National Survey on Drug Use and Health (most recent available as of this writing) 50.1% of young adults aged 18-25 reported consuming alcohol in the past 30 days, and 29.2% reported binge drinking in the past 30 days (Substance Abuse and Mental Health Services Administration, 2023). Furthermore, alcohol use is more prevalent among certain subpopulations of young adults. Female young adults have a slightly higher prevalence of past-30-day alcohol use (55%) than males (51.8%), as well as past-30-day binge drinking (females=30.5%; males=27.8%). Drinking behaviors also vary by racial/ethnic identity including past-30-day alcohol use (Hispanic=51.8%; non-Hispanic white=57.2%; non-Hispanic Black=39.8%; non-Hispanic Asian=32.9%; two or more racial/ethnic identities = 47.2%) and past-30-day binge drinking (Hispanic=30%; non-Hispanic white=33.6%; non-Hispanic Black=21.8%; non-Hispanic Asian=17.2%; two or more racial/ethnic identities=29.6%). Studies have found that young adults who identify as LGBTQ+ are more likely to engage in drinking behavior with 55.7% of LQBTQ+ young adults reporting past-30-day alcohol use as compared to 50.1% of general population of young adults (Substance Abuse and Mental Health Services Administration, 2022). Additionally, young adults from lower socioeconomic backgrounds may be more likely to engage in alcohol use with 54.2% of young adults at 200% or more the national poverty level reporting past-30-day alcohol use and 30.2% of young adults at 200% or more the national poverty level reporting past-30-day binge drinking (Substance Abuse and Mental Health Services Administration, 2023).

***Etiology of Alcohol Use***

The etiology of alcohol use among young adults aged 18-25 is complex and multifaceted and is best conceptualized in a bio-psycho-social paradigm (Engel, 1977; Zucker et al., 1986; Skewes et al, 2013). While the focus of the present dissertation is on alcohol marketing, which firmly falls in the "social" realm, this section will briefly review the relevant other etiological factors. Then, this section will provide more in-depth coverage of social factors, specifically alcohol marketing.

Biologically, research suggests that increased activity in the mesolimbic dopaminergic pathway (i.e., the reward system pathway) is associated with susceptibility to alcohol use and problematic drinking (Solinas et al., 2019). Additionality, a reduction in inhibitory activity of the neurotransmitter GABA in the prefrontal cortex is associated with impulsivity, which is also associated with drinking behaviors (Jupp et al., 2013; Silveri et al., 2013; Fish et al., 2022). Many studies suggest that young adults are notoriously impulsive (Schreiber et al., 2012; Reynolds et al., 2019). Taken together, mesolimbic dopamine and prefrontal cortex GABA are influential in the biological etiology of alcohol use among young adults. Finally, genetic factors have been found to play a role in the development of drinking habits, specifically AUD, with a meta-analysis suggesting that AUD is approximately 50% heritable (Verhulst et al., 2015).

Psychological factors of stress and anxiety are significant contributors to alcohol use among young adults (Temmen et al., 2020). College students, for example, often face high levels of stress related to academic performance and social expectations, which can lead to increased alcohol consumption (Krieger et al., 2018). Personality differences also influence alcohol use among young adulthood to some degree, with aggressive and extroverted young adults more likely to drink (Samek, et al, 2017).

Socially, peer influence and social norms are often cited as significant factors that contribute to the initiation and continuation of alcohol use among young adults (Bainter, 2018; Cleveland et al., 2018). Young adulthood is a time when peer influence and social norms are particularly relevant because of the desire for social validation and the formation of attitudes and beliefs that persist throughout life (Arnett, 2014). Other social factors such as family history, parental alcohol use, and availability of alcohol can also play a role in the development of alcohol use among young adults (Zucker, 2015). Individuals who grow up in households with a history of alcohol use may be more likely to drink and to develop alcohol use disorders themselves.

Alcohol-related media's role in shaping alcohol consumption patterns is another notable social determinant. A significant body of scholarly research has evaluated the relationship between alcohol marketing and drinking behaviors (Anderson et al., 2009; Smith et al., 2009; Jernigan et al. 2017). The research suggests a clear link: increased exposure to alcohol advertising often leads to both the initiation and escalation of alcohol consumption, especially among young people. A systematic review by Anderson et al. (2009) analyzed 13 longitudinal studies conducted between 1990-2008 totaling over 38,000 individuals aged 10-26 and found consistent evidence that exposure to alcohol marketing and promotional activities is associated with (1) the onset of alcohol use among young people and (2) increases in the quantity consumed by existing drinkers. Notably, Anderson et al. (2009) report that across all media types examined (i.e., TV commercials, film depictions, sponsorship, advertisements at bars/nightclubs, branded merchandise) with the sole exception of outdoor printed advertisements (e.g., storefront signs), alcohol media exposure was directly associated with alcohol use. In the exception of outdoor media, although no direct effects on alcohol consumption were found, the study in question did find an increase in intentions to drink within the next month (Pasch et al., 2007). Similarly, another systematic review of prospective cohort studies by Smith and Foxcroft (2009) investigating seven studies totaling over 13,000 individuals aged 10-26 with baseline survey data collected as far back as 1985 and follow-up data as recent as 2001 found that exposure to alcohol advertising increased the likelihood of alcohol initiation and, for those already drinking, increased their volume of consumption. Specifically, Smith and Foxcroft (2009) suggest a consistent (i.e., replicated across multiple studies), temporal (i.e., exposure to alcohol ads come before drinking), and dose response (i.e., the more ad exposure, the higher the level of drinking) relationship. Work by Synder et al. (2006), which was included in Smith and Foxcroft (2009), suggest that that, on average, each additional alcohol advertisement exposure event among TV, radio, billboard, and magazine/newspaper ads is associated with a 1% increase in the number of alcoholic beverages consumed (Snyder et al., 2006). Jernigan et al. (2017) further underscored these conclusions in their more recent systematic review of 12 longitudinal studies conducted between the years 2008-2016 totaling over 35,000 individuals aged 10-29. Their research also supported the hypothesis that exposure to alcohol marketing is positively associated with the likelihood of young people starting drinking (odds ratio up to 1.69 for alcohol initiation among those exposed to alcohol marketing compared to unexposed) and with severity of drinking (odds ratio up to 2.15 for hazardous drinking among those exposed to alcohol marketing compared to unexposed). Taken together, Anderson et al. (2009), Smith and Foxcroft (2009), and Jernigan et al. (2017) systematically review 31 years of empirical evidence evaluating the relationship between alcohol marketing and alcohol use and come to the consensus that exposure to alcohol marketing increases alcohol use. While the exact causal mechanisms that link alcohol media exposure to alcohol use remain under ongoing investigation, research suggests that exposure to such media modifies youth and young adults' beliefs about alcohol (such as reducing the perceptions that alcohol is harmful, or increasing the perception that drinking alcohol will contribute to pleasant outcomes), portraying alcohol as widely available and socially acceptable, and implying that drinking is a necessary component to becoming a successful adult (Blume et al. 2003; Sudhinaraset et al., 2016; Collins, et al, 2017; Martino, et al, 2016). In other words, a large body of research suggests that the effects of traditional alcohol marketing are increased alcohol use in the population, and the mechanisms might be through modifying beliefs and increasing social acceptability.

The literature also indicates potential interventions to mitigate the effects of alcohol marketing on consumption, namely policy-level regulation. Norway is one of the few countries globally that has both a total ban on alcohol advertising and has accessible data to allow for assessment. Studies indicate that this ban, which was instituted in 1975, has significantly and consistently decreased consumption in Norway by an average of 7.4% annually (Rossow, 2021). Though this one example shows a consistent, strong effect of an advertising ban, partial restrictions of alcohol advertising are more common. France, for example, enacted its Loi Évin legislation in 1991 which serves one of the strongest models for partial restricting of alcohol marketing to date (Loi Évin, 1999). Loi Évin had the following main provisions: (1) alcohol advertising is permitted only in specific mediums like adult press, certain radio hours, points of sale, and direct mail, but not on youth-targeted media or places like TV, cinemas, or sporting events; (2) content must be factual without associating alcohol with positive or evocative themes; and (3) all ads must display a health warning (Friant-Perrot et al., 2022). While the French legislation was associated with an 11.36% reduction in overall population-level alcohol consumption in the first 8-

years following enactment (from 14.88 liters of alcohol per inhabitant 15 years and older in 1991 to 13.19 L in 1999) (Cogordan et al, 2014), by 2016, lobbying from alcohol producers and retailers had sufficiently weakened the framework. An analysis by Gallopel-Morvan et al. (2017) showed that Loi Évin no longer protected young people from alcohol marketing sufficiently, citing intense lobbying by the alcohol industry leading to (1) the allowance of alcohol billboard advertisements in 1994, (2) the allowance of online alcohol advertising in 2009 despite the fact that the internet is the most popular medium among youth, (3) a large legislative shift in 2015 which stated that alcoholic beverages associated with cultural heritage (specifically, beer, cider, wine, vodka, whiskey) are now exempt from the Évin Law's restrictions, and (4) general lack of enforcement (i.e., the French government not sanctioning organizations that allow alcohol media in areas prohibited under Loi Évin such as at sporting events).

In the United States, alcohol marketing regulation is limited, and legal precedent thus far has favored unregulated alcohol marketing due to protected freedom of commercial speech under the first amendment of the constitution (Merrill, 1976). Two governmental agencies currently oversee alcohol marketing: the Alcohol and Tobacco Tax and Trade Bureau (TTB), and the Federal Communications Commission (FCC). Broadly speaking the TTB oversees labeling and non-broadcast advertisements and requires that advertisements cannot be false, misleading, or make certain health-related claims (Alcohol and Tobacco Tax and Trade Bureau, 2023). The FCC oversees broadcast media and does not have specific regulations beyond that of the TTB but retains the authority to intervene if an advertisement is deemed to be against the public interest (Federal Communications Commission, 1999). Both agencies have language in their policies that deter advertising to minors, but the language is vague, and enforcement is lacking. While some states do have additional regulations, such as restrictions on outdoor advertising, advertising in college communities, and prohibitions on price advertising (National Institute on Alcohol Abuse and Alcoholism, 2023), the US typically relies on industry self-regulation, predominantly by three industry bodies: the Distilled Spirits Council, the Beer Institute, and the Wine Institute (Federal Trade Commission, 2013).

American alcohol marketing is comparable to the position of tobacco marketing in the mid-to-late 20th century: ubiquitous and largely unregulated. However, theoretical lessons can be drawn from tobacco marketing regulation to infer the potential and demonstrate precedent for alcohol marketing regulation in the United States. The Federal Cigarette Labeling and Advertising Act of 1965 (Federal Trade Commission, 2023) required all cigarette packages to carry a health warning, the Public Health Cigarette Smoking Act of 1969 (U.S. Congress, 1970B) banned cigarette advertising on television and radio, and the Master Settlement Agreement of 1998 (National Association of Attorneys General, 1998) banned the sponsorship of sports/athletic entities and other events as well as marketing efforts towards youth (such as the use of cartoon characters and billboard advertising). Tobacco marketing regulations have been implemented in response to the recognized public health risks of tobacco use and have been shown to have contributed to the overall success of tobacco control efforts (Saffer et al., 2000; World Health Organization, 2023). The lessons from tobacco suggest that there is a strong case - both theoretically and legally - for stricter alcohol marketing regulations in the U.S. to mitigate the impact of alcohol consumption.

In conclusion, the scientific literature consistently shows a significant association between alcohol marketing and alcohol use, particularly among youth and young adults. Exposure to alcohol marketing increases the likelihood of initiating drinking and an elevated level of consumption among those who already drink. However, the empirical work examining the relationship between alcohol media and alcohol use has been mostly focused on traditional forms of alcohol media with relatively long histories (i.e., print media, broadcast media, sponsorships, endorsements, and product placement). Compared to the century-long empirical history of studying and regulating traditional alcohol media, there has been less attention focused on more recent forms of alcohol media technology such a social media which was invented less than 25 years ago. Furthermore, there is little longitudinal research examining alcohol marketing on social media. Interventions such as advertising restrictions are evidence to potentially attenuate these effects, suggesting their inclusion in a comprehensive prevention strategy may mitigate the harmful consequences of alcohol use. Compared to other western countries such as France and Norway, the United States is lacking in interventive efforts to reduce alcohol advertising. The next section reviews what is known about the effects of social media marketing of alcohol on alcohol use.

### 2.B.2: Relationship Between Alcohol Marketing on Social Media and Alcohol Use Among Young Adults

Much of the research on the effect of alcohol marketing on alcohol use has been conducted on longstanding marketing strategies: printed advertisements, television commercials, film depictions, sponsorships, and endorsements. *Key Point 1* illustrates that the alcohol industry has adapted its marketing as communication technology has advanced. As such, alcohol marketing has emerged on social media. The marketing of alcohol products on social media is a significant area of concern, particularly given that young adults are among the most frequent users of social media (Auxier et al., 2021; Shannon et al., 2022).

Several studies have indicated a moderate correlation between exposure to alcohol-related content on social media and subsequent alcohol use. For example, Curtis et al. (2018) investigated the link between young adults' exposure to (typically defined as the frequency and duration of viewing content) alcohol marketing on social media and their alcohol consumption and related problems in a meta-analysis of 19 articles published 2011-2016 with over 9,000 individuals, and found a statistically significant meta-correlation between exposure to alcohol-related content on social media and alcohol consumption ($r = 0.36$, 95% C.I. = 0.29 - 0.44, $p < 0.001$) and alcohol-related issues ($r = 0.37$, 95% C.I. = 0.21 - 0.51, $p < 0.001$). While a substantial finding, the meta-analysis was not able to determine directionality in the relationship between alcohol-related social media activity and alcohol use (i.e., which comes first) because most of the included studies were cross-sectional.

In a more recent systematic review with a wider scope (i.e., investigating the broad relationship between social media use and drinking), Alhabash et al. (2022) identified 204 global studies published 2009-2019 and reported that an overwhelming majority (93.1%) found a positive relationship between social media use and alcohol consumption (broadly defined, for example, as simple as the correlation between time spent on social media and self-reported alcohol use). Of the studies reviewed, 108 included measurements of exposure to alcohol marketing on social media alongside large batteries of other measures, and 9 studies were designed with the exclusive purpose of studying alcohol marketing on social media. The authors suggested that up to 2019 there is evidence of a strong correlation between alcohol marketing on social media and drinking; however, there is insufficient evidence to infer causal links between the two due to methodological homogeneity (i.e., mostly survey based, cross-sectional, or qualitative), and they called for experimental and longitudinal studies focused on research questions related to the temporal association between exposure to alcohol marketing on social media and drinking (i.e., does exposure to alcohol marketing on social media precede drinking behavior in a meaningful way?).

Recognizing the lack of causal evidence in the literature, Hendriks et al. (2021) conducted a 6-week longitudinal experimental study examining the influence of alcohol-related social media content on subsequent alcohol consumption among adolescents and young adults, and they reported that experimentally induced exposure to alcohol-related social media posts was predictive of both the likelihood and quantity of next-day drinking. Specifically, each additional alcohol-related social media post viewed increased the log odds of consuming alcohol the next day by 0.27 ($b=0.27$, 95% C.I. = 0.18-0.35), and also predicted an increase in the number of drinks consumed. These experimental findings underscore the significant impact of alcohol-related social media content on drinking behaviors, suggesting an urgent need for interventions. However, there remains a gap in longitudinal population-level research investigating the temporal relationship between alcohol industry marketing content specifically and alcohol use, particularly studies that utilize objectively obtained content from alcohol industry accounts rather than general alcohol-related social media content or self-reported exposure measures.

In summary, while the landscape of alcohol marketing has evolved with the growth of communication technology, its implications for alcohol use, especially among young adults, remain a critical area of research. Traditional marketing channels have paved the way for social media marketing, heightening concerns about its influence on drinking behaviors. A large body of evidence shows a correlation between alcohol marketing on social media and drinking. However, the predominant methodological approach in these studies is cross-sectional in nature which fails to provide definitive conclusions about causality. The findings by Hendriks et al. (2021) present compelling experimental evidence on the causal impact of social media content on drinking behaviors, but further longitudinal, population-level research is imperative. Addressing this research gap will not only enhance our understanding but will also guide effective policy and intervention strategies aimed at mitigating the potential harms associated with alcohol consumption triggered by social media marketing.

**2.B.3: Societal Disruption caused by the COVID-19 Pandemic and the Effects on Alcohol Use and Social Media Use**

The COVID-19 pandemic was declared a global health emergency by the World Health Organization in March 2020. In an attempt to control the spread of COVID-19, nations implemented measures such as lockdowns (i.e., stay-at-home orders, curfews, quarantines), travel restrictions, and social distancing which, in America, widely extended through at least the end of May 2020 (Centers for disease Control and Prevention, 2023). While these actions were designed to protect health, they had some unexpected consequences, particularly in how people consumed alcohol and used social media. *Key Point 2* illustrates that the alcohol industry has a long history of distributing its products to, and exploiting, vulnerable populations. Therefore, it is important to examine changing drinking patterns and social media use during this time of societal trauma, and to consider how the alcohol industry may have used COVID-19 in their social media marketing to promote their products.

Despite many drinking establishments such as bars and restaurants being closed during the COVID-19 lockdowns, Grossman et al. (2020) reported that 60% of Americans increased their alcohol consumption during this time. This trend is reflected in economic data which report a 54% increase in national sales of alcohol and 262% increase in online sales during the early period of COVID-19 lockdowns (i.e., late March/early April 2020) compared to the same time period one year earlier (Pollard et al., 2020). The vast majority of drinking during the COVID-19 lockdowns occurred at home (Foster et al., 2021), and research has linked drinking during this time to increases in psychological despair caused by the pandemic (Koob et al., 2020). This uptick in at-home drinking was not without consequences. More people experienced problems related to alcoholic liver disease (Moon et al., 2021), hospitals reported an increase in acute cases related to alcohol such as accidents, alcohol poisoning, and delirium tremens (Sharma et al., 2021), and there was a significant increase in alcohol-related deaths (White et al., 2022). Specifically, among young adults, White et al., (2022) used U.S. mortality data from the national Center for Health Statistics from 2019 to 2020 and reported a 29.8% increase in alcohol-related deaths among Americans ages 16-20, 25.6% increase among those age 21-24, and 37% increase among age 25-34; and the biggest increase occurred during April 2020 and May 2020 which mirrors the period of highly restrictive lockdowns in the U.S.

Parallel to the increases in drinking during the COVID-19 lockdowns, the digital world became a primary source of social connection. People suffering from isolation sought refuge in online communities, particularly on social media platforms such as Facebook, Instagram, and Twitter (Ellis et al., 2020; Pandey et al, 2020; Sikorska et al., 2021; Stuart et al., 2021). Quantifying this digital migration, a recent systematic review suggested that people experienced an average of 50-70% increase in internet usage during the pandemic, the heaviest of users experienced up to 17.5 hours of screen time per day, and up to 50% of that time was spent on social media platforms (Pandya et al., 2021). While these platforms seemingly offer a sense of community, a recent meta-analysis including 14 studies investigating social media use during the COVID-19 pandemic and mental health concluded that 2 or more hours of daily social media exposure was associated with more anxiety symptoms (pooled OR = 1.55, 95% C.I. = 1.30-1.85) and more depressive symptoms (pooled OR = 1.43, 95% C.I. = 1.30-1.85) as compared to less than 2 hours of daily social media exposure (Lee et al., 2022). However, it is important to note that the direction of this effect cannot be disentangled from these studies, meaning it remains unclear whether increased social media use contributes to heightened anxiety and depressive symptoms, or if individuals experiencing these symptoms are more likely to engage in prolonged social media use.

In light of the pronounced increase in alcohol consumption during the lockdown phase of the COVID-19 pandemic and the parallel surge in social media usage, it is crucial to evaluate the potential intersection of these trends. While lockdown measures were essential for reducing the spread of COVID-19 in the early phase of the pandemic, they fostered an environment conducive to greater alcohol consumption, coupled with increased social media use. Furthermore, given the alcohol industry's historical propensity to market to vulnerable populations (i.e., *Key Point 2*), it is important to evaluate the alcohol industry's possible use of the pandemic-fueled digital migration to increase their commercial activities. In this context, the term 'vulnerable populations' can indeed extend to virtually everyone, as the pandemic's widespread impact meant that nearly all individuals were exposed to heightened stressors, making them susceptible to the influence of alcohol marketing. The marked rise in alcohol-related incidents and fatalities, especially among young adults during the stringent lockdown months (i.e., March-May 2020), coupled with the association of extensive social media exposure paints a concerning picture. This calls for a pressing need for research that evaluates the relationship between alcohol-related social media marketing during the pandemic and the proliferation of messages related drinking behaviors that are contained within alcohol industry marketing, and to implement safeguards against similar proliferation of unhealthy drinking messages in future crises. The next section reviews what is currently known about alcohol marketing on social media during the COVID-19 pandemic.

### 2.B.4: Alcohol Marketing on Social Media During the COVID-19 Pandemic

During the COVID-19 lockdowns, the digital migration prompted corporations to significantly increase their digital marketing spend. Specifically, between February to June 2020, corporate expenditure on social media marketing rose by 74% (Moorman et al., 2021). This period saw a surge in social media usage as well as psychological despair (Anwar et al., 2020; Pfefferbaum et al., 2020; Koob et al., 2020, Pandey et al, 2020; Panchal et al, 2023; Pandya et al., 2021, Lee et al., 2022), presenting a ripe opportunity for the alcohol industry, which has a long well-established history of creating new marketing strategies alongside new developments in communication technology and distributing their product to vulnerable people (i.e., *Key Point 1* and *Key Point 2)* consistently across two millennia. Since the inception of social media platforms, alcohol companies have established a social media presence (Hastings, 2009; Moreno et al., 2014; Jernigan et al., 2017), and the COVID-19 pandemic provided the opportunity for them to further amplify their social media marketing campaigns given the increases in social media use. Economic data might suggest they were successful, indicating a 54% increase in national sales of alcohol and 262% increase in online sales during the early period of COVID-19 lockdowns (Pollard et al., 2020), although no empirical study directly links alcohol marketing during the pandemic to increased sales.

There is growing concern about the impact of social media marketing of alcohol. Research indicates that exposure to such marketing, typically defined as the frequency and duration of viewing alcohol-related advertisements or content on social media platforms, is associated with increased alcohol consumption (Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022). Worryingly, the COVID-19 pandemic was also marked by a significant increase in drinking and a surge in alcohol-related health problems (Grossman et al., 2020; Koob et al., 2020; Foster et al., 2021; Moon et al., 2021; White et al., 2022). Given that during the early period of the COVID-19 pandemic both alcohol use and social media use increased, it may be that increased exposure to alcohol marketing on social media during this period was also associated with increased risk of alcohol use. Some studies have investigated the nature and extent of alcohol marketing on social media during the pandemic, aiming to understand the key messages used by the alcohol industry during this period (Colbert et al., 2020; Huckle et al. 2021; Martino et al., 2021; Gerritsen et al., 2021; Kennedy et al., 2022). A small but emerging body of evidence suggests that social media marketing of alcohol during the pandemic was highly prevalent with one study suggesting that consumers saw one social media alcohol advertisement every 35 seconds (Foundation for Alcohol Research & Education, 2020), and these ads centered on several primary themes:

- *isolation drinking* (e.g., intensified encouragement of home drinking)
- *community support* (e.g., virtual gatherings with alcohol present)
- *promotion of hygiene* (e.g., social distancing/washing hands/wearing masks while consuming alcohol)
- *home or curbside delivery* (e.g., using a web application or calling orders in)

- *promotion of corporate social responsibility* (e.g., alcohol brands making sanitization products or calling for support of essential job workers).

The alcohol industry's promotion of these themes may have encouraged or reinforced related behaviors within the community. However, current research has not yet established a direct link between alcohol industry marketing on social media during the COVID-19 pandemic and community-level drinking-related social media activity during this period. If such an association exists, it raises significant concern due to the lasting impact of media portrayals associated with the alcohol industry (Key Point 3). Historical patterns suggest that unhealthy drinking behaviors initiated or strengthened during the pandemic might persist into the future.

Preventing the alcohol industry from exploiting advancements in communication technology (Key Point 1) to target vulnerable populations (Key Point 2) and induce long-term changes in drinking behaviors (Key Point 3) is crucial. Achieving this requires the public health community to critically understand how these *Three Key Points* were manifested during the COVID-19 pandemic. An effective approach could involve analyzing alcohol marketing content on Twitter during this period and modeling how it proliferated across the platform. Insights from this analysis could inform strategies to proactively prevent similar harmful marketing practices by alcohol companies in future episodes of significant societal disruption.

Overall, during the initial period of the COVID-19 pandemic there was (1) an increase in alcohol marketing on social media, (2) an increase of people using social media, and (3) an increase in alcohol use, morbidity, and mortality. Examining the nexus of these three occurrences has important implications for public health because it is possible that the first (i.e., increase in marketing on social media) is related to the last (i.e., an increase in alcohol use). Social media platforms are an ideal place to examine this because researchers can identify the content of social media marketing (e.g., promoting drinking at home) and model the spread of that information from the industry to the community (e.g., demonstrating that messages about drinking at home generated by the alcohol industry preceded increases in social media content about drinking at home generated by the community). The inspiration of this work is rooted in previous public health work that sought to protect populations from predatory industry practices that harm health (e.g., the entire multi-decade saga that led to the Tobacco Master Settlement Agreement (National Association of Attorneys General, 1998), or the sanctioning and prosecuting of the Sackler family for Perdue Pharma's role in the ongoing opioid epidemic (Keef, 2017; Case et al., 2020). It is important for society to understand the phenomenon of alcohol marketing on social media during the COVID-19 pandemic so that we can be aware of - and potentially protect ourselves from - harmful marketing practices by corporations, especially during major societal disruptions such as the COVID-19 pandemic. Stringently regulating alcohol industry's marketing practices - especially on social media and especially during times of great societal disruption - could be an area worth attention, but more research needs to be conducted.

### 2.B.5: The Broader context of alcohol-related content on social media during the COVID-19 Pandemic

The COVID-19 pandemic has not only been associated with changes in alcohol marketing practices, but also with altered social behaviors and communication patterns on social media platforms. Evidence suggests that content related to alcohol use on social media changed during the pandemic, potentially reflecting shifts in drinking behaviors and societal attitudes toward alcohol consumption. This section reviews several empirical studies that demonstrate such changes in the broader context of alcohol-related content on social media during the pandemic.

First, Wanchoo et al. (2023) leveraged Reddit data to investigate shifts in discussions about diet, physical activity, substance use, and smoking during the pandemic. This study reported an increase in discussions about coping with stress and anxiety, alcohol misuse, and harm-reduction strategies starting from late March 2020, coinciding with the implementation of stay-at-home orders. Second, Ward et al. (2021) analyzed tweets referencing alcohol-induced blackouts before and during the pandemic. This study reported that more tweets about alcohol-induced blackouts were posted between March 13 - April 24 in 2020 compared to the same period in 2019. Next, Stone et al. (2022) explored differences in emotional, psychological, and social sentiment between alcohol-related tweets and non-alcohol tweets during the onset of the pandemic. They found that tweets about alcohol posted after the pandemic declaration were more authentic, contained more affiliation language, and were higher in emotionality, as compared to tweets about alcohol before the pandemic. This finding suggest that tweets included discussion about alcohol's role as a coping mechanism and a greater willingness to share personal experiences related to alcohol consumption on social media during the pandemic. Additionally, Litt et al. (2021) studied the association between alcohol-specific social norms on social media, posting behavior, and drinking to cope during the pandemic. Their study found that perceiving peers to be posting about using alcohol to cope with pandemic-related stress was associated with an increased likelihood of individuals making similar posts and drinking to cope themselves. These findings highlight the role of social norms and online behavior in influencing personal alcohol use. Lastly, Ahmed et al. (2024) examined social media content related to risky drinking the COVID-19 pandemic. They found that discussions about alcohol on Twitter reflected a culture where risky drinking and binge drinking are normalized.

Collectively, these studies demonstrate that content related to alcohol use on Twitter changed during the COVID-19 pandemic, with increased discussions about using alcohol to cope with stress and anxiety. The broader context of alcohol-related content on social media during the COVID-19 pandemic is relevant to the alcohol industry's marketing practices on social media because it is possible that their marketing practices reflect society (as opposed to society up-taking marketing messages). In other words, did alcohol marketing influence ideas about drinking during the pandemic, or did the social context of drinking during the pandemic influence alcohol marketing? This notion raises important questions about the potential bidirectional effects of relationship between alcohol marketing and drinking.

### 2.B.6: The Current Gap in the Literature, Media Effects Theories, and the Public Health Significance of the Present Dissertation

In sum, the review of the literature has suggested *Three Key Points:* (1) the alcohol industry's success is linked to advances in communication technology, (2) the alcohol industry has a long history of distributing its products to, and exploiting, vulnerable people, and (3) the media portrayals of the alcohol industry are lasting. Further, alcohol causes many diseases and social issues and is widely used despite its known consequences. Lastly, marketing by the alcohol industry can influence alcohol use. However, there remain critical gaps in the literature that further research can illuminate. This section highlights two specific gaps in the literature, reviews relevant media effects theories that might help fill these gaps, and frames the public health significance of the present dissertation.

### ***The Current Gap in the Literature***

*Gap 1:* Historical perspectives and more than three decades worth of empirical research on traditional alcohol media (i.e., graphic media, culture as media, and broadcast media) has suggested a causal link (i.e., consistent - replicated across multiple studies, temporal - exposure to alcohol media come before drinking, and dose response - the more alcohol media exposure, the higher the level of drinking) between traditional alcohol marketing exposure and alcohol use (Standage, 2006; Gately, 2008; Anderson et al., 2009; Smith and Foxcroft, 2009; Phillips, 2014; Jernigan et al., 2017). However, less empirical research has been conducted evaluating the prospective relationship between alcohol marketing on social media and alcohol use. While a decade-and-a-half of research indeed suggests that alcohol marketing on social media is potentially detrimental to public health because of associations between alcohol marketing on social media and alcohol use, there is only one study that begins to suggest causation, which was experimental (i.e., highly controlled, limited generalizability, and with a relatively small number of participants) (Hendriks et al., 2021). **More longitudinal (i.e., occurring over a long period of time), ecological (i.e., across multiple levels of the social ecological model, linking industry activity to community or individual activity), naturalistic (i.e., occurring in a natural environment without laboratory constraints), population-level (i.e., large, diverse sample size) research is needed to temporally link alcohol industry marketing activities on social media to drinking behavior (e.g., demonstrating that changes in industry marketing precedes changes in drinking and/or indicators of drinking such as posting about drinking on social media). Furthermore, more research is needed that investigates potential bidirectional effects in the relationship between alcohol**

**marketing and alcohol use to help fully elucidate the complex nature of alcohol marketing on social media.**

*Gap 2*: Historical perspectives suggest the alcohol industry adapts to societal disruptions in order to grow its commercial activities, and a small body of empirical literature suggests that this happened again during the COVID-19 pandemic (Colbert et al., 2020; Huckle et al. 2021; Martino et al., 2021; Gerritsen et al., 2021; Kennedy et al., 2022). The current existing research has identified several strategies/themes that alcohol marketers used on social media during the COVID-19 pandemic; however, this research is not extensive, has limited scope in terms of the number and types of advertisements investigated, and has only been conducted in a few countries around the world, and lacks sufficient information beyond broad themes/topics (i.e., isolation drinking, community support, promotion of hygiene, home or curbside deliver, and promotion of corporate social responsibility). **There is a need to validate these themes in other datasets and explore additional themes that may emerge in different contexts and regions, as well as examine the underlying mechanisms of communication that were used in these topics (e.g., emotional appeal, cognitive framing, and calls to action).**

### *Media Effects Theories: Media Influences People*

Media's ability to shape public opinions and modify behaviors is widely acknowledged (Valkenburg et al., 2016). For over a century, scholars of communication, psychology, sociology, political science, and education, among other fields, have debated the extent of this influence, as well as who is most affected, when, how, and why. There are several prominent media effects theories, including the hypodermic needle model, cultivation theory, social cognitive theory, and the agenda-setting theory. The hypodermic needle model suggests that media messages are like a drug injected into a passive audience, causing them to adopt the attitudes and behaviors portrayed in the media (Lasswell, 1971). Cultivation theory suggests that exposure to media messages can shape an individual's perception of reality and lead to the development of shared cultural values (Potter, 2014). Social cognitive theory suggests that individuals learn new behaviors and attitudes through observing valued others, and mass media presents a wide variety of people to observe (Bandura, 2001). Finally, agenda-setting theory suggests that media messages can influence the salience of issues and topics in the public sphere (Cook et al., 1983).

The Differential Susceptibility to Media Effects Model (DSMM) builds on these theories by suggesting that media effects are conveyed through the response states elicited within viewers in reaction to consuming media (Valkenburg et al., 2013). DSMM suggests that three media response states - cognitive, emotional, and excitative responses - explain the relationship between media use and media effects (see Illustration 29). For example, a common thread of 21st-century media effects research examines the relationship between consuming propaganda (i.e., media use) and religious radicalization (i.e., media effects). DSMM suggests that consuming radicalizing propaganda (i.e., media use) causes cognitive (e.g., religious framing of a topic), emotional (e.g., distress about opposition to a topic), and excitative (e.g., suggestions that violence is effective, urgent, and the only way to help) response states that help explain the subsequent radicalization (i.e., media effects) (Neumann et al., 2018; Wolfowicz et al., 2022). Similarly, DSMM can be applied to understand the effects of social media marketing for consumer products, such as alcohol. For example, when individuals encounter alcohol-related content on social media platforms, DSMM posits that this exposure (i.e., media use) triggers cognitive (e.g., processing information about the product's quality, price), emotional (e.g., evoking feelings of excitement, relaxation, or social connection associated with alcohol consumption), and excitative (e.g., stimulating a desire to purchase or consume the product) response states that help explain subsequent alcohol consumption, changes in attitudes towards drinking, or greater brand loyalty (i.e., media effects) (Noel et al., 2018).

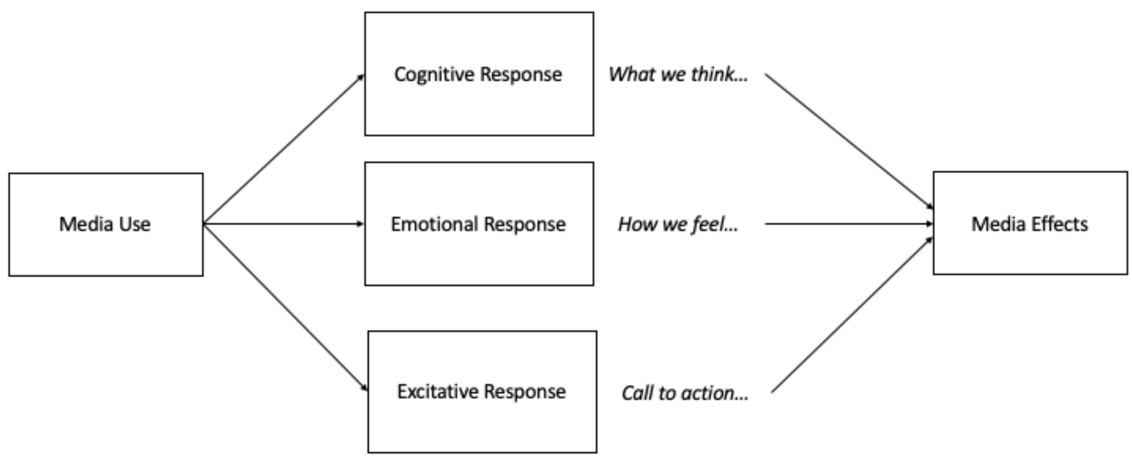



Illustration 29. Adapted Conceptual Model for the DSMM

The DSMM's assertion that media influences cognitive, emotional, and excitative response states is grounded in the idea that messages - especially those conveyed through language - carry information, evoke emotional responses, and possess the potential to shape behavior (Katz et al., 1973; Ellis et al., 1993). Messages inherently include a cognitive component because they transmit information or ideas, whether explicit (e.g., stating a fact) or implicit (e.g., making an assumption or suggestion). Whenever a message is communicated, some form of information is conveyed. Similarly, emotional responses - including neutral emotionality - are intrinsic to the communication process. When an individual receives a message, certain emotions are elicited, and even the absence of an emotional reaction is informative, as it provides insight into the message's impact or lack thereof. Furthermore, the potential for messages to influence behavior is an essential aspect of communication. Since time progresses linearly, actions occur after message transmission. Whether a message directly leads to subsequent actions or not carries its own significance, and the absence of behavioral change is still meaningful, reflecting a message's success or failure in motivating action.

These concepts (cognitive, emotional, and excitative response states leading to media effects) align with several broader theories of communication and media. The Transmission Model of Communication (Ellis et al., 1993) emphasizes the linear process of sending and receiving information between a source and a receiver. Additionally, Uses and Gratifications theory (Katz et al., 1973) suggests that individuals actively select media to consume based on their psychological needs and desires. Katz et al. (1973) assert that people are not passive recipients of media content, but rather they engage with media to fulfill specific cognitive or emotional gratifications, further shaping their behavioral responses.

DSMM is an integrative theoretical framework of mass media that can describe media effects and explain (a) how and why media is influential and (b) how media effects can be enhanced or counteracted. Under the DSMM framework, we can expect any given corpus of messages to contain cognitive, emotional, and excitative content, as media inherently encapsulates these elements. Analyzing media through the lens of the DSMM will allow for a better understanding of how different types of messages engage these response states, and how they might contribute to behavioral outcomes. This theory will guide research questions in this dissertation by providing a framework for understanding the mechanisms through which media, particularly social media, influences behaviors, and how these effects can be studied, measured, and potentially mitigated.

### *Media Effects Theories: Alternative Pathway, People Influence Media*

While traditional media effects theories often focus on the media-to-individual pathway, it is crucial to consider the bidirectional nature of idea transmission, especially in the age of social media. With a particular focus on the theory of sociogenesis, this section explores theories that explain how ideas can originate from individuals, spread to the community, and subsequently influence media.

Sociogenesis, a concept rooted in sociology and developmental psychology, provides a compelling framework for understanding how social phenomena emerge from collective human activities (Valsiner et al., 2000). In the context of social media, sociogenesis explains how user-generated content and community-driven trends can shape the broader media landscape such as industry practices and marketing strategies. This bottom-up process of idea transmission is particularly relevant in the dynamic landscape of social media, where user engagement and viral content can rapidly influence industry decisions.

For example, the phenomenon of "clean eating" in social media marketing of food products was initially popularized by health-conscious individuals sharing their dietary choices on platforms like Instagram (Baker et al., 2020; Walsh et al., 2020). This trend exemplifies sociogenesis in action. As users shared posts about whole foods, plant-based diets, and "natural" ingredients, a collective understanding of "clean eating" emerged within online communities. Food industry marketers, recognizing this increased interest, began incorporating "clean" messaging into their product development and advertising strategies (Negowetti, et al, 2022). This adaptation demonstrates how ideas originating at the community level can ascend the social ecological model to influence industry-wide practices.

Complementing sociogenesis, the theory of cultural evolution offers another perspective on the individual-to-media pathway (Mesoudi, 2011). This theory posits that cultural ideas and practices evolve through processes similar to biological evolution, including variation, selection, and transmission. In the social media context, user-generated content represents cultural variations, with the most engaging or resonant ideas being "selected" through likes, shares, and comments. These selected ideas then transmit across the platform and potentially to industry actors, who may adopt and amplify them.

The concept of "memetics," introduced by Richard Dawkins (1976), provides a third theoretical lens for understanding individual-to-media idea transmission. Memes, defined as units of cultural information that spread through imitation, are particularly relevant in the social media ecosystem. Internet memes, often originating from individual users or small communities, can rapidly gain traction and influence broader cultural narratives. Industry actors, recognizing the power of these memes to capture attention and convey messages succinctly, may co-opt or reference them in their marketing strategies.

The individual-to-media pathway, as illustrated by these theories, underscores the importance of considering bidirectional idea transmission in social media research. While the Differential Susceptibility to Media Effects Model (DSMM) provides a robust framework for understanding media-to-individual effects, integrating theories like sociogenesis, cultural evolution, and memetics can offer a more comprehensive understanding of how ideas circulate in the social media ecosystem.

By examining both pathways - media-to-individual and individual-to-media - researchers can gain a more nuanced understanding of the complex dynamics of idea transmission on social media. This bidirectional approach acknowledges the agency of social media users as both consumers and creators of content and recognizes the adaptive nature of industry actors in response to community-driven trends and phenomena.

### *The Public Health Significance of the Present Dissertation*

Alcohol use has a large public health burden on society, including on young adults. In fact, deaths of despair (that is, deaths from suicide, liver failure, alcohol poisoning, and drug overdose) are commonly preceded by or co-occurring with alcohol abuse, and the recent surge in deaths due to these causes are responsible for repeated annual decreases in life expectancy in the United States, an alarming trend that had not happened for a century prior to the 2$^{nd}$ half of the 2010's (Case et al., 2020). Furthermore, the global burden of disease studies have attributed more than 1 out of every 20 years of life lost to death or disability to alcohol use and suggested that alcohol abuse is the #1 risk factor for death and disease among individuals aged 15-49 (GBD 2016 Alcohol Collaborators 2018; World Health Organization, 2019). Alcohol costs the world 132.6 million years of quality life annually, and many people who die or become disabled from alcohol-related diseases suffer immensely. To put the cause of death due to alcohol use in perspective against the top two direct causes of death and disability worldwide: cancers (of all variety) accounts for 244.6 million quality life years lost annually, and heart disease accounts for 203.7 million quality life years lost annually, and both of which chronic alcohol abuse is a major risk factor for developing (Mattiuzzi et al., 2019). Based in this evidence, more research about what causes the detrimental effects of alcohol is needed.

Despite the well-known consequences of drinking, the prevalence of alcohol use is high, particularly among young adults. Over half of young adults report alcohol use, and nearly one-third report binge drinking, in the past 30 days (Substance Abuse and Mental Health Services Administration, 2023). The high rates of alcohol use among young adults are particularly detrimental to public health because not only are young adults who drink at risk for acute alcohol-related issues such as violence, traffic accidents, alcohol poisoning, or psychiatric emergencies like suicide (Rehm et al., 2003; Krieger et al., 2018), but also because young adulthood is a critical developmental period for the establishment of behavioral patterns that often endure for a lifetime (Arnett, 2014). Thus, young adults who drink are also at elevated risk for chronic alcohol-related issues across the life span such as liver failure, many cancers of the digestive tract, chronic mental health disorders, dysfunctional relationships, and unemployment (Klingemann et al., 2001; Rehm et al., 2010; Friedmann, 2013; Park et al, 2015; Castillo-Carniglia et al., 2019; McCrady et al., 2021; Daviet et al., 2022). Therefore, young adulthood is a crucial period for reducing the use of alcohol, particularly problematic alcohol use.

Alcohol marketing is associated with the growth of the alcohol industry and with alcohol use, both historically and presently. As detailed in Chapter 2, Part A, the history of the alcohol industry is associated with genocide, slavery, and predatory capitalism. Throughout history the alcohol industry has created marketing campaigns that appear to be aimed at distributing its products to soldiers, people in poverty, and people experiencing trauma; and currently, the alcohol industry has successfully infused itself into American culture with omnipresent multichannel marketing. Furthermore, robust research has causally linked alcohol marketing to alcohol use and related death and disability and has suggested that youth and young adults are particularly vulnerable to alcohol marketing

(Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017). Therefore, when examining the detrimental effects of alcohol among young adults, a focus on the alcohol industry is needed, specifically their marketing practices.

Tracing the evolution of communication technologies, alcohol marketers have consistently used the most advanced forms of communication technology to increase their commercial activities. As detailed in Chapter 2, Part A, the alcohol industry appears to have leveraged each historical development in communication technology (e.g., writing, printing, broadcast, internet) to increase its commercial activities. The most recent paradigm shifting advancement in communication technology has been the advent of social media, which is used by the great majority of young adults (Auxier et al., 2021; Shannon et al., 2022). Furthermore, early research into the relationship between alcohol marketing on social media and alcohol use suggests a positive correlation (Curtis et al., 2018; Alhabash et al., 2022), and one experimental study suggests causation (Hendriks et al., 2021); however, more research is needed in this area, as suggested by *Gap 1*. Therefore, when examining how alcohol marketing contributes to the detrimental effects of alcohol among young adults, it is important to focus attention on social media.

The alcohol industry has a long history of adapting marketing campaigns and practices to societal disruption and traumatic events (e.g., war, social and political changes, economic depression) to increase their commercial activities, as detailed in Chapter 2, Part A. Furthermore, a small body of literature suggests that this happened again during the COVID-19 pandemic (Colbert et al., 2020; Huckle et al. 2021; Martino et al., 2021; Gerritsen et al., 2021; Kennedy et al., 2022); however, the effects and mechanisms of alcohol marketing on social media during the COVID-19 pandemic are incompletely understood, as suggested by *Gap 2*. Therefore, when examining how alcohol marketing on social media contributes to the detrimental effects of alcohol among young adults, the COVID-19 pandemic, and the marketing that occurred during it, provides a useful period to examine the themes alcohol marketing used and how that may be related to alcohol use.

Furthermore, the public health significance of the present dissertation is particularly strong because the marketing practices by the alcohol industry during COVID-19 likely echo the history of marketing practices by the alcohol industry (i.e., using advances in communication technology - *Key Point 1* - to advertise alcohol to people who are vulnerable due to pandemic-related reasons - *Key Point 2* - to evoke new and persistent changes in drinking behavior - *Key Point 3*). Said another way, this dissertation is significant to public health because it examines how alcohol marketing can use events in society to promote its products and thus could be used as an example for other future events. The COVID-19 pandemic is just an example.

Therefore, the present dissertation aims to examine the effects and mechanisms of alcohol marketing on social media during the COVID-19 pandemic in a series of two studies. The purpose of this dissertation is to (1) identify themes that the alcohol industry used during the COVID-19 pandemic, (2) explore the cognitive, emotional, and excitative mechanisms through which the media effects of marketing related to these themes were elicited, and (3) model the spread of these themes between the industry and the community. The hope of this dissertation is to provide the necessary empirical evidence to the powers that be (i.e., policy makers) to protect their constituents from harmful commercial practices on social media by the alcohol industry during future disasters.

## PART C: DISSERTATION AIMS

### Study 1

Study 1 validated known themes that the alcohol industry used during the COVID-19 pandemic (Martino et al., 2021), explored additional themes that emerged, and explored the cognitive, emotional, and excitative mechanisms that were embedded in these themes. Framed by the DSMM (Valkenburg et al., 2013), Study 1 documented and described alcohol advertisements on Twitter related to isolation drinking, community support, promotion of hygiene, home or curbside delivery, the promotion of corporate social responsibility, and any additional prominent themes that emerged (henceforth, "*these themes*") from January 2019 - July 2021. Twitter is the ideal platform to conduct this research because Twitter was a primary method of communication by government officials (e.g., the President of the United States, many governors), health organizations (e.g., the Centers for Disease Control and Prevention), scientists, celebrities, and others during the COVID-19 pandemic in America (Haman et al., 2020; Rufai et al., 2020). Twitter is also an ideal platform to conduct this research because the primary information in tweets is 280 characters or less of text, which enables the writer to provide short, brief, high-level communication. Furthermore, Twitter has a unique user-generated categorization function called a hashtag ("#") which enables these brief messages to become interconnected with other messages about the same topic (Murthy, 2018). While tweets can be embedded with images, videos, or links to longer written content, the primary function of Twitter is the 280 character or less text which often summarizes or contextualizes content contained in attachments. The simple, text-based interface makes Twitter an ideal platform to conduct research at a large scale. From a database of >20,000 tweets generated by consumer product brands spanning January 2019 - July 2021 (the Prevention Research Lab Twitter Marketing

Surveillance Study [PRL-TMS]), Study 1 identified a sample of tweets from alcohol brands that contain information related to *these themes*. Using Natural Language Processing (NLP) software, Study 1 enumerated the tweets and then analyzed them in terms of their cognitive, emotional, and excitative composition.

Aim 1: Validated previously documented themes used by the alcohol industry on social media during the COVID-19 pandemic to market their products in a sample of alcohol companies targeted towards Americans, explored additional themes that may emerge, and assessed their prevalence before during and after early phases of the COVID-19 pandemic (i.e., March-May 2020).

Hypothesis: Themes related to isolation drinking, community support, promotion of hygiene, home or curbside delivery, the promotion of corporate social responsibility, and/or other additional themes will be present in tweets from alcohol brands during the early phases of the COVID-19 pandemic (i.e., March-May 2020), and they will be less prevalent before and after this period.

Aim 2: Explored the cognitive, emotional, and excitative composition of tweets related to *these themes* during the early phases of the COVID-19 pandemic (i.e., March-May 2020). Aim 2 was exploratory and descriptive in nature and intended to describe the linguistic composition of each theme. There was no hypothesis associated with Aim 2.

By validating previous documented themes in an American dataset, exploring additional themes that emerged, and exploring the cognitive, emotional, and excitative composition of tweets generated by the alcohol industry during COVID-19, Study 1 fostered a description of the media effects mechanisms used by the alcohol industry on Twitter during COVID-19 through the lens of the DSMM.

## Study 2

Study 2 was a longitudinal, ecological, naturalistic, population-level study to temporally link alcohol industry marketing activities on social media to community social media activity related to drinking behavior during the COVID-19 pandemic (i.e., demonstrating that alcohol industry marketing activity on Twitter preceded changes in community activity on Twitter related to drinking, and/or that community activity on Twitter related to drinking behaviors preceded changes in alcohol industry marketing activity on Twitter). The data for Study 2 was comprised of repeated measures from the same Twitter accounts over time. Research has shown that behaviors can be predicted from social media activity, such as predicting drinking behavior from social media posts about drinking (Young, 2014; Lane et al., 2023). Therefore, monitoring community activity related to drinking on Twitter is a good approximation of community drinking behavior.

Study 2 examined how ideas about specific themes that are prominent within a sample of Twitter accounts operated by alcohol companies spread across the social media community. For example, *isolation drinking* (e.g., intensified encouragement of home drinking) and *home or curbside delivery* (e.g., using a web application or calling orders in). From a database of >19,000,000 tweets containing hashtags related to COVID-19 from nearly 6,000,000 unique Twitter users spanning March-July 2020 (the Computational Media Lab COVID Database [CML-COVID]; Dashtian et al., 2021), Study 2 identified a sample of tweets that contain information related to *alcohol delivery and isolation drinking* that occurred during the early phases of the COVID-19 pandemic when lockdown measures were widely active in America and concurrently alcohol-related death and disease surged (i.e., March 2020 - May 2020), and then enumerated them. After enumerating the number of industry-level tweets and the number of community-level tweets related to *alcohol delivery and isolation drinking* March-May 2020, Study 2 built a cross-lagged panel model of these data to examine how ideas about *alcohol delivery and isolation drinking* originated and spread on Twitter (e.g., see Illustration 30).

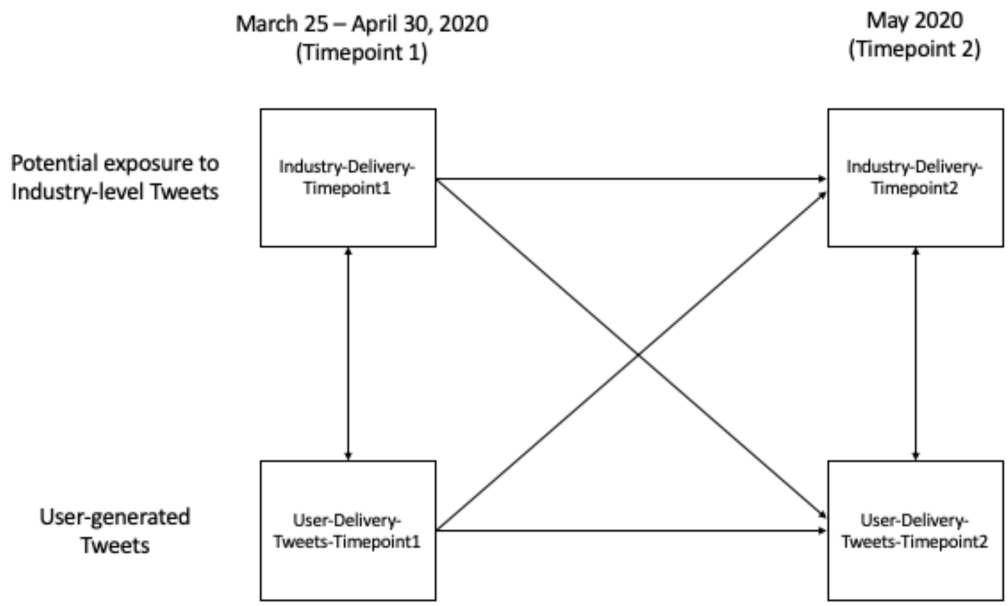



Illustration 30. Conceptual Model for Cross-lagged Panel Analyses in Study 2

Aim 3: Determined if ideas about alcohol delivery and isolation drinking transmitted from the industry level to the community level.

Hypothesis: Increased activity related to alcohol delivery and isolation drinking at the industry-level preceded increases in activity related to alcohol delivery and isolation drinking at the community-level.

Aim 4: Determined if ideas about alcohol delivery and isolation drinking transmitted from the community level to the industry level.

Hypothesis: Increased activity related to alcohol delivery and isolation drinking at the community-level preceded increases in activity related to alcohol delivery and isolation drinking at the industry-level.

By examining if ideas about *alcohol delivery and isolation drinking* transmitted from the industry level to the community level, or vice versa, Study 2 provided longitudinal, ecological, naturalistic, population-level evidence temporally linking alcohol industry marketing activities on social media to community social media activity related to drinking behavior during the COVID-19 pandemic.

Together, Study 1 and Study 2 describe the effects and mechanisms of the alcohol industry on Twitter during COVID-19 as stipulated by DSMM, and they temporally link alcohol industry activity on Twitter to community activity on Twitter related to drinking behaviors during the COVID-19 pandemic.

# Chapter 3: Research Methods

## STUDY 1

### 3.1.1: Study 1 Overview

Throughout history, the alcohol industry has adapted its marketing campaigns when major societal events occurred (e.g., war, disaster, social and political changes, economic depression). A small number of recent studies suggest that the industry may have used the COVID-19 pandemic to its advantage in this way (Foundation for Alcohol Research & Education, 2020; Colbert et al., 2020; Huckle et al. 2021; Martino et al., 2021; Gerritsen et al., 2021; Kennedy et al., 2022). This is concerning because during the COVID-19 pandemic, alcohol-related death and disease increased (Grossman et al., 2020; Koob et al., 2020; Foster et al., 2021; Moon et al., 2021; White et al., 2022). The existing studies about alcohol marketing on social media during the COVID-19 pandemic show that alcohol brands used social media to promote themes like drinking at home, virtual happy hours, drinking while social distancing and wearing masks, home deliver and curbside pickup of alcohol, and corporate socially responsible. However, these studies are limited; they did not include many advertisements, were only done in a few countries, and did not go beyond identifying general themes.

To build onto this work, the present study examined how the alcohol industry used Twitter during the COVID-19 pandemic to determine if these themes were used in response to the COVID-19 pandemic. The present study also explored the potential emergence of other themes. It is important to examine themes that the alcohol industry used in their social media marketing during the COVID-19 pandemic more deeply because it is possible that the proliferation of these themes could be associated with increases in drinking behaviors during the COVID-19 lockdown period, which could then be related to increases in drinking, which in turn could lead to alcohol-related death and disease.

Twitter is an ideal communication medium to use because it was an important form of mass communication during the COVID-19 pandemic and many important figures, including government officials, health experts, and celebrities, used it to communicate with the public (Haman et al., 2020; Rufai et al., 2020). Furthermore, Twitter has been shown to be a valuable resource for sociological research, offering real-time, extensive data from diverse users (Murthy, 2018). Twitter facilitates the study of various social phenomena, including activism, political discourse, and public discussion. Although Twitter's user demographic tends to skew towards younger, urban, and tech-savvy individuals in certain regions, some argue that this can be advantageous for specific research areas, such as analyzing digital communication patterns (Murthy, 2024). Twitter functions as a digital public forum where individuals express their thoughts and engage in discussions. Its features, like hashtags, combined with its global reach, help researchers to examine collective identities, social structures, and shifts in public sentiment. The platform excels at capturing both individual and group dynamics, providing insights into contemporary social trends such as solidarity, polarization, and civic engagement. Despite limitations in its user base representation, Twitter's widespread adoption and accessibility make it very useful for social analysis (Murthy, 2024). Researchers can leverage this platform to gain unique perspectives on how people interact, form opinions, and participate in broader societal conversations, contributing to a more nuanced understanding of our interconnected world.

This study used a dataset of more than 20,000 tweets from January 2019 to July 2021 collected by the Prevention Research Lab at UT-Austin (PRL-TMS). This study, informed by the Differential Susceptibility to Media Effects Model (DSMM) (Valkenburg et al., 2013), analyzed tweets from the alcohol industry using topic modeling, a Natural Language Processing (NLP) methodology (Nunez-Mir et al., 2016; Grootendorst, 2022), and Linguistic Inquiry and Word Count (LIWC) computer software (Chung et al., 2012; Boyd et al., 2022) to understand how the alcohol industry tweets might make people think, feel, and act.

The hypothesis for this study was that themes related to isolation drinking, community support, promotion of hygiene, home or curbside delivery, the promotion of corporate social responsibility, and/or other additional themes would be present in tweets from alcohol brands during the early phases of the COVID-19 pandemic (i.e., March-May 2020), and they would be less prevalent before and after this period. To identify themes, this study applied a topic model technique called BERTopic (Grootendorst, 2022) to the PRL-TMS data. Topic modeling is like sorting a big pile of documents into different buckets based on the main themes or "topics" they talk about. Computer algorithms search for words and phrases that appear together frequently in documents and groups them into these topics, and then researchers look for the similarity between the words in each pile of documents and labels the topic, using various techniques such as having a small number of human research assistants code each theme or having a vast array of artificial intelligence (A.I.) agents classify the words and phrases (Li et al., 2023; Chae et al., 2023). BERTopic combines transformer-based language models with classical clustering techniques and is effective for use on large samples of diverse documents (such as tweets) due to its ability to create coherent topics and its flexibility in handling a varying number of topics (Grootendorst, 2022; Egger et al., 2022). For example, in a collection of news articles, the BERTopic algorithm might identify topics like "basketball," "football," "baseball," "congress," "president," "movies," "TV," and "music" based on the words and their patterns in the articles. Social and psychological researchers can analyze these granular topics to derive higher-order themes that reflect broader social dynamics and psychological constructs (Braun et al., 2006; Blei et al., 2003; Nowell et al., 2017). This process involves human interpretation to ensure the themes maintain coherence and interpretability while capturing meaningful patterns in the data (Egger et al., 2022). In this hypothetical context of news articles, the initial granular topics might be interpreted and grouped under higher-order themes such as "sports," "politics," and "entertainment." This study mapped the themes that emerged from the BERTopic model onto the known themes identified in previous research (Martino et al., 2021), while also exploring the potential for new themes.

Iterative approaches to interacting with the outputs of transformer-based topic models (as well as other machine learning topic models) is an active area of ongoing research, but preliminary work has provided a framework called the Theme and Topic system (Gillies et al., 2022) as guidance. In the Theme and Topic system, Themes are defined as meaningful patterns in qualitative data that emerge through human interpretation of text, and Topics are defined as statistically derived patterns of word co-occurrences in a text identified through computational analysis. Topics can be found autonomously, and Themes emerge via human interpretation. By using topic modeling and human thematic interpretation in parallel, researchers can identify robust thematic connections across very large datasets. This creates a research methodology where machine learning augments human insights rather than replacing them. Study 1's application of BERTopic was informed by the Theme and Topic system.

To evaluate the prevalence of themes, this study classified all of the PRL-TMS data into the identified themes (including a "not belonging to any theme" classification), and compared the proportional frequency (i.e., the number of tweets with these themes present divided by the total number of tweets in a period) of tweets with these themes from January 2019 - January 2020 as compared to March 2020 - May 2020 and June 2020 - July 2021 using proportional z-tests. A proportional z-test is a statistical test used to compare proportions which is typically used to determine if there are significant differences between the observed proportions in two or more groups.

This study also conducted an exploratory analysis, without specific hypothesis testing, to describe the communication mechanisms used in these themes as stipulated by the DSMM. The entire collection of tweets (i.e., the corpus) for each theme was analyzed using LIWC (Chung et al., 2012; Boyd et al., 2022). LIWC functions as an advanced word counter, helping researchers understand the frequency of specific word types to gain insights into emotional states, social concerns, and thinking styles reflected in written content. It serves as a language magnifier, revealing underlying patterns of expression. LIWC groups words into psychologically meaningful categories, making it particularly valuable due to its ability to capture nuanced emotional expressions beyond simple polarity assessments (Tausczik et al., 2010). Its strength lies in its psychological research foundation, allowing for a theoretically informed analysis that considers the multifaceted nature of human emotion (Bantum et al., 2009). Studies have demonstrated LIWC's efficacy across various domains, including social media analysis (De Choudhury et al., 2017), consumer reviews (Yin et al., 2014), and political discourse (Tumasjan et al., 2010). LIWC's ability to provide insights into both explicit and implicit emotional content makes it particularly useful for detecting subtle sentiment shifts and underlying psychological states that may not be apparent through traditional techniques (Kahn et al., 2007). Furthermore, LIWC's standardized approach facilitates cross-study comparisons and longitudinal analyses, enhancing its value for researchers seeking to understand sentiment trends over time or across different contexts (Pennebaker et al., 2003). While LIWC has limitations, such as its reliance on a predefined lexicon, its psychological grounding and extensive validation make it a robust and insightful tool for sentiment analysis in both academic research and practical applications.

Identifying the themes within tweets from the alcohol industry and understanding the thoughts, feelings, and actions these tweets elicited during COVID-19, informed on the alcohol industry's use of Twitter during the pandemic. Determining the prevalence of these themes before, during, and after the early lockdown period of the pandemic helped us to understand if these themes were adapted specifically to the changing societal context of the pandemic. This study provided an example of how the alcohol industry used social media to market products, particularly during a time when the nation was focused on a singular event.

### 3.1.2: Sample

The sample for this study included all the tweets generated by alcohol companies in the PRL-TMS dataset. There were a total of 5,872 tweets from 11 alcohol companies between January 2019 - July 2021. A summary of the tweets included in the study are presented in Table 1 and Figure 1. On average, companies tweeted 534 times throughout the study, or 17 times per month.

### 3.1.3: Measures

The hypothesis for this study was: Themes related to isolation drinking, community support, promotion of hygiene, home or curbside delivery, the promotion of corporate social responsibility, and/or other additional themes would be present in tweets from alcohol brands during the early phases of the COVID-19 pandemic (i.e., March-May 2020), and they would be less prevalent before and after this period. To identify themes, the following measures were used:

**Dictionary of Words For Each Theme.** The topic model yielded a number of latent topics within the corpus of tweets. Each topic was composed of k words. Study 1 created a dictionary for each theme that contained each array of k topic words within that theme.

Aim 2 was exploratory and descriptive in nature and intended to describe the linguistic composition of each theme. There was no hypothesis associated with Aim 2. To address Aim 2, the following measures were used.

**Corpus of tweets for each theme.** All tweets that were classified within each theme were concatenated to develop independent corpuses.

**Proportion of words that meet linguistic characteristics in LIWC for each theme corpus.** Linguistic Inquiry and Word Count software (LIWC-22) was used to quantify the proportion of words within each thematic corpus that matched predefined linguistic and psychological categories. LIWC analyzes text across more than 100 validated variables that capture psychological, emotional, and sociolinguistic dimensions of language. For each theme corpus, the percentage of words falling into LIWC-defined categories was calculated. Key variables included summary scores (Analytical Thinking, Clout, Authenticity, and Emotional Tone) as well as specific word categories related to cognitive processes (e.g., insight, causation), psychological states (e.g., anger, anxiety, sadness), personal concerns (e.g., work, money, health), and informal language use (e.g., swear words, fillers, netspeak). Each LIWC variable is derived from empirically constructed dictionaries developed over decades of research in psycholinguistics and semantic analysis.

### 3.1.4: Procedure

PRL-TMS was conducted between February 2021 and May 2022. Trained research assistants were provided a list of verified Twitter accounts for various consumer products and were instructed to audit the tweets for these accounts generated between January 2019 and July 2021. Each research assistant reviewed every tweet for the given account within those dates and captured information using a data entry form. For each tweet, they recorded the timestamp of their data entry, the name of the company, the date of the tweet, and the type of tweet (e.g., text, text+image, text+video). If it was a retweet, they identified the original author. They also included the verbatim text from the tweet, and the verbatim text from any accompanying image. If there was a video, they transcribed the dialogue. For images or videos, they described in plain language what was visible in the ad, specified the setting or context of the ad, and determined if people were displayed. When people were present, they noted the apparent racial or ethnic identities represented, observed if they were wearing masks, checked if they were socially distanced, and described their facial expressions or general demeanor. The data collection form also had an entry field for additional notes about the tweet, the name of the research assistant working on the entry, and a direct link to the tweet.

Data entry was reviewed and verified weekly by a research coordinator who also provided oversight for all data collection activities. Weekly meetings were held with all research assistants, the research coordinator, and the principal investigator to review progress and address any concerns about the data. The present study focused specifically on the 'verbatim text from the tweet' data. This study was reviewed by the UT-Austin institutional review board and determined to not be human-subject research.

### 3.1.5: Analysis

To prepare the data for topic modeling, a comprehensive preprocessing pipeline was implemented. The preprocessing implementation was carried out using the NLTK library (Bird et al., 2009) in Python (version 3.9.12) (Python Software Foundation, 2022). First, all non-alphabetic characters were removed from all tweets, and the text was converted to lowercase. Then, the tweets were tokenized using the NLTK library's word_tokenize function. To reduce noise and focus on meaningful content, stopwords were removed using a combination of NLTK's English stopwords and a custom list of domain-specific stopwords which included brand terms that frequently appeared in the tweets (e.g., budlight), and time-reference terms that were common on twitter but not informative of topics (am, pm, Monday, January).

To identify themes, Study 1 used the BERTopic algorithm (Grootendorst, 2022) for topic modeling, informed by the Themes and Topics system (Gillies et al., 2022). BERTopic was chosen for its ability to create coherent topics and its flexibility in handling a varying number of topics (Grootendorst, 2022; Egger et al., 2022). The implementation was carried out using the BERTopic library in Python (version 3.9.12) (Python Software Foundation, 2022).

The topic modeling process consisted of five key steps:

1. **Embedding:** The SentenceTransformer model "all-MiniLM-L6-v2" was used to convert the preprocessed tweets into dense vector representations. This model was

selected for its balance of performance and computational efficiency (Wang et al., 2020).

2. **Dimensionality Reduction:** The high-dimensional embeddings were reduced to a lower-dimensional space using Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018). UMAP was configured with variable settings during hyperparameter fine tuning for the following: n_neighbors, n_components, min_dist, and metric, and used the cosine metric.
3. **Clustering:** The reduced embeddings were clustered using the HDBSCAN algorithm (Campello et al., 2013). HDBSCAN was chosen for its ability to handle clusters of varying densities and its robustness to noise. The algorithm was configured with variable settings during hyperparameter fine tuning for min_cluster_size and used the 'eom' (Excess of Mass) cluster selection method.
4. **Topic Extraction:** Topics were extracted from the clusters using a class-based TF-IDF approach (Ramos, 2003). This method helped identify the most representative words for each cluster while accounting for their relative importance across all clusters.
5. **Topic Reduction:** The initial topics were further refined through a topic reduction step, which aimed to merge similar topics and remove less coherent ones (Grootendorst, 2022).

The BERTopic implementation was engineered to output several items for each iteration: a notation of hyperparameters that were used, a list of topics with representative words, an intertopic distance visualization, and a topic similarity matrix. A hyperparameter optimization process was conducted using the BERTopic model to identify the optimal parameters for UMAP and HDBSCAN that would yield the most coherent topics. The hyperparameter optimization process consisted of five key steps:

1. **Parameter Space Definition:** The parameter spaces for UMAP and HDBSCAN were defined with a range of values for each parameter (i.e., UPMAP = number of neighbors ranging from 5 to 35, number of components ranging from 3-10, minimum distance ranging from 0.01-0.5; HDBSCAN= min cluster size ranging from 5-35).
2. **Iterations:** A total of 50 iterations were performed. In each iteration, parameters were randomly sampled from the defined spaces for UMAP and HDBSCAN (Bergstra et al., 2012).
3. **Topic Range Evaluation:** For each hyperparameter set, models were evaluated with a range of maximum topic solutions from 2 to 25.
4. **Model Evaluation:** With 50 hyperparameter optimization iterations of 24 maximum topic solutions, a total of 1,200 models were evaluated. The coherence score for each model was calculated using the c_v coherence measure from Gensim's Coherence Model (Röder et al., 2015). This measure evaluates the degree of semantic similarity between high-scoring words in the topic.
5. **Selection of Best Model:** The parameter set and topic number that resulted in the highest coherence score was identified as the optimal configuration, and selected as the final topic model. The proposed model selection process with coherence score as the primary metric for model evaluation was chosen due to coherence score's correlation with human interpretability of topics (Newman et al., 2010), as well as precedence in similar research, including a recent study that used coherence

score from BERTopic models to evaluate themes in tobacco marketing on social media (Lee et al., 2024).

After identifying the optimal number of topics, the representative topic words for each topic were examined for underlying semantic or contextual connections and synthesized using 5 thousand iterations of evaluation by the large language model (LLM) GPT-4o (OpenAI, 2024). Using LLMs to define topics is an emerging area of research, but preliminary research suggests that the approach is robust (Li et al., 2023). The process involved prompting the OpenAI API to name a topic based on the context of the topic words, and looping through all identified topics with 5000 independent API pings for each. The LLM approach was democratic, meaning that in the case of multiple A.I. interpretations, the topic definition with the most "votes" out of 5000 iterations was selected.

Two researchers reviewed the BERTopic output, sample tweets that mapped onto each topic, and the LLM interpretation of the topics. These materials were viewed through the lens of the current literature regarding alcohol marketing on social media during the COVID-19 pandemic (e.g., Martino et al., 2021). The researchers evaluated these materials for relevance and consistency and identified higher-order themes through discussion. Specifically, the primary researcher checked a sample of tweets for each topic, confirmed the LLM interpretation of the topic words was relevant to the sample, and checked for overlapping thematic connections between topics. The senior researcher reviewed the human interpretation proposed by the primary researcher, and together, through discussion, they reached a consensus on the identified themes. This process of leveraging topic model outputs from machine learning frameworks and qualitative human interpretation was in alignment with recommendations from the Theme and Topic system as well as methods from existing literature (Gillies et al., 2022; Lee et al., 2024).

Hypothesis 1 was supported if the methodology identified themes which synthesized into isolation drinking, community support, promotion of hygiene, home or curbside delivery, the promotion of corporate social responsibility, and/or other additional themes that may emerge.

To evaluate the prevalence of themes, this study classified all of the tweets in each of the specified periods (i.e., period 1 = January 2019 - January 2020; period 2 = March-May 2020; period 3 = June 2020 - July 2021) into a theme using the dictionary of words for each theme identified in the measures section above as well at the interpretation of the theme generated by the LLM and human interpretation. The classification method allowed each tweet to be labeled as related to one of the identified themes, or "not belonging to any theme." Thus, every tweet received a classification. Two approaches for classifying each tweet into a theme were piloted, and the one with the best performance compared to a subset of 250 human-labeled tweets was selected. First, a rule-based classification algorithm using Boolean search logic was created where tweets were mapped against each theme's dictionary words based on logical operators (i.e., AND, OR, NOT). For example, a document containing a few or most of the dictionary words for a theme (AND logic) was considered more relevant to the theme compared to those containing one or none of the dictionary words, and thus the highly theme-relevant tweet would be classified as belonging to that theme. Second, a LLM approach was piloted where a prompt instructed an A.I. agent (using GPT-4o (OpenAI, 2024)) to evaluate each tweet and classify it into one of the identified themes or "not belonging to any theme." The LLM approach involved prompting the OpenAI API to classify a tweet based on its verbatim text, and looping through all tweets in the corpus with independent API pings for each. The method with the best performance compared to the human-labeled subset was scaled to the entire corpus of tweets.

Proportions of tweets related to each theme were calculated for each period (i.e., the number of tweets related to a theme divided by the total number of tweets in that period). Periods were compared using proportional z-tests. Hypothesis 1 was supported if the pandemic lockdown period (March -May 2020) had significantly higher proportions of tweets related to each theme as compared to the pre-pandemic period (January 2019 - January 2020) and later in the pandemic (June 2020 - July 2021). Proportional z-tests were conducted using Stata 17 (StataCorp, 2021).

Aim 2 was exploratory and descriptive, designed to characterize the linguistic composition of each theme. No specific hypothesis was associated with this aim. To address Aim 2, the study analyzed each theme-specific corpus of tweets using LIWC. For each corpus, the proportion of words matching LIWC's predefined linguistic categories was calculated. These proportions were used to identify and rank the most prevalent cognitive, emotional, and excitative processes across themes.

## STUDY 2

### 3.2.1: Study 2 Overview

During the early stages of the COVID-19 pandemic, when lockdowns were common, there was a noticeable surge in alcohol-related illnesses and deaths, including among young adults (Grossman et al., 2020; Koob et al., 2020; Pollard et al., 2020; Auxier et al., 2021; Foster et al., 2021; Moon et al., 2021; Sharma et al., 2021; White et al., 2022).

This period also saw a rise in social media activity, particularly on platforms like Twitter, of which young adults are the most frequent users (Ellis et al., 2020; Pandey et al, 2020; Pandya et al., 2021; Sikorska et al., 2021; Stuart et al., 2021; Lee et al., 2022; Shannon et al., 2022). Alcohol marketing has been linked to alcohol use, particularly among young adults (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017), and including marketing on social media platforms such as Twitter (Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022). Some research has suggested that the alcohol industry used specific themes in their marketing campaigns during the COVID-19 pandemic lockdowns that are potentially harmful to public health, such as isolation drinking and using home or curbside delivery (Martino et al., 2021). However, no research to date has linked social media marketing from the alcohol industry using COVID-19-related themes with activity related to these themes in the broader social media community. Therefore, this study investigated how the alcohol industry's marketing strategies on social media during the lockdown phases of the COVID-19 pandemic might have influenced public behavior and attitudes towards drinking during this time. This study also investigated if the general public's tweets about alcohol might have influenced the alcohol industry's marketing strategies.

This study drew from a database of over 19 million tweets from about 6 million unique users on Twitter, the Computational Media Lab COVID-19 Twitter Dataset (CML-COVID; Dashtian et al., 2021). CML-COVID contains tweets with several specific hashtags, including #coronavirus, #mask, and #covid. When the study first began collecting data, #coronavirus was the only hashtag included, then later #mask and #covid were added. From all the CML-COVID tweets, this study used a sample of 66,874 Twitter accounts based in the United States that tweeted 486,786 times with the #coronavirus hashtag in

English during the early (March 25, 2020 - April 30, 2020) and later (May 2020) phases of the COVID-19 pandemic lockdowns. These tweets provided a snapshot of what people were discussing and how they were feeling during the lockdowns. This study also drew measures from another database of over 5,000 tweets from 11 well-known alcohol brands like Budweiser, Jack Daniel's, and White Claw Hard Seltzer, among others, that occurred during the same timeframe and were collected in a study called the Prevention Research Lab Twitter Marketing Surveillance Study (PRL-TMS). The PRL-TMS tweets were referred to as 'industry-level' tweets. Specifically, this study used the total count of industry-level tweets that were related to alcohol delivery and isolation drinking during the early and later phases of the COVID-19 pandemic lockdowns. This study compared potential exposure to industry-level tweets about alcohol delivery and isolation drinking with user-generated tweets from CML-COVID, which were referred to as 'community-level' tweets.

This study was structured to have two main aims. First, to see if the alcohol industry's tweets about alcohol delivery and isolation drinking) influenced similar activity among the general Twitter community. Second, to see if the general Twitter community's tweets about alcohol delivery and isolation drinking influenced similar activity among the alcohol industry.

To address these aims, this study used a statistical technique known as cross-lagged panel modeling. This study asked the research question: "If industry tweets about alcohol-related ideas increase in the early phase of the COVID-19 pandemic lockdowns, do we see those ideas reflected by the general public during the later phase of the COVID-19 pandemic lockdowns?" Exploring the other direction of effect, this study also asked, "If the general public's tweets about alcohol-related ideas increase in the early phase of the

COVID-19 pandemic lockdowns, do we see a similar increase in tweets containing those ideas from the alcohol industry during the later phase of the COVID-19 pandemic lockdowns?" The hypotheses for this study were that potential exposure to alcohol industry tweets related to alcohol delivery and isolation drinking during the early phase of the COVID-19 pandemic lockdowns was correlated with generating such tweets during the later phase of the lockdowns. Additionally, the hypotheses for this study were that generating tweets related to alcohol delivery and isolation drinking during the early phase of the COVID-19 pandemic lockdowns was correlated with potential exposure to such tweets from the alcohol industry during the later phase of the lockdowns.

It is important to note that this study had some limitations. For example, it only looked at English tweets from the US, and it could not control for all factors, such as the impact of users' behaviors that were not represented in their tweets. Also, this study could not be sure if every tweet represented a real person's thoughts (some could be from automated accounts, or 'bots'). Based on empirical research, it is estimated that 8-18% of Twitter users are bots (Fukuda et al., 2022). However, bot accounts often exhibit distinctive behavioral patterns, such as tweeting extremely frequently (i.e., dozens or hundreds of times per day), tweeting rapidly (i.e., many tweets within the window of several seconds), or tweeting repetitively (i.e., relaying identical content from the same account or across other accounts) (Pastor-Galindo et al., 2022; De Clerck et al., 2024; Mouronte-Lopez et al., 2024). These behavioral patterns could be leveraged for simple bot detection and mitigation strategies, which this study integrated into its preprocessing pipeline. While not comprehensive, filtering out tweets that were reminiscent of bot activity was a straightforward way to attempt to address the bot account limitation.

This study was important because if it found that industry tweets about delivery of alcohol and isolation drinking led to more community tweets on these topics, it could suggest that alcohol marketing on social media had a real impact on public behavior during the COVID-19 pandemic lockdowns. Additionally, the study could reveal that the tweets generated by the general public led to or reinforced the alcohol industries marketing strategies. These findings could be crucial for public health, potentially leading to discussions on stricter regulations for alcohol marketing on social media, especially during times of societal stress like pandemics.

This study aimed to fill a gap in current research by providing longitudinal, real-world evidence of how alcohol marketing on social media might influence drinking behaviors during challenging times, and how the general public may influence industry activity. The results could be significant for public health policies and the regulation of alcohol marketing.

### 3.2.2: Sample

The sample for this study was drawn from CML-COVID, which is a publicly available Big Data dataset composed of 19,298,967 tweets from 5,977,653 global users containing hashtags related to the COVID-19 Pandemic, collected between March 2020 and June 2020 (Dashtian et al., 2021). This data was gathered via Netlytic 2, a tool that utilizes the Twitter REST API for data collection. In total, the dataset constitutes around 15 gigabytes of raw data.

The present study collected all of the CML-COVID tweets containing the #coronavirus hashtag, filtered them to include only English-language content, and removed duplicates, yielding 3,500,547 tweets. This dataset was then refined to include only users with at least two tweets (n = 2,665,987), and further filtered to retain only U.S.-based accounts based on user location data (n = 1,026,503). The analysis was limited to tweets posted between March 25 and May 31, 2020, capturing the early lockdown period in the United States (n = 823,036 tweets from 88,877 users). To facilitate temporal comparison, the data was segmented into two periods: early lockdown (March 25-April 30, 2020) and later lockdown (May 1-31, 2020). To enhance data quality, the study implemented three complementary bot mitigation strategies: (1) removing users who posted more than twice within the same minute (removing 1,105 users from Time 1 and 844 users from Time 2), (2) retaining only users with at least one tweet in both time periods (n = 66,874), and (3) excluding users with tweet counts ≥3 standard deviations above the mean (removing 571 users with an average of 161.60 tweets [SD = 204.24]). These procedures yielded a final sample of 66,303 unique users and 486,786 tweets. Users' average tweet count was 7.34 (SD = 9.48) across the study period.

### 3.2.3: Measures

To address the hypotheses, the following measures were used:

**Counts of tweets related to alcohol delivery and isolation drinking at the industry level from March 25, 2020-April 30, 2020.** Study 1 identified industry tweets related to alcohol delivery and isolation drinking. For this measure, the total count of these tweets from March 25, 2020 to April 30, 2020 was used as a continuous variable.

**Counts of tweets related to alcohol delivery and isolation drinking at the industry level during May 2020.** Study 1 identified industry tweets related to alcohol delivery and isolation drinking. For this measure, the total count of these tweets during May was used as a continuous variable.

**Counts of tweets related to alcohol delivery and isolation drinking for each Twitter account in the study from March 25, 2020-April 30, 2020.** The number of tweets related to alcohol delivery and isolation drinking that occurred from March 25, 2020 to April 30, 2020 was also calculated for each unique Twitter account and used as a continuous variable. The same classification approach used in Study 1 was used here (i.e., a LLM classification model).

**Counts of tweets related to alcohol delivery and isolation drinking for each Twitter account in the study during May 2020.** Similarly, the number of tweets related to alcohol delivery and isolation drinking that occurred during May 2020 was calculated for each unique Twitter account and used as a continuous variable.

**Total number of "Friends" (i.e., the number of accounts that a particular user follows) for each Twitter account in the study.** CML-COVID included the variable user_friends_count which was sampled from the Twitter application program interface. A friend was defined as someone that a Twitter account follows. Number of friends on Twitter has been shown to be associated with the volume, variety, and velocity of content that appears on one's Twitter feed (Sasaki et al., 2015). This study collected the number of friends for each Twitter account in the study from the initial time they appeared (i.e., their first entry).

**Potential exposure to industry-level tweets related to alcohol delivery and isolation drinking from March 25, 2020-April 30, 2020.** CML-COVID did not have a list of all the users' friends (only the total count), therefore given the present data, it was impossible to directly measure if a user followed any of the 11 alcohol companies that comprised the industry-level tweets. Furthermore, one could be exposed to tweets created by an industry Twitter account without following them, for example through the paid ads or retweet mechanisms. Therefore, this study created an index of potential exposure to the industry-level tweets. Given that number of friends on Twitter had been shown to be associated with the volume, variety, and velocity of content that appeared on one's Twitter feed, and given that we had a count of the number of times each user tweeted during the study period, the number of friends multiplied by the number of tweets during a given period was an approximation of overall exposure to Twitter content during that period. That is to say, each time a user logged on to write a tweet, they were potentially exposed to a certain volume of content which was correlated in size to their friend count. Furthermore, multiplying this number (number of friends * number of tweets in a given period) by the total number of industry-level tweets about a specific theme that existed on Twitter estimated the total potential of exposure to these specific themes. That is to say, each time a user logged on and was exposed to their pool of potential content, there were a specific number of tweets generated by the alcohol industry about a specific theme that it was possible to be exposed to.

This measure was rooted in justification from the field of informatics. The concept of network density measures connectivity of networks as the ratio of actual connections that exist between users to the total possible connections within the network (Luarn et al., 2016; Bhattacharya et al., 2023). This concept is crucial for understanding the structure and effectiveness of social networks in facilitating information dissemination. In high density social networks, information can spread quickly and easily, making things like viral marketing or spread of misinformation more likely. Conversely, low density social networks have fewer connections and slower information spread. Sasaki et al. (2015) demonstrated that number of friends on Twitter increases network density, making exposure to information more likely compared to low density networks. In other words, if a user had more friends on Twitter, then they had higher potential to be exposed to information.

This measure was also rooted in justification from the field of public health. Point-of-sale marketing surveillance of tobacco products has a substantial literature. Several approaches to measuring exposure to such marketing exist. One robust method integrates objective audits of retail stores (i.e., counting the number of tobacco ads at a tobacco retail outlet) with self-report measures from individuals who have access to those stores (i.e., asking "How many times per week do you visit retail stores such as gas stations, drug stores, or grocery stores?"). This method was used by Henriksen (2010) and replicated in similar research (Pasch et al., 2023). To calculate an index of exposure, researchers multiplied the number of times a participant visited stores by the number of tobacco advertisements counted at the stores they had access to. It is important to note here that this research was unable to confirm whether the participants actually viewed the ads at the store, but rather the measurement calculated by Henriksen (2010), and others, indicated that they were exposed to an environment that contained the ads, effectively making it an index of potential exposure.

The measure herein (i.e., potential exposure to industry-level tweets related to isolation drinking) was conceptually identical. The number of tweets that a user generated was analogous to the number of times a participant visited a store. The number of friends that a user had was analogous to the type and size of the store. The number of industry-level tweets that were generated during the study period was analogous to the objective count of tobacco ads that existed at the retail stores participants had access to. In both instances, multiplying these signals together created an index of potential exposure. Indeed, social media platforms generate much of their revenue via advertisements (Raffoul et al., 2023), and these platforms have been suggested as "the storefront of the future" (Wiese, 2021). When users visited "the store of Twitter" they were potentially exposed to a pool of advertisements, and the measure herein was a methodological way to quantify potential exposure to a given segment of products (i.e., alcohol) on Twitter.

Therefore, the equation [number of friends * number of tweets from March 25, 2020-April 30, 2020 * number of industry-level tweets about alcohol delivery and isolation drinking from March 25, 2020-April 30, 2020] provided a variable that represented the potential exposure to industry-level tweets related to alcohol delivery and isolation drinking from March 25, 2020-April 30, 2020.

**Potential exposure to industry-level tweets related to alcohol delivery and isolation drinking during May 2020.**Similarly, this study calculated a variable that represented the potential exposure to industry-level tweets related to alcohol delivery and isolation drinking during May 2020.

### 3.2.4: Analysis

Cross-lagged panel analysis was the statistical technique used to examine associations between variables across different time points (Kuiper et al., 2018). This method was particularly useful in understanding how variables may influence each other over time. This study built such a model for tweets related to alcohol delivery and isolation drinking across two timepoints. Timepoint 1 was early in the COVID-19 pandemic lockdown (March 25, 2020 - April 30, 2020), and Timepoint 2 was later in the COVID-19 pandemic lockdown (May 2020). The model focused on four variables: Potential Exposure to Industry-level Tweets at Timepoint 1, Number of Tweets Generated by users at Timepoint 1, Potential Exposure to Industry-level Tweets at Timepoint 2, Number of

Tweets Generated by users at Timepoint 2. The aim was to explore the bi-directional influences between Potential Exposure to Industry Tweets and User Tweets across two timepoints.

The GSEM package in Stata 17 (Stata Corp, 2021) was used to construct these models. The models included: autoregressive paths (i.e., paths that assessed the stability of each variable over time), cross-sectional covariation paths (i.e., the covariation between potential exposure to industry tweets and user tweets at the same timepoint), and cross-lagged paths (i.e., paths that examined the influence of one variable at Timepoint 1 on the other variable at Timepoint 2). The hypotheses were supported if the estimate for the path between potential exposure to industry tweets at timepoint 1 and user tweets at timepoint 2 was positive and significant, and vice versa (i.e., user tweets at timepoint 1 and potential exposure to industry tweets at timepoint 2).

# Chapter 4: Study 1

**Title:** Alcohol Marketing on Twitter During the COVID-19 Pandemic: Theme Identification, Prevalence Comparison, and Sentiment Analysis

**Intended Journal:** Cyberpsychology, Behavior, and Social Networking / PLOS One


## ABSTRACT

**Introduction:** During the COVID-19 pandemic, increased social media use and targeted digital marketing by alcohol companies potentially heightened consumer exposure to alcohol advertising. This study aimed to identify themes used in tweets by alcohol brands early in the pandemic, determine if the prevalence of these themes differed across time, and characterize the sentiment used in the themes.

**Method:** We analyzed 5,872 tweets from 11 prominent alcohol companies on Twitter between January 2019 and July 2021. Themes were identified using a BERTopic modeling approach informed by the Themes and Topics system. We leveraged the GPT-4o Large Language Model for topic interpretation, alongside human evaluation for thematic validation. Theme prevalence was assessed by comparing tweets from pre- (January 2019 - January 2020), peri- (March 2020 - May 2020), and post-pandemic response periods (June 2020 - July 2021) using proportional z-tests. Sentiment analysis using Linguistic Inquiry and Word Count (LIWC) assessed cognitive, emotional, and excitative characteristics of thematic content, informed by the Differential Susceptibility to Media Effects Model (DSMM).

**Results:** Five primary themes emerged: Alcohol Delivery and Isolation Drinking, Restaurant Support, Social Media Promotions, Sports, and Gaming. Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions themes had significantly higher prevalence during the peri-pandemic response period compared to pre- and post-pandemic response periods ($p<0.003$). Alcohol Delivery and Isolation Drinking, and Social Media Promotions had significantly higher prevalence during the post-pandemic response period compared to the pre-pandemic response period ($p<0.003$). The Sports and Gaming themes remained statistically stable across periods. Sentiment analysis revealed that Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions utilized stronger emotional and excitative appeals, while Sports and Gaming were more cognitively oriented and neutral.

**Discussion:** The findings suggest that alcohol companies adopted marketing content on social media that emphasized emotionally engaging and action-oriented messaging about ordering delivery alcohol, drinking while staying home, engaging with alcohol brands on social media, and supporting restaurants; and that some of this content remained in use at higher rates after the early pandemic period. Future research linking these themes with consumer behavior is needed. Understanding these dynamics is essential to developing effective policies and protective measures against exploitative marketing strategies during future crises.

## INTRODUCTION

The alcohol industry has a history of adapting its marketing strategies during crisis events and major societal changes, such as wars, natural disasters, and economic downturns. For example, after World War II, returning soldiers brought back drinking habits shaped by their wartime experiences (Herrmann et al., 1996; Shephard, 2001; National World War II Museum, 2021). In response, alcohol companies used patriotic-themed advertisements in print and broadcast media to target this group (Samuel, 2001; Scott, 2009; Richards, 2022). During this time, the country also experienced increases in consumption of alcohol per capita (a trend that extended into the 1960's and accelerated in the 1970's) (Landberg, 2009). Such instances of strategic adaptation - numerous throughout history - raise concerns about the how the alcohol industry may use crisis events to sustain and expand its market.

Emerging evidence suggests that this pattern of adaptive alcohol marketing continued during the COVID-19 pandemic (Foundation for Alcohol Research & Education, 2020; Colbert et al., 2020; Gerritsen, et al., 2021; Huckle et al., 2021; Martino et al., 2021; Kennedy et al., 2022). Further, widespread stay-at-home orders confined the great majority of Americans to home, and pushed many aspects of daily life online, which led to a surge in social media usage during the early pandemic (Fernandez et al., 2020; Pandey et al., 2020; Al-Habaibeh et al., 2021; Meier et al., 2021; Pham et al., 2022, Cho et al., 2023; Kelly et al., 2023). The confluence of increased social media usage and targeted marketing by alcohol companies during the pandemic may have resulted in greater exposure to alcohol marketing. Further, alcohol-related mortality (e.g., death from acute alcohol poisoning) and morbidity (e.g., instances of liver-related hospitalizations) both increased during the early pandemic (Grossman et al., 2020; Koob et al., 2020; Foster et al., 2021; Moon et al., 2021; White et al., 2022). Given the well-established link between alcohol marketing exposure and increased alcohol use (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017; Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022), this rise in marketing exposure may have contributed to higher levels of consumption, elevating the risk of negative health outcomes. This is especially concerning given that a large proportion of social media users are youth or young adults, populations that are both particularly vulnerable to marketing and, in some cases, under the legal drinking age ( Arnett, 2014; Packer et al., 2022). Lastly, this marketing adaptation is important to study because it demonstrates a recurring pattern of adaptation in response to crisis events; thus, there is opportunity for future action to mitigate harmful marketing adaptation during future crises.

Research examining alcohol marketing on social media during the COVID-19 pandemic has identified several emergent themes, including isolation drinking (e.g., intensified encouragement of home drinking), community support (e.g., virtual gatherings with alcohol present), promotion of hygiene (e.g., social distancing/washing hands/wearing masks while consuming alcohol), home or curbside delivery (e.g., using a web application or calling orders in), and promotion of corporate social responsibility (e.g., alcohol brands making sanitization products or calling for support of essential job workers) (Martino et al., 2021; Gerritsen et al., 2021). However, this research is not extensive, has a relatively small timeframe under investigation (i.e., only during the early lockdown phases of the pandemic without comparing to pre- or post- time periods), is limited in the number and types of advertisements, has only been conducted in a few countries around the world, and lacks sufficient information beyond broad themes/topics.

Given these limitations in the existing literature, there is a need to validate and expand upon previous findings using different datasets and methodologies. It is also crucial to test if these themes differed across time so that we can quantify the degree to which these themes were adapted specifically to COVID-related conditions. Further, previous research has largely focused on identifying themes without examining the linguistic structures and tones through which they were communicated. By incorporating sentiment analysis, we can investigate the tones in which these themes were conveyed - such as whether they were framed positively, negatively, or neutrally. This sentiment dimension is critical because linguistic structures such as emotional valence in marketing messages significantly influence consumer behavior (Vrtana, 2023; Zhao et al., 2025). Additionally, fitting these findings to a theoretical model can provide a more comprehensive understanding of how marketing may have reinforced or altered drinking behaviors during the pandemic. Reproducing and expanding upon this body of knowledge will not only strengthen existing evidence but also enhance our understanding of alcohol marketing strategies during times of crisis and their potential impacts.

Understanding the themes and sentiment contained within alcohol industry marketing content on social media during the COVID-19 pandemic and tracing the prevalence of these themes over time is crucial for public health. The alcohol industry's ability to rapidly adapt its messaging in response to the pandemic may have influenced drinking behaviors, shaped social norms around alcohol consumption, and exacerbated health risks. By examining these adaptations during a crisis event, the present study provides insight into the alcohol marketing playbook during such times which enables the development of mitigation strategies during future crises. Therefore, the purpose of the present study was two-fold. First, we validated previously documented themes used by the alcohol industry on social media during the COVID-19 pandemic and explored if additional themes were present. Using BERTopic (an advanced natural language model; Grootendorst, 2022), we analyzed marketing content on Twitter from alcohol companies targeting American consumers. We then analyzed if theme prevalence varied across three time periods: pre-pandemic (pre-; January 2019-January 2020), early pandemic lockdown (peri-; March 2020-May 2020), and later pandemic phases (post-; June 2020-July 2021). Second, we explored the cognitive, emotional, and excitative composition of content related to these themes during the peri-pandemic phase (March-May 2020). This analysis utilized Linguistic Inquiry and Word Count (LIWC; Boyd et al., 2022), a text analysis tool that categorizes words into psychological and linguistic dimensions, alongside the Differential Susceptibility to Media Effects Model (DSMM; Valkenburg et al., 2013), which provides a framework for understanding how media messages influence individuals through various response states.

We hypothesized that themes related to isolation drinking, community support, hygiene promotion, delivery services, corporate social responsibility, and/or potentially other emerging themes would be more prevalent in alcohol brand tweets during the early phase of the COVID-19 pandemic (March-May 2020) compared to pre-pandemic and later pandemic periods. In utilizing LIWC and DSMM for sentiment analysis, this study is the first to explore the communications mechanisms underlying these themes, and thus this exploratory analysis has no specific hypothesis. Understanding the language used by the alcohol industry in social media marketing during the COVID-19 pandemic can inform public health policy and industry regulation, helping anticipate and mitigate similar marketing patterns during future crises.

## METHOD

Twitter (now, X) emerged as a crucial communication platform during the COVID-19 pandemic, serving as a primary channel for information dissemination by government officials, health experts, and influential figures (Haman et al., 2020; Rufai et al., 2020). Beyond its role in pandemic communication, Twitter has established itself as a valuable resource for sociological research, offering real-time data from diverse users (Murthy, 2018). While Twitter's user demographic tends to skew toward younger, urban, and technologically engaged populations, this characteristic can be advantageous for specific research domains, particularly in analyzing digital communication patterns (Murthy, 2024). The platform's functionality as a digital public forum, combined with its active advertising environment makes it particularly suitable for research examining themes and sentiment among industry actors. Thus, the present study utilizes the platform Twitter for investigation.

**Sample and Data Collection**

The dataset comprised 5,872 tweets from 11 alcohol companies posted between January 2019 and July 2021, collected as part of the Prevention Research Lab at UT-Austin Twitter Marketing Study (PRL-TMS) (Table 1, Figure 1). PRL-TMS is a larger study of over 20,000 tweets from consumer product companies that aimed to document and describe marketing practices before, during, and after the onset of the COVID-19 pandemic; the data from the present study is a subset of alcohol companies. Alcohol companies were selected to include a variety of popular beverage types (i.e., beer, spirit, hard seltzer) with at least 10,000 followers. Data collection occurred between February 2021 and May 2022, with trained research assistants auditing verified Twitter profiles of selected alcohol companies. Research assistants systematically documented each tweet using a standardized data entry form. Weekly quality control reviews were conducted by a research coordinator, with regular team meetings including research assistants and the principal investigator. This study was reviewed by the UT-Austin institutional review board and determined to be non-human-subject research.

| Company (n=11) | Number of Tweets (n= 5,872) | Number of Followers (n=1,199,052) (single cross-section during study period) |
|---|---|---|
| Bud Light | 1,142 | 248,718 |
| Truly Hard Seltzer | 755 | 17,762 |
| Malibu | 695 | 46,824 |
| Jägermeister | 653 | 88,688 |
| Samuel Adams Beer | 594 | 81,707 |
| Brooklyn Brewery | 592 | 125,605 |
| Jack Daniel's | 535 | 198,896 |
| Budweiser | 409 | 220,140 |
| Bacardi | 233 | 99,687 |
| Absolut Vodka | 147 | 30,526 |
| White Claw Hard Seltzer | 117 | 40,499 |

Table 1. Summary of tweets included in Study 1

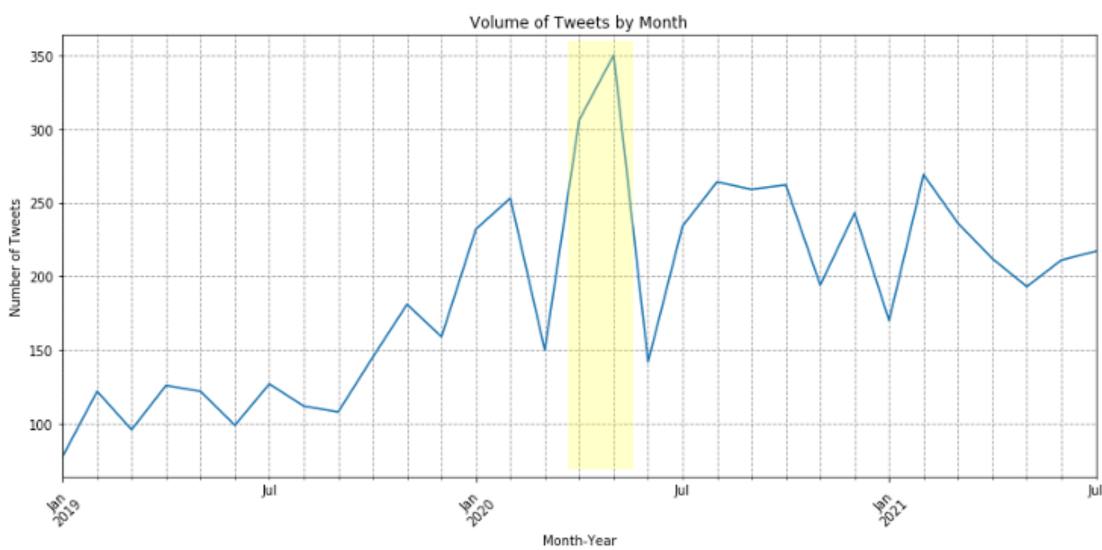



Figure 1. Month-over-month volume of tweets included in Study 1, COVID-19 Lockdown Period Highlighted (n= 5,872)

## Analysis

The present study focused on coding the content of the alcohol company tweets using three primary analytic approaches: theme identification, prevalence comparison, and sentiment analysis.

### *Theme Identification*

To identify themes, this study used the BERTopic algorithm (Grootendorst, 2022) for topic modeling, informed by the Themes and Topics framework (Gillies et al., 2022), wherein researchers apply human interpretation to identify themes from algorithm-derived topics. This framework is important because it helps ensure that the analysis is thoughtful and structured, rather than relying solely on automated results. It makes the findings easier to understand by organizing topics into broader themes that align with real-world concepts. It also adds depth by allowing researchers to consider the context and nuances of the data, reducing the risk of misinterpretation. BERTopic was chosen for its ability to create coherent topics and its flexibility in handling a varying number of topics (Grootendorst, 2022; Egger et al., 2022). The implementation was carried out using the BERTopic library in Python (version 3.9.12) (Python Software Foundation, 2022), and is available in Appendix 1.

The topic modeling process consisted of several key steps:

**Tweet Preprocessing.** Using Python (version 3.9.12) and the NLTK library (Bird et al., 2009), tweets underwent systematic preprocessing: removal of non-alphabetic characters, conversion to lowercase, tokenization using NLTK's *word_tokenize* function, and removal of standard and domain-specific stopwords (including brand terms and temporal references).

**Embedding.** The SentenceTransformer model "all-MiniLM-L6-v2" was used to convert the preprocessed tweets into dense vector representations. This model was selected for its balance of performance and computational efficiency (Wang et al., 2020).

**Dimensionality Reduction.** The high-dimensional embeddings were reduced to a lower-dimensional space using Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018). UMAP was configured with variable settings during hyperparameter fine tuning for the following: n_neighbors, n_components, min_dist, and metric, and uses the cosine metric.

**Clustering.** The reduced embeddings were clustered using the HDBSCAN algorithm (Campello et al., 2013). HDBSCAN was chosen for its ability to handle clusters of varying densities and its robustness to noise. The algorithm was configured with variable settings during hyperparameter fine tuning for min_cluster_size and uses the 'eom' (Excess of Mass) cluster selection method.

**Topic Extraction.** Topics were extracted from the clusters using a class-based TF-IDF approach (Ramos, 2003). This method helps identify the most representative words for each cluster while accounting for their relative importance across all clusters.

**Topic Reduction.** The initial topics were further refined through a topic reduction step, which aims to merge similar topics and remove less coherent ones (Grootendorst, 2022).

The BERTopic implementation was engineered to output several items for each iteration: a notation of hyperparameters that were used, a list of topics with representative words, and a coherence score. A hyperparameter optimization process was conducted using the BERTopic model to identify the optimal parameters for UMAP and HDBSCAN that yield the most coherent topics. The hyperparameter optimization process consisted of several key steps:

**Parameter Space Definition**. The parameter spaces for UMAP and HDBSCAN were defined with a range of values for each parameter (i.e., UPMAP = number of neighbors ranging from 5 to 35, number of components ranging from 3-10, minimum distance ranging from 0.01-0.5; HDBSCAN= min cluster size ranging from 5-35).

**Iterations**. A total of 50 iterations were performed. In each iteration, parameters were randomly sampled from the defined spaces for UMAP and HDBSCAN (Bergstra et al., 2012).

**Topic Range Evaluation**. For each hyperparameter set, models were evaluated with a range of maximum topic solutions from 2 to 25.

**Model Evaluation**. With 50 hyperparameter optimization iterations of 24 maximum topic solutions, a total of 1,200 models were evaluated. The coherence score for each model was calculated using the *c_v* coherence measure from Gensim's Coherence Model (Röder et al., 2015). This measure evaluates the degree of semantic similarity between high-scoring words in the topic.

**Selection of Best Model**. The parameter set and topic number that resulted in the highest coherence score was identified as the optimal configuration and selected as the final topic model. Using coherence score as the primary metric for model evaluation was chosen due to coherence score's correlation with human interpretability of topics (Newman et al., 2010) and use in similar research, including a recent study that used coherence score from BERTopic models to evaluate themes in tobacco marketing on social media (Lee et al., 2024).

The principal output from the BERTopic model is lists of topics defined by their representative words. These representative words must be evaluated for underlying semantic or contextual connections and synthesized into an understandable topic name. For example, a hypothetical representative word array of ‘teacher, student, classroom, math, history, science’ may be synthesized into the topic name ‘education.’ The synthesizing process was done using 5 thousand iterations of evaluation by the large language model (LLM) GPT-4o (OpenAI, 2024). Using LLMs to name topics is an emerging area of research, but preliminary research suggests that the approach is robust, and particularly good at naming topics without inherent bias that may result from a human with an a priori hypothesis naming the topics (Li et al., 2023). The process involved prompting the OpenAI API to name a topic based on the context of the topic words and looping through all identified topics with 5000 independent API pings for each. The LLM approach was democratic, meaning that in the case of multiple A.I. interpretations, the topic definition with the most “votes” out of 5000 iterations was selected. The implementation was carried out using the OpenAI library in Python (version 3.9.12) (Python Software Foundation, 2022), and is available in Appendix 2.

Two researchers reviewed the BERTopic output, sample tweets that mapped onto each topic, and the LLM interpretation of the topics. These materials were viewed through the lens of the current literature regarding alcohol marketing on social media during the COVID-19 pandemic (e.g., Martino et al., 2021; Gerritsen et al., 2021). To transition from BERTopic-generated topics to qualitative thematic interpretations, the researchers first evaluated each topic by carefully examining representative tweets and comparing these against the machine-generated topic labels. This interpretive step involved assessing the coherence, meaning, and contextual relevance of topic terms provided by the LLM within the broader context of alcohol marketing literature. Researchers then considered whether the algorithmically-derived topic labels accurately captured the thematic essence of the sampled tweets or required refinement or adjustment.

Following this initial interpretation, the researchers collaboratively discussed the interpretations to establish preliminary thematic categories. This involved grouping topics based on conceptual similarity, shared underlying meanings, or relationships indicated by topic keywords and sample tweet content. For example, grouping the topics 'Virtual NFL Draft Party with Celebrity Hosts' and 'Home Sports Challenge Campaigns' into one 'Sports' theme. The grouping process allowed the researchers to identify overarching narratives, patterns, or messaging.

To validate the identified higher-order themes, the primary researcher performed additional checks, selecting representative tweets within each theme to ensure alignment between individual tweets, the LLM-generated labels, and the thematic interpretations derived by the researchers. The senior researcher reviewed and critiqued these thematic interpretations, providing a secondary check for interpretive validity. Differences in interpretation were reconciled through discussion, allowing the researchers to achieve consensus on clearly defined, higher-order themes that accurately reflected the underlying data.

Ultimately, this systematic and collaborative process of leveraging outputs from the BERTopic machine learning framework and integrating qualitative interpretation and validation aligned closely with methodological recommendations from the Theme and Topic system and established approaches in the existing literature (Gillies et al., 2022; Lee et al., 2024).

### *Prevalence Comparison*

To evaluate the prevalence of themes across the pre- (January 2019 - January 2020), peri- (March 2020 - May 2020), and post-pandemic response (June 2020 - July 2021) periods, this study used GPT-4o (OpenAI, 2024) to classify all of the tweets during each of the specified periods into one of the identified themes, or "not belonging to any theme." Thus, every tweet received a classification. The process involved prompting the OpenAI API to classify text (i.e., a verbatim tweet) into a list of options (i.e., the identified themes + "not belonging to any theme"), and looping this procedure through an entire corpus of text (i.e., the dataset of tweets). LLMs are well suited for text classification with recent review suggesting remarkable performance and versatility across many text classification tasks (Fields et al., 2024). Furthermore, during prompt engineering, researchers compared the LLM classifications to a sample of 250 human-labeled classifications to ensure >85% agreement. The implementation was carried out using the OpenAI library in Python (version 3.9.12) (Python Software Foundation, 2022), and is available in Appendix 3.

The proportions of tweets classified into each theme was calculated for each period (i.e., the number of tweets related to a theme divided by the total number of tweets in that period), and then periods were compared using proportional z-tests. Proportional z-tests were conducted comparing the pre-pandemic response period (January 2019 - January 2020) and the post-pandemic response period (June 2020 - July 2021) to the peri-pandemic response period (March 2020 - May 2020), using Stata 17 (StataCorp, 2021). Additionally, a proportional z-test was conducted comparing the pre-pandemic response period (January 2019 - January 2020) to the post-pandemic response period (June 2020 - July 2021). A Bonferroni correction was applied to the $\alpha$ value, adjusted proportionally to the number of identified themes multiplied by three, to account for the three tests conducted on each theme (Bland et al., 1995). The Bonferroni correction is justified because when conducting multiple statistical tests on a large dataset, the probability of observing at least one significant result by random chance alone increases dramatically with each additional test (known as the multiple comparisons problem), making it necessary to adjust the significance threshold to maintain the overall error rate at the desired level and prevent the proliferation of false positive findings that would otherwise occur when analyzing numerous variables simultaneously.

### *Sentiment Analysis*

To describe the communication mechanisms used in these themes during the early lockdown phases of the pandemic as framed by the DSMM (Valkenburg et al., 2013), the entire collection of tweets for each theme was analyzed using LIWC (Chung et al., 2012; Boyd et al., 2022). LIWC functions as a linguistic magnifying glass, helping researchers understand the frequency of specific word types to gain insights into emotional states, social concerns, and thinking styles reflected in written content. The prominent cognitive, emotional, and excitative processes that LIWC yields were reported for each theme.

## RESULTS

### Sample Characteristics

Alcohol companies ranged in frequency of tweets from 117 to 1142 with an average of 534 tweets throughout the study period or approximately 17 tweets per month. Based on a single cross-sectional audit, the 11 companies had nearly 1.2 million Twitter followers around the early pandemic period (March/April 2020) (Table 1, Figure 1).

## Topic Model Performance

Across 1,200 candidate models, the configuration with the following hyper parameters had the highest coherence score (0.768): iteration=25; UPMAP: number of neighbors=15, number of components=4, minimum distance=0.01; HDBSCAN= min cluster size=20; maximum topic solution=23 (Figure 2). This optimal configuration yielded seven topics.

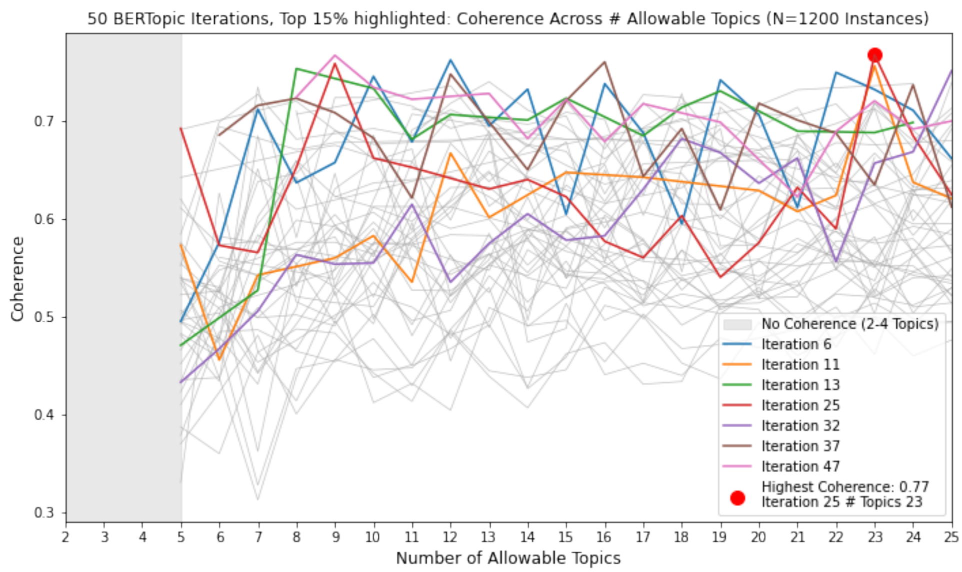



Figure 2. Topic Model performance

## Theme Identification

The seven topics were synthesized using LLM and human interpretation into five themes: Alcohol Delivery and Isolation Drinking, Restaurant Support, Social Media Promotions, Sports, and Gaming (Table 2).

**Prevalence Comparison**

Given that five themes were identified, and three proportional tests were needed per theme, the alpha level was set at $\alpha=0.0033$ per the Bonferroni correction. The prevalence of Alcohol Delivery and Isolation Drinking tweets significantly increased from the pre- to peri-pandemic response period ($\Delta$ = 91.2%; $z=6.03$; $p<0.001$), significantly decreased from the peri- to post-pandemic response period ($\Delta$ = -31.6%; $z=-4.13$; $p<0.001$), but remained significantly higher comparing the post-pandemic to the pre-pandemic response period ($\Delta$ = 30.8%; $z=2.98$; $p=0.0029$). Similarly, Restaurant Support tweets significantly increased from the pre- to peri-pandemic response period ($\Delta$ = 189.5%; $z=4.89$; $p<0.001$), and significantly decreased from the peri- to post-pandemic response period ($\Delta$ = -69.1%; $z=-6.18$; $p<0.001$). There was no significant difference between pre- and post-pandemic in Restaurant Support tweets ($\Delta$ = -30.8%; $z=-0.50$; $p=0.62$). Social Media Promotion tweets also significantly increased from the pre- to peri-pandemic response period ($\Delta$ = 150%; $z=7.80$; $p<0.001$), significantly decreased from the peri- to post-pandemic response period ($\Delta$ = -35.8%; $z=-4.62$; $p<0.001$), and the prevalence remained significantly higher in the post-pandemic period as compared to the pre-pandemic period ($\Delta$ = 60.6%; $z=4.59$; $p<0.001$). Sports and Gaming tweets did not differ across the any of the study periods (Sports pre- to peri-pandemic response period ($\Delta$ = 39%; $z=2.59$; $p=0.01$), peri- to post-pandemic response period ($\Delta$ = -21.9%; $z=-2.17$; $p=0.03$), and pre- to post-pandemic response period ($\Delta$ = 8.5%; $z=0.083$; $p=0.41$); Gaming pre- to peri- ($\Delta$ = 80.8%; $z=1.80$; $p=0.07$), peri- to post-($\Delta$ = -20%; $z=-0.79$; $p=0.43$), and pre- to post- ($\Delta$ = -45.46%; $z=1.40$; $p=0.16$) (Table 3).

| Theme | Constituent Topics (LLM Interpretation) | Representative Words | Example Tweets |
|---|---|---|---|
| Alcohol Delivery and Isolation Drinking | Pandemic Alcohol Delivery and Enjoyment | sunshine, drizly*, bit, day, order, happy, part, original, drink, get<br><br>*delivery app company, acquired by Uber | Pro tip: Put your beer in a coffee mug and no one knows you're drinking on the video conference.<br><br>Bring the beachside vibe *inside* with Malibu delivered to your door in 60 minutes or less Get $5 off your first order, courtesy of Drizly, with code 'Sunshine20'! |
| Restaurant Support | COVID-19 Restaurant Support and Donation Initiatives | restaurant, donate, support, worker, help, fund, restaurantstrong, strong, covid, donation | Yeah, we miss bars too. But we can still support restaurant and bar workers today. Order takeout and we'll buy your first round when they reopen.<br><br>ATTENTION TWITTER: The people who serve us every day are depending on our support. Restaurants and bars, let us know that you're #OpenForTakeout, so we can spread the word. |
| Social Media Promotions | Virtual Bar Tours & Live Instagram Events<br><br>Pandemic Social Engagement & Promotions | live, instagram, tune, tonight, ig, go, dive, tour, bar, edition<br><br>whassup, say, tag, win, merch, friend, chance, need, collection, week | Going live now! doing a little thing with @budlight tonight on IG and facebook live. come hang 6pm pst / 9pm est #divebartour<br><br>Win fresh Bud Whassup crewnecks for your happy hour crew! RETWEET + FOLLOW for the chance to outfit your crew. |
| Sports | Virtual NFL Draft Party with Celebrity Hosts<br><br>Home Sports Challenge Campaigns | draft, boo, commish, nfl, youtube, drafterparty, robgronkowski, guest, host, camillekostek<br><br>challenge, team, house, sport, league, submit, home, athlete, chance, video | TONIGHT!!!! @nfl @budlight<br><br>Want to win a @trulyseltzer putting mat? Show us your #dailyninesetup. Photo, video, putting, chipping, whatever. Show us. |
| Gaming | Charity Gaming Events by Alcohol Brands | royale, twitch, tv, charity, money, tonight, dkm, therealgavinlux, snellzilla, final | Excited to announce the Bud Light Seltzer Charity Royale. 24 pro athletes compete in Call of Duty: Warzone to raise money for awesome local causes.<br><br>Live right now on http://Twitch.tv/BudLight. Tune in! |

Table 2. Theme Structure and Representative Content

| Theme | Pre-Pandemic Response (%) | Peri-Pandemic Response (%) | Post-Pandemic Response (%) |
|---|---|---|---|
| Alcohol Delivery and Isolation Drinking | 9.1* | 17.4† | 11.9*† |
| Restaurant Support | 1.9* | 5.5† | 1.7* |
| Social Media Promotions | 6.6* | 16.5† | 10.6*† |
| Sports | 8.2 | 11.4 | 8.9 |
| Gaming | 1.1 | 2.0 | 1.6 |

* = Statistical difference from Period 2, $p < 0.0033$
† = Statistical difference from Period 1, $p < 0.0033$

Table 3. Theme Prevalence Across Periods

**Sentiment Analysis**

Linguistic markers derived from LIWC provided a descriptive analysis of five themes throughout the peri-pandemic response period in terms of their cognitive, emotional, and excitative characteristics. The entire output of applying LIWC to each theme corpus is available in Appendix 4, and a summary is reported in this section.

***Alcohol Delivery and Isolation Drinking***

Relative to the other themes, the Alcohol Delivery and Isolation Drinking theme exhibited high analytical thinking (85.63%; i.e., more logical, structured, evidence-based reasoning) with moderate clout (76.92%; i.e., more confidence, dominance, social authority) and relatively low authenticity (36.78%; i.e., a more guarded, strategic, or detached communication style.). The notably higher tone score (80.16%; i.e., emotional positivity) suggests predominantly positive emotional valence. The theme demonstrated relatively higher use of perceptual processes (9.64%; i.e., words related to seeing, hearing, feeling, and other sensory experiences) and spatial references (5.75%; i.e., words related to location, direction, distance, and spatial relationships), with minimal use of negative emotions (0.22%; i.e., words such as angry sad, hate) or conflict-related language (0%; i.e., words such as oppose, fight, win). The relatively higher presence of food-related terms (4.89%; i.e., words such as eat, drink) and physical references (5.84%; i.e., words related to the body or physical actions) suggests content possibly focused on sensory or experiential descriptions.

***Restaurant Support***

The Restaurant Support theme presented the highest clout score (94.38%) and high analytical thinking (89.68%), coupled with the highest tone score (98.49%) across all themes. It showed relatively lower authenticity (9.14%), suggesting formal or carefully constructed content. The text was characterized by relatively stronger social references (17.04%; i.e., words related to social interaction, relationships, communities) and higher affiliation markers (7.77%; i.e., words related to group membership, cooperation, unity), with higher lifestyle-related content (11.02%; i.e., words related to daily habits, routines, personal interests, ways of living). The complete absence of negation words (0%; i.e., words related to contradiction, opposition, or refusal) and negative emotions (0%) reinforces its overwhelmingly positive orientation.

### *Social Media Promotions*

The Social Media Promotions theme maintained relatively high analytical thinking (88.89%) and clout (93.8%), with moderate authenticity (32.37%). It showed relatively stronger perception-related content (12.04%) and motion references (2.73%; i.e., words related to movement, direction, and physical action), suggesting dynamic or descriptive content. The theme demonstrated relatively balanced use of temporal references (past: 0.50%, present: 2.70%, future: 1.49%) and maintained relatively higher positive emotional tone (88.52%) while keeping negative emotions minimal (0.20%).

### *Sports*

The Sports theme showed the second-highest analytical thinking score (92.32%) with relatively higher clout (87.37%), but a relatively lower tone score (67.22%) compared to the previous themes. It maintained substantial perceptual processes (10.26%) and spatial references (6.28%), with increased use of numbers (5.85%) suggesting more technical or detailed content. The theme showed moderate social references (11.61%) and achievement-related content (2.31%; i.e., words related to success, goal-setting, progress, and performance), with balanced temporal orientation.

### *Gaming*

The gaming theme exhibited the highest relative analytical thinking score (98.37%) but lower clout (71.90%) and tone (51.75%) compared to other themes. It showed the highest use of big words (26.99%; i.e., words six or more letters in length) and numbers (10.02%), suggesting technical or specialized content. The theme demonstrated relatively reduced function word usage (29.86%; i.e., words are essential for grammar, cohesion, and sentence flow) but increased technical references (3.48%; i.e., words related to STEM, precision, expertise) and perception-related content (11.04%), indicating possibly more complex or specialized subject matter.

### *Through the DSMM Lens*

Building on the LIWC analysis through the lens of the DSMM, we can explore how these themes might employ certain cognitive, emotional, and excitative processes when consumed. From a cognitive processing perspective, the consistently high analytical scores (ranging from 85.63% to 98.37%) suggest content that elicits substantial cognitive engagement, with Gaming potentially eliciting the highest cognitive stimulation due to its elevated use of big words (29.86%) and technical language (3.48%). The emotional processing dimension showed an interesting gradient across themes, with Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions displaying high positive emotional tone (ranging from 80.16%-98.49%) and minimal negative affect, suggesting content that might be more emotionally engaging and accessible. This emotional content notably decreased in The Sports and Gaming themes (67.22% and 51.75% respectively), indicating a shift toward more neutral, technically-oriented content. Regarding excitative processes, the varying levels of perceptual references (particularly high in Social Media Promotions at 12.04%) and motion-related content (also peaking in Social Media Promotions at 2.73%) suggest different levels of arousal potential. Restaurant Support’s high social and affiliation markers (17.04% and 7.77% respectively) might facilitate stronger state excitation through social engagement, while Gaming's elevated technical content and reduced social markers might demand more focused, less arousal-dependent processing.

## DISCUSSION

The purpose of the present study was two-fold. First, to validate previously documented themes used by the alcohol industry on social media during the COVID-19 pandemic to market their products in a sample of alcohol companies targeted towards Americans, explore additional themes that may emerge, and asses the prevalence of these themes before (January 2019 - January 2020), during (March 2020 - May 2020), and after (June 2020 - July 2021) the early lockdown phases of the COVID-19 pandemic. Five themes were identified: Alcohol Delivery and Isolation Drinking, Restaurant Support, Social Media Promotions, Sports, and Gaming. Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions were significantly more prevalent during the peri-pandemic response period (March 2020 - May 2020) compared to pre-and post-pandemic response periods, while Sports and Gaming were more stable.

The identification of five themes and their variation across time supports the study hypothesis. The findings align with existing research but also expands to new understandings potentially related to regional (i.e., U.S.) and platform (i.e., Twitter/X) differences. Specifically, this study reproduced Alcohol Delivery and Isolation Drinking, and Restaurant Support themes similar to previous research (Martino et al., 2021; Gerritsen et al., 2021). The significant increase in Alcohol Delivery and Isolation Drinking ($\Delta$ = 91.2%) during the peri-pandemic period demonstrates a clear shift in marketing focus. Although there was a subsequent decrease during the post-pandemic period ($\Delta$ = -31.6%), Alcohol Delivery and Isolation Drinking remained significantly higher post-pandemic compared to pre-pandemic ($\Delta$ = 30.8%). The consistency of these findings suggests that the alcohol industry focused marketing efforts on delivery services and at-home drinking activities during the early phases of the pandemic across regions and contexts, a period when alcohol-related morbidity and mortality also increased (Grossman et al., 2020; Koob et al., 2020; Foster et al., 2021; Moon et al., 2021; White et al., 2022). Exposure to these themes in marketing content may have contributed to spreading unhealthy ideas such as delivering alcohol to your home for consumption in isolation, though future research modeling the spread of these ideas between the industry and community is needed.

Similarly, Restaurant Support showed a substantial increase during the peri-pandemic period ($\Delta$ = 189.5%), highlighting the industry's use of corporate social responsibility (CSR) strategies previously documented within the alcohol industry (Martino et al., 2021; Babor et al., 2013; Mialon et al., 2018). While supporting struggling restaurants and bars during the pandemic was essential and pro-social in helping mitigate substantial losses across the food and hospitality industries (Song et al., 2021), CSR also serves alcohol companies via reputation management and direct advancement of commercial interests simultaneously (Mialon et al., 2018). Importantly, despite the substantial initial increase, Restaurant Support messaging significantly decreased post-pandemic (69.1%), returning to below pre-pandemic levels ($\Delta$ = -30.8%). By positioning themselves as champions of restaurant recovery while simultaneously promoting home delivery and isolation drinking, alcohol companies created a dual narrative legitimizing their pandemic response through apparent social responsibility and expanding their direct-to-consumer channels, exemplifying CSR's function in deflecting regulatory scrutiny while advancing market objectives (Yoon et al., 2013).

This study identified the Social Media Promotion theme, which, although potentially similar to the Community Support theme identified in previous research (Martino et al., 2021), represents a distinct and unique theme. Tweets classified under this theme frequently mentioned events such as virtual happy hours and livestreamed gatherings, like Community Support themes in previous research. However, these tweets commonly contained explicit calls to action encouraging engagement with a brand's social media channels, thereby increasing exposure to marketing content and the likelihood of alcohol use. Social Media Promotions showed a notable surge ($\Delta$ = 150%) during the peri-pandemic period, aligning with broader trends of increased social media usage driven by people's pursuit of virtual connections amid physical isolation (Nabity-Grover et al., 2020). Despite a significant decrease post-pandemic ($\Delta$ = 35.8%), Social Media Promotion remained significantly elevated compared to pre-pandemic levels ($\Delta$ = 60.6%). The intersection of intensified alcohol marketing and heightened social media use is especially concerning given the established association between social media use and increased risks for mental health problems, including depression, anxiety, and problematic substance use behaviors (U.S. Department of Health and Human Services, 2023). Furthermore, these findings parallel troubling societal trends of declining mental health among youth and young adults (Thomas et al., 2022; Haidt, 2024; McGorry et al., 2024), the demographic most active on social media platforms.

This study also found evidence of themes not previously identified as prominent within alcohol industry marketing the COVID-19 pandemic. Specifically, Gaming and Sports. The identification of Gaming and Sports themes could reflect both platform-specific and regional characteristics. Twitter's young adult user base in the U.S. (Murthy, 2024) combined with the established propensity for alcohol companies to use sports and gaming marketing campaigns make this finding expected (Westberg et al., 2018; Kelly et al., 2021). Nevertheless, this finding highlights the importance of considering platform demographics and regional contexts when studying alcohol marketing adaptations.

The temporal patterns in theme prevalence provide important insights into the alcohol industry's strategic marketing adaptations during the COVID-19 pandemic. The significant increases in Alcohol Delivery and Isolation Drinking ($\Delta$ = 91.2%), Restaurant Support ($\Delta$ = 189.5%), and Social Media Promotions ($\Delta$ = 150%) during the peri-pandemic period demonstrate a clear shift in marketing focus. While there were subsequent significant decreases in these themes during the post-pandemic period (decreases of 31.6%, 69.1%, and 35.8% respectively), our statistical analyses reveal important long-term changes. Notably, Alcohol Delivery and Isolation Drinking remained significantly higher in the post-pandemic compared to pre-pandemic period ($\Delta$ = 30.8%), as did Social Media Promotions ($\Delta$ = 60.6%). In contrast, Restaurant Support tweets showed no significant difference between post-pandemic and pre-pandemic periods ($\Delta$ = -30.8%). These patterns suggest durable changes in certain marketing strategies even after the acute lockdown phases ended. The magnitude of changes that occurred from the pre- to peri-pandemic response period suggests a deliberate and substantial reallocation of marketing resources in response to pandemic conditions. In contrast, the statistical stability of Sports and Gaming themes throughout the study period indicates these were likely part of ongoing marketing strategies rather than pandemic-specific adaptations. This pattern of selective theme adaptation shows how the alcohol industry maintained certain marketing foundations while strategically amplifying specific messages in response to the unique circumstances of the pandemic, with some of these adaptations persisting well beyond the initial pandemic response. This finding contributes to the existing literature by demonstrating longitudinal patterns of alcohol marketing themes before, during, and after the initial pandemic response period, whereas existing research focuses mostly on the presence of those themes during one period.

The second purpose of this study was to explore the cognitive, emotional, and excitative composition of content related to these themes during the early phases of the COVID-19 pandemic (i.e., March-May 2020). Sentiment analysis suggested that Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions had higher emotionality, whereas Sports and Gaming had higher scores on cognitive dimensions. Restaurant Support and Social Media Promotions had higher scores on excitative dimensions (i.e., calls to action).

The finding that the Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions themes carried stronger emotional appeals is concerning for several reasons. First, these themes emerged during a period of heightened psychological vulnerability brought on by pandemic-related stress (Pfefferbaum et al., 2020; Panchal et al, 2021). Research indicates that emotion-laden advertisements have more persuasive impact (Hamelin et al., 2017), and that online advertisers exploit psychological vulnerabilities of consumers using their marketing communication (Strycharz et al., 2021). Hence, the introduction of these themes during the early phases of the pandemic when stress and anxiety were high - coupled with linguistic structures that engage emotional triggers - raises ethical questions.

Second, the emotional pull of these campaigns often alluded to the hardships faced by essential workers in the restaurant industry. For example, one alcohol company tweeted "Yeah, we miss bars too. But we can still support restaurant and bar workers today. Order takeout and we'll buy your first round when they reopen. Find out more at [link]…" One interpretation of this is that the alcohol industry, consciously or not, harnessed the sympathy generated by restaurant support messaging to direct attention to other emotionally laden themes (i.e., Alcohol Delivery and Isolation Drinking, Social Media Promotions) aimed at broadening commercial activities.

Furthermore, the Restaurant Support and Social Media Promotions themes contained the strongest calls to action, reminiscent of subverting action towards charitable donation and directing the user towards engaging with their social media page. For example, one alcohol company tweeted "[Brand] Mojito time everyone! Have you put your creative spin on #The CoconutChallenge yet?! @[BrandTwitterProfile] is extending their pledge to support businesses most impacted by COVID-19 and your dancing can do even more good!..." These tactics, in essence, repurpose genuine empathy and goodwill towards struggling businesses to advance corporate interests, ultimately exploiting a population already made vulnerable by the pandemic.

Concurrent to alcohol companies launching these campaigns, alcohol-related deaths among young people increased significantly compared to the previous year. Specifically, deaths among Americans aged 16-20 rose by 29.8%, those aged 21-24 saw a 25.6% increase, and deaths among individuals aged 25-34 increased by 37% (White et al., 2022). Given the link between alcohol marketing exposure and increased alcohol use (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017; Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022), it is possible that exposure to marketing messages such as the ones found in this study could have been associated with the increased use of alcohol. Future research which objectively identifies exposure to marketing messages and prospectively examines the association with alcohol use is needed.

Beyond conventional public health effects (e.g., increased alcohol use), the marketing strategies employed by the alcohol industry during the early phases of the pandemic could also have economic effects. The food and beverage delivery industry (i.e., delivery apps) has grown immensely since the pandemic, with alcohol delivery commanding a significant market share (Duthie et al., 2023). Delivery apps have been shown to redistribute tips from restaurant workers to gig drivers as well as redistribute thin profit margin from restaurant owners to Big Tech companies (e.g., Uber, DoorDash) (Chen et al., 2022). Thus, while originally marketed alongside supporting restaurants, in the long term, delivery apps could ultimately hurt the restaurant industry. The use of delivery apps has also been linked to unfavorable health behaviors (e.g., eating high-calorie/low-nutrient foods) and is used by vulnerable populations such as those experiencing food insecurity (Buettner et al., 2023); and thus, in the long-term delivery apps could contribute to widening health disparities. This also aligns with the commercial determinants of health framework wherein a significant proportion of individual health outcomes are driven by the activity of commercial entities, which are entirely beyond the person's control (Lacy-Nichols et al., 2023).

A key strength of this study is its longitudinal design, covering extensive 'pre-' (January 2019–January 2020) and 'post-' (June 2020–July 2021) windows surrounding the acute pandemic response period (March–May 2020). This extended observation period allows for more robust identification of thematic shifts in alcohol marketing over time. Specifically, having approximately one full year of baseline (pre-pandemic lockdown) data and over a year of post-pandemic lockdown data provides a clear context to determine whether observed thematic changes were transient adaptations limited to the early pandemic period or represented longer-term shifts in marketing strategy. This design significantly enhances the study's capacity to draw more confident conclusions regarding the durability and evolution of marketing themes, compared to designs with narrower or non-existent comparative timeframes.

The findings from this study have important implications for contemporary alcohol marketing practices, particularly in the context of evolving digital environments and delivery services. The demonstrated persistence of themes such as Alcohol Delivery and Isolation Drinking and Social Media Promotions beyond the immediate COVID-19 lockdown period suggests a durable shift toward social and direct-to-consumer strategies within the alcohol industry. These insights highlight the necessity for ongoing regulatory attention to social media platforms, where persuasive emotional and action-oriented appeals can rapidly reach and influence consumers, potentially exacerbating problematic drinking behaviors. Furthermore, given the increasing prominence of app-based delivery services that make alcohol readily available, understanding these thematic trends can guide the development of targeted public health interventions and policies designed to mitigate potential harms of alcohol industry marketing, especially strategically adaptive campaigns.

**Limitations**

This study has several limitations that should be acknowledged. First, the dataset consisted exclusively of tweets from 11 alcohol companies with substantial social media followings (>10,000 followers), potentially overlooking marketing strategies employed by smaller or regional companies with fewer followers and less widespread popularity. Consequently, findings may not be generalizable to the broader alcohol industry, periods beyond the study timeframe, or other digital platforms beyond Twitter. Additionally, Twitter's demographics (typically younger, urban, and technologically adept) may influence the types of themes identified, potentially omitting strategies targeted at older or rural populations.

Second, not all data collection occurred concurrently with the posting of tweets, meaning that any tweets deleted prior to data capture were inevitably missed. This could result in an incomplete representation of the marketing strategies deployed during the study period.

Third, although the BERTopic modeling approach combined with GPT-4o and human interpretation offers a rigorous framework inspired by the Themes and Topics system (Gillies et al., 2022), synthesizing information from topic modeling and AI-generated outputs into themes retains inherent subjectivity. Despite efforts to minimize bias through extensive iterative human-AI validation, subtle biases stemming from language complexity and contextual nuances may still exist.

Fourth, the sentiment analysis tool employed (LIWC; Chung et al., 2012; Boyd et al., 2022), while insightful regarding cognitive, emotional, and excitative dimensions of communication, uses a dictionary-based approach that might overlook nuanced meanings better captured by alternative sentiment analysis methods. Thus, future research should explore complementary sentiment analysis techniques to validate the present findings. Additionally, the sentiment analysis conducted was exploratory and descriptive, providing aggregate rankings of various linguistic variables rather than statistical tests. More detailed analyses may raise different findings.

Lastly, this study cannot establish associations between exposure to alcohol marketing and subsequent consumption behaviors or health outcomes. Future investigations using longitudinal or experimental designs, along with direct observations of marketing exposure and individual-level data, are essential to empirically test these associations, including examination of potential moderating factors such as psychological vulnerability or preexisting drinking patterns.

## Conclusion

This study provides critical insights into the strategic adaptations of alcohol marketing on social media during the COVID-19 pandemic, highlighting significant shifts in themes such as Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions. These adaptations leveraged emotional and excitative appeals, potentially exploiting psychological vulnerabilities heightened during crisis conditions. Understanding these dynamics underscores the need for regulatory scrutiny and public health preparedness to mitigate potentially harmful industry practices during future crises.

# Chapter 5: Study 2

**Title:** Alcohol Information Dynamics During Crisis: A Longitudinal Analysis of Bidirectional Influence Between Alcohol Industry Marketing and User-Generated Content on Twitter During the COVID-19 Pandemic

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## ABSTRACT

**Introduction:** During the COVID-19 pandemic, alcohol consumption and related mortality increased among young adults in the U.S., coinciding with increased social media use. Given the documented influence of alcohol marketing on drinking behaviors, this study examines the bidirectional association between alcohol industry social media marketing and user-generated content related to alcohol delivery and isolation drinking on Twitter during early pandemic lockdowns.

**Method:** We conducted a longitudinal analysis combining industry tweets from 11 major alcohol brands (n=704 tweets) and user-generated tweets from 66,303 U.S. Twitter users (n=486,786 tweets), collected during two periods: early during the pandemic lockdown (March 25-April 30, 2020) and later in the lockdown (May 1-31, 2020). Tweets were classified using GPT-4o to identify content related to alcohol delivery and isolation drinking. Potential exposure to alcohol brand tweets was quantified as a standardized composite score integrating user friend count, tweet frequency, and alcohol brand tweet volume. A cross-lagged panel model assessed bidirectional temporal associations.

**Results:** Analysis revealed an asymmetrical association. Early potential exposure to alcohol brand tweets significantly predicted subsequent user-generated content about alcohol delivery and isolation drinking ($\beta = 0.05$, $p<0.01$), indicating each standard

deviation increase in potential exposure corresponded to a 5% greater likelihood of generating related content later. Conversely, early user-generated content did not significantly predict subsequent potential exposure to alcohol brand tweets ($\beta = 0.01$, $p=0.639$). High temporal stability emerged for both potential exposure ($\beta = 0.85$, $p<0.01$) and user-generated content ($\beta = 2.76$, $p<0.01$).

**Discussion:** Findings highlight the unidirectional influence of alcohol industry marketing on user-generated content during crisis periods. The absence of a bidirectional influence underscores industry-driven messaging rather than an industry response to public discourse. Given established links between alcohol marketing and drinking behaviors, these results advocate for strengthened regulatory oversight of digital alcohol advertising during public health crises. Future research should explore the dynamics of alcohol information over extended periods and across diverse crisis contexts.

## INTRODUCTION

During the COVID-19 pandemic, alcohol consumption and related mortality increased substantially in the United States, particularly among young adults (Grossman et al., 2020; White et al., 2022). Specifically, from 2019 to 2020, alcohol-related deaths increased by 29.8% among Americans aged 16-20, 25.6% among those aged 21-24, and 37% among individuals aged 25-34 (White et al., 2022). Simultaneously, social media usage intensified as stay at home orders and physical distancing measures (i.e., lockdowns) transformed daily communication patterns (Fernandez et al., 2020; Pandey et al., 2020; Al-Habaibeh et al., 2021; Meier et al., 2021; Pham et al., 2022; Cho et al., 2023; Kelly et al., 2023), with platforms like Twitter (now X) becoming essential channels for pandemic-related information exchange and social connection (Haman et al., 2020; Rufai et al., 2020). Given these intersecting trends, understanding the potential role social media played in shaping drinking behaviors during the pandemic is critical for informing future public health strategies.

The convergence of increased alcohol consumption and heightened social media usage raises important public health concerns regarding alcohol industry marketing practices during this time. Research consistently demonstrates that exposure to alcohol marketing influences drinking behaviors, with young adults being particularly susceptible (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017; Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022). Furthermore, several studies indicate that the alcohol industry adapted marketing strategies during the early COVID-19 period (March-May 2020), prominently featuring pandemic-related themes such as alcohol delivery services and at-home drinking (Foundation for Alcohol Research & Education, 2020; Colbert et al., 2020; Gerritsen et al., 2021; Huckle et al., 2021; Martino et al., 2021; Kennedy et al., 2022). To this end, our previous research analyzed over 5,000 tweets from major alcohol brands between January 2019 and July 2021, revealing "Alcohol Delivery and Isolation Drinking" as the dominant theme during initial lockdown months, comprising nearly 20% of tweets. Furthermore, sentiment analysis of tweets related to alcohol delivery and isolation drinking revealed high levels of emotionality, suggesting these campaigns may have leveraged emotional vulnerability to promote alcohol consumption during a period of heightened pandemic-related psychological distress (Pfefferbaum et al., 2020; Panchal et al., 2021). This practice is consistent with established tactics among online advertisers who routinely exploit psychological vulnerabilities of consumers in their marketing communication (Strycharz et al., 2021).

Centering marketing campaigns on alcohol delivery and isolation drinking represents a significant public health concern for several reasons. First, messaging that promotes rapid alcohol access could have encouraged or exacerbated problematic drinking behaviors during a period of heightened psychological distress, as increased isolation, anxiety, and economic uncertainty during the COVID-19 pandemic have been consistently linked with greater alcohol use (Clay et al., 2020; Pollard et al., 2020; Barbosa et al., 2021). Further, given that lockdown periods restricted traditional sources for obtaining alcohol, marketing that emphasized alcohol delivery may have encouraged drinking among individuals who would otherwise have abstained.

Second, these marketing strategies could indicate a structural adaptation by the alcohol industry in response to temporary regulatory changes, potentially resulting in permanent shifts in alcohol accessibility. For example, the temporary policy in Texas permitting restaurants and alcohol retailers to offer alcoholic beverages "to-go" was subsequently made permanent (Texas Legislature, 2021), highlighting the enduring impacts of pandemic-era regulatory adjustments on alcohol distribution and consumption norms. These permanent regulatory changes allow increased access to alcohol even in situations where distance or time of day may preclude traveling to an outlet to purchase alcohol.

Third, the industry's marketing shift aligned with documented increases in user-generated social media content that helped to normalize alcohol consumption as a coping mechanism during the pandemic. Multiple studies have reported significant increases in online discussions about problematic drinking and using alcohol as a coping strategy. For example, Wanchoo et al. (2023) identified elevated Reddit activity related to alcohol-based coping mechanisms, and Ward et al. (2021) noted an increase in Twitter discussions concerning alcohol-induced blackouts during the initial lockdown period compared to previous years. Similarly, Stone et al. (2022) observed a shift toward more authentic and emotionally charged tweets about alcohol use, indicating greater openness regarding personal drinking experiences. Litt et al. (2021) further demonstrated that perceiving that peers were posting on social media about using alcohol to cope with the pandemic was associated with both an increased likelihood of making similar posts and increased drinking among adults. Additionally, Ahmed et al. (2024) found that Twitter discussions during the COVID-19 pandemic normalized risky drinking behaviors such as binge drinking, especially among young adult males aged 25-34, and frequently associated alcohol consumption with coping, celebrations, and entertainment.

Collectively, these findings suggest that increased online dialogue about alcohol during COVID-19 may have coincided with alcohol industry marketing adaptations, raising important questions about how changes in alcohol industry marketing practices may have affected social media posts about alcohol use and also how social media posts about alcohol use may have affected alcohol industry marketing practices. Social media activity has been shown to predict drinking behaviors (Young, 2014; Lane et al., 2023); thus, increased online posts during the pandemic may serve as a meaningful proxy for increased alcohol consumption. Understanding the dynamics between industry-driven marketing and user-generated social media content during the pandemic could help inform the future development of safeguards against potentially exploitative marketing practices during future public health crises.

Media effects theories provide a framework for understanding how industry marketing and user-generated social media content may have bidirectional effects. The Differential Susceptibility to Media Effects Model (DSMM) suggests that media exposure triggers cognitive, emotional, and excitative responses that influence attitudes and behaviors (Valkenburg et al., 2013). Applied to alcohol marketing, exposure to alcohol-related content can inform consumers about drinking options, emotionally engage them through positive associations, and stimulate desire for consumption behaviors. Additionally, the theory of sociogenesis highlights the bidirectional nature of media influence, wherein user-generated content and community trends can shape industry marketing practices (Valsiner et al., 2000; Baker et al., 2020; Walsh et al., 2020; Negowetti, et al, 2022). In this context, user generated content about drinking behaviors may be adapted by industry actors and transformed into a media campaign.

Despite growing evidence of strategic shifts in alcohol marketing and changes in alcohol-related discussions on social media during the pandemic (Foundation for Alcohol Research & Education, 2020; Colbert et al., 2020; Gerritsen et al., 2021; Huckle et al., 2021; Litt et al., 2021; Martino et al., 2021; Ward, et al., 2021; Kennedy et al., 2022; Stone et al., 2022; Ahmed et al., 2024), significant gaps remain in understanding the bidirectional association between alcohol industry messaging and user-generated social media content. While previous research has documented marketing adaptations by the alcohol industry and changes in social media discussions about alcohol, no studies have systematically examined the longitudinal associations between alcohol industry marketing and user-generated content during the pandemic. Therefore, the purpose of the present study is to investigate the bidirectional association between potential exposure to alcohol brand tweets and user-generated content related to alcohol delivery and isolation drinking during COVID-19 lockdown.

To assess the bidirectional association between alcohol brand tweets and user-generated content about alcohol delivery and isolation drinking, the present study employed a longitudinal design using Twitter data collected during two distinct pandemic periods (T1 - early in the pandemic lockdown, March 25-April 30, 2020; and T2 - later in the pandemic lockdown, May 1-May 31, 2020). Using a publicly available dataset of over 19 million tweets with hashtags related to the COVID-19 pandemic (Dashtian et al., 2021) as well as a dataset of tweets from 11 verified alcohol brand Twitter accounts, we identified a sample of 66,303 U.S. Twitter users who posted in English at least once during both time periods. For each user, we quantified: the frequency of posts related to alcohol delivery and isolation drinking during both periods; and a measure of potential exposure to alcohol brand tweets about alcohol delivery and isolation drinking incorporating both individual-level factors (i.e., friend count, overall tweet volume) and industry-level metrics (i.e., frequency of themed marketing messages during respective time periods) during both periods. We then constructed a cross-lagged panel model to assess temporal associations between potential exposure to alcohol brant tweets at T1 and user-generated content at T2, and conversely, between user content at T1 and subsequent potential exposure to alcohol brand tweets at T2, while controlling for stability paths of both variables as well as cross-sectional correlations between both variables.

We hypothesized positive bidirectional associations, anticipating that increased potential exposure to alcohol brand tweets would predict subsequent increased user-generated content about alcohol delivery and isolation drinking, and conversely, that greater user-generated alcohol content would predict subsequent increased potential exposure to alcohol brand tweets.

## METHOD

During the COVID-19 pandemic, Twitter (now, X) became an essential medium for communication, heavily utilized by public health officials, governmental representatives, and prominent public figures to disseminate timely information (Haman et al., 2020; Rufai et al., 2020). In addition to its pandemic-specific significance, Twitter has proven beneficial for sociological research by providing scientists with access to immediate data generated by diverse user groups (Murthy, 2018). The platform's demographics (younger, urban, and digitally proficient) offer advantages for targeted research, especially when studying online communication dynamics (Murthy, 2024). Moreover, Twitter's dual capacity as both a digital public square and an active advertising hub makes it particularly relevant for exploring marketing trends. Given these attributes, Twitter was chosen as the analytical medium for the current research.

### Sample and Data Collection

The present study combined user-generated content from the Computational Media Lab COVID-19 Twitter Dataset (CML-COVID; Dashtian et al., 2021) with alcohol brand tweets from the Prevention Research Lab Twitter Marketing Surveillance Study (PRL-TMS; Study 1). The CML-COVID dataset was collected via Netlytic 2 utilizing the Twitter REST API and comprises over 19 million tweets from nearly 6 million global users containing COVID-19-related hashtags, collected between March and July 2020. PRL-TMS data included 5,872 tweets from 11 alcohol companies posted between January 2019 and July 2021, collected as part of a larger study comprising over 20,000 tweets from consumer product companies, aiming to document and describe marketing practices before, during, and after the onset of the COVID-19 pandemic. The PRL-TMS data analyzed here is a subset specifically from alcohol companies. Alcohol companies were selected to represent a variety of popular beverage types (i.e., beer, spirit, hard seltzer) with at least 10,000 followers. Data collection occurred between February 2021 and May 2022, with trained research assistants auditing verified Twitter profiles of selected alcohol companies and systematically documenting each tweet using a standardized data entry form. The present study uses the verbatim text from each tweet.

For user-generated data from CML-COVID, we extracted tweets with the #coronavirus hashtag and filtered to include only English-language content, yielding 3,500,547 tweets. The #coronavirus hashtag was selected because it was the earliest data available in CML-COVID; other pandemic-related hashtags were added later. We refined this dataset to include only users with at least two tweets (n = 2,665,987), then filtered to retain only U.S.-based accounts based on user location data (n = 1,026,503). We further limited our analysis to tweets posted between March 25 and May 31, 2020, capturing the lockdown period in the United States (n = 823,036 tweets from 88,877 users). To facilitate temporal comparison, we segmented the data into two periods: early lockdown (March 25-April 30, 2020) and later lockdown (May 1-31, 2020). This division allowed us to examine how potential exposure to content in the early lockdown period might influence content generation in the later period, and vice versa.

Bot activity is known to be problematic on Twitter (Fukuda et al., 2022). Because problematic twitter accounts often exhibit distinctive behavioral patterns, such as tweeting extremely frequently (i.e., dozens or hundreds of times per day), tweeting rapidly (i.e., many tweets within the window of several seconds), or tweeting repetitively (i.e., relaying identical content from the same account or across other accounts) (Pastor-Galindo et al., 2022; De Clerck et al., 2024; Mouronte-Lopez et al., 2024), and to enhance data quality, we implemented three complementary strategies. First, we removed users who posted more than twice within the same minute (removing 1,105 users from Time 1 and 844 users from Time 2), Second, we retained only users with at least one tweet in both time periods (n = 66,874). Finally, we excluded users with tweet counts ≥3 standard deviations above the mean (removing 571 users with an average of 161.60 tweets [SD = 204.24]). These procedures yielded a final sample of 66,303 unique users and 486,786 tweets. Average tweet count per user was 7.34 (SD = 9.48) across the study period, and they had an average friend count of 3235.19 (SD = 12824.25). A flow chart of the data collection procedures for user-generated content is shown in Figure 3.

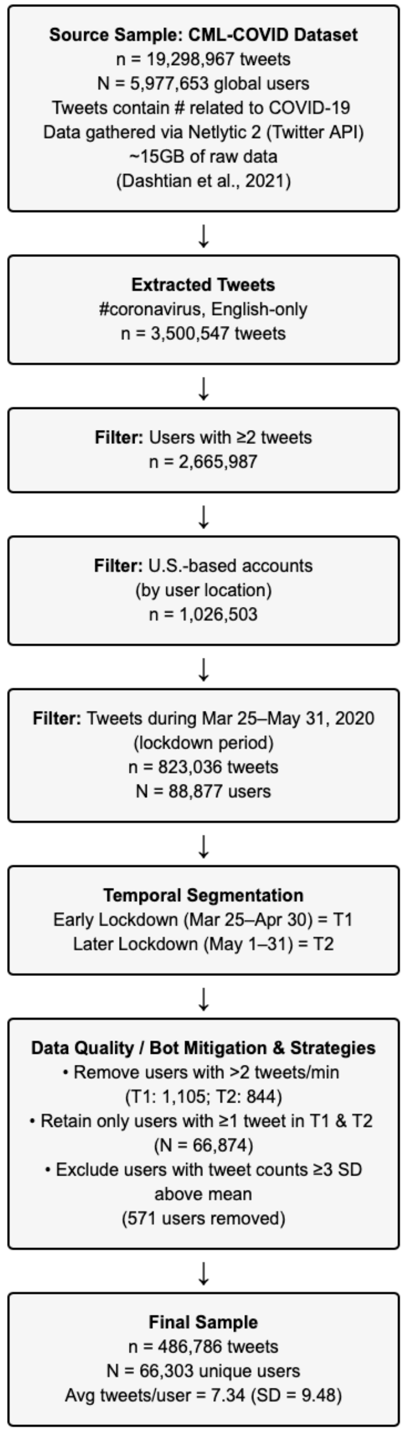


Figure 3. Flow chart of data collection procedures for user-generated content in Study 2

For alcohol brand data, we extracted alcohol company tweets from PRL-TMS posted between March 25 - May 31, 2020 (n=704). The distribution of these alcohol brand tweets is summarized in Table 4. The methods for the present study were reviewed by the UT-Austin institutional review board and determined to be non-human-subjects research.

| Brand (n=11) | Number of Tweets (n= 703) | Number of Followers (n=1,199,052) (single cross-section during study period) |
|---|---|---|
| Bud Light | 193 | 248,718 |
| Budweiser | 148 | 17,762 |
| Malibu | 107 | 46,824 |
| Truly Hard Seltzer | 67 | 88,688 |
| Jagermeister USA | 49 | 81,707 |
| Jack Daniel's | 45 | 125,605 |
| Samuel Adams Beer | 35 | 198,896 |
| Brooklyn Brewery | 30 | 220,140 |
| White Claw | 11 | 99,687 |
| Absolut Vodka | 11 | 30,526 |
| Bacardi | 7 | 40,499 |

Table 4. Summary of alcohol brand tweets included in Study 2 (March 25 - May 31, 2020)

**Tweet Classification**

Given that previous research identified “Alcohol Delivery and Isolation Drinking” as the most prevalent theme tweeted about by alcohol brands during the COVID-19 pandemic lockdowns (e.g., Martino et al., 2021; Study 1), we classified each tweet as related to "Alcohol Delivery and Isolation Drinking,” other prominent themes (see Appendix 3), or “not belonging to any theme” using the Large Language Model (LLM)

GPT-4o (OpenAI, 2024). The process - adapted from our previous study (Study 1) - involved prompting the OpenAI API to classify text (i.e., a verbatim tweet) into a list of options (i.e., a list of potential themes + "not belonging to any listed theme"), and looping this procedure through an entire corpus (i.e., the dataset of tweets). LLMs are ideal for text classification with systematic review suggesting excellent performance across many text classification tasks (Fields et al., 2024). Furthermore, during prompt engineering, researchers compared the LLM classifications to a sample of 250 human-labeled classifications to ensure >85% agreement. The implementation was carried out using the OpenAI library in Python (version 3.9.12) (Python Software Foundation, 2022), and is available open source in Appendix 3. There were a total of 126 classifications as related to Alcohol Delivery and Isolation Drinking in the alcohol brand data, and 536 classifications in the user-generated data.

## Measures

### *Potential Exposure to Alcohol Brand Tweets Related to Alcohol Delivery and Isolation Drinking*

Given that CML-COVID does not provide direct information about which alcohol brands users followed, we created an index of potential exposure based on established methods in both informatics and public health research (Henrikson et al., 2010; Sasaki et al., 2015; Luarn et al., 2016; Bhattacharya et al., 2023; Pasch et al., 2023). This measure incorporated three components: (1) the number of friends a user had on Twitter (user_friends_count from the Twitter API), (2) the user's tweet frequency during the relevant period (a proxy for platform engagement), and (3) the total number of alcohol brand tweets about “Alcohol Delivery and Isolation Drinking” during that period (T1=56;

T2=70). Multiplying each of these components together creates an index of potential exposure.

The underlying rationale for this approach draws from network density theory, which demonstrates that users with more connections have higher potential exposure to information within their network (Luarn et al., 2016; Bhattacharya et al., 2023). Similarly, research by Sasaki et al. (2015) established that friend count on Twitter correlates with the volume, variety, and velocity of content appearing in a user's feed. Further, this approach also parallels established methodologies in point-of-sale tobacco marketing surveillance (Henriksen, 2010; Pasch et al., 2023), where researchers multiply the self-reported frequency of store visits by the number of objectively audited tobacco advertisements present in a store to create an index of potential tobacco marketing exposure at retail outlets. Our measure of potential exposure to alcohol brand tweets related to alcohol delivery and isolation drinking is adapted from this previous work. The number of tweets that a user generated is analogous to the number of times a participant visited a store. The number of friends that a user has is analogous to the type and size of the store. The number of alcohol brand tweets that were generated during the study period is analogous to the objective count of tobacco ads that existed at the retail stores participants had access to. In both instances, multiplying these components together creates an index of potential exposure. Indeed, social media platforms generate much of their revenue via advertisements (Raffoul et al., 2023), and these platforms have been suggested as "the storefront of the future" (Wiese, 2021). When users visit "the store of Twitter" they are potentially exposed to a pool of advertisements, and the measure herein is a methodological way to quantify potential exposure to a given segment of products (i.e., alcohol).

Therefore, the equation [number of friends * number of tweets from March 25, 2020-April 30, 2020 * number of alcohol brand tweets about alcohol delivery and isolation drinking from March 25, 2020-April 30, 2020] created a variable that represents the potential exposure to alcohol brand tweets related to alcohol delivery and isolation drinking during the early pandemic lockdown period (i.e., T1).

Additionally, the equation [number of friends * number of tweets from May 1, 2020-May 31, 2020 * number of alcohol brand tweets about alcohol delivery and isolation drinking from May 1, 2020-May 31, 2020] created a variable that represents the potential exposure to alcohol brand tweets related to alcohol delivery and isolation drinking later in the pandemic lockdown period (i.e., T2).

To address skewness in these exposure variables while preserving interpretability and fitting convention in the field of public health, potential exposure variables at both time points were standardized using z-score transformation.

***User-Generated Tweets Related to Alcohol Delivery and Isolation Drinking***

For each user, we calculated the number of tweets classified as related "Alcohol Delivery and Isolation Drinking" during each time period. Due to the highly skewed nature of these variables (most users had zero related tweets), and the fact that very few users produced more than one related tweet, we treated them as binary variables (0 = no related tweets, 1 = one or more related tweets) with the following distributions: T1: 65,984 users with 0 tweets, 319 users with 1 or more tweets; T2: 66,131 users with 0 tweets, 172 users with 1 or more tweets.

## Analysis

To examine the bidirectional associations between potential exposure to alcohol brand tweets and user-generated tweets, we employed a cross-lagged panel model using generalized structural equation modeling (GSEM) in Stata 17 (StataCorp, 2021). This analytical approach allows simultaneous modeling of autoregressive effects (i.e., stability of each variable over time), cross-sectional correlations (i.e., associations between variables at each time point), and cross-lagged effects (i.e., the influence of one variable at Time 1 on the other variable at Time 2, and vice versa) of mixed variable types (Rabe-Hesketh et al., 2022).

Given the mixed variable types (standardized continuous potential exposure variables and binary user-generated indicators), we specified appropriate link functions in our model. The continuous variables used identity links (Gaussian family), while the binary variables used logit links (Bernoulli family) to account for their dichotomous nature (Kuiper, 2018; Wooldridge, 2020; StataCorp, 2023). We employed robust standard errors (vce(robust)) to account for potential heteroscedasticity and non-normality in the data (White, 1980). All analyses were conducted in Stata 17, with a significance threshold of $p < 0.05$ for hypothesis testing.

## RESULTS

The cross-lagged panel model examining the bidirectional associations between potential exposure to alcohol brand tweets and user-generated content about alcohol delivery and isolation drinking during the COVID-19 pandemic converged to a log pseudolikelihood of -147146.67. The model is visualized in Figure 4.

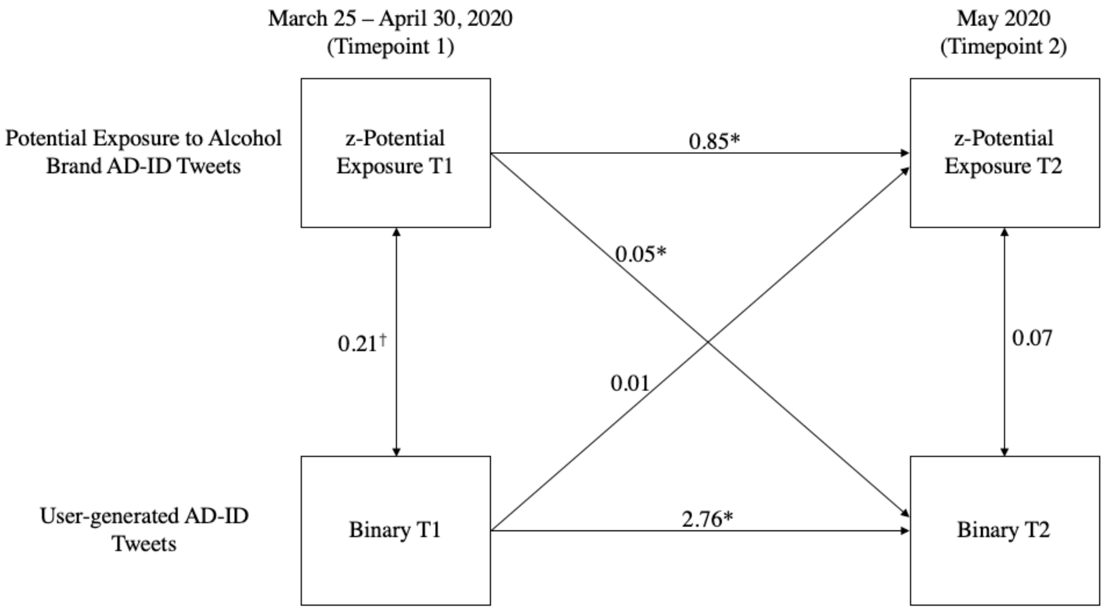


Figure 4. Cross-lagged panel model examining the bidirectional associations between potential exposure to alcohol brand tweets and user-generated content about Alcohol Delivery and Isolation Drinking (AD-ID) during the COVID-19 pandemic. * = $p < 0.05$ † = $p = 0.054$

**Cross-Sectional Associations**

At Time 1 (early lockdown period), a marginally significant positive association emerged between potential exposure to alcohol brand tweets and user-generated tweets about alcohol delivery and isolation drinking ($\beta = 0.21$, $p = 0.054$). This suggests that individuals with higher potential exposure were potentially more likely to generate related content themselves during the initial pandemic lockdown phase, although the p-value is marginal. At Time 2 (later lockdown period), the cross-sectional association between variables was not statistically significant ($\beta = 0.07$, $p = 0.12$).

### Stability over Time

Both variables demonstrated high stability across the study period. Potential exposure exhibited strong temporal consistency ($\beta = 0.85$, $p < 0.01$), indicating that individuals' relative exposure levels remained largely stable from early to later lockdown periods. User-generated content about alcohol delivery and isolation drinking also showed temporal consistency ($\beta = 2.76$, $p < 0.01$), suggesting that individuals who discussed these topics during early lockdown were significantly more likely to continue doing so in the later period (OR=15.8).

### Bidirectional Cross-Lagged Effects

The bidirectional pathways between potential exposure and user-generated content revealed an asymmetrical pattern of associations. Potential exposure in the early lockdown period positively predicted subsequent user-generated content about alcohol delivery and isolation drinking ($\beta = 0.05$, $p < 0.01$). This statistically significant effect suggests that higher initial potential exposure was associated with higher odds of generating related content later (OR=1.05), even after controlling for prior content generation patterns. User-generated content related to alcohol delivery and isolation drinking in the early lockdown period did not significantly predict subsequent potential exposure ($\beta = 0.01$, $p = 0.639$).

## DISCUSSION

This study investigated the bidirectional association between alcohol brand tweets and user-generated content about alcohol delivery and isolation drinking on Twitter during the COVID-19 pandemic lockdowns. Building on previous research which suggested alcohol companies strategically emphasized these topics during the pandemic (Foundation for Alcohol Research & Education, 2020; Colbert et al., 2020; Gerritsen et al., 2021;

Huckle et al., 2021; Martino et al., 2021; Kennedy et al., 2022; Study 1), we examined how industry messaging and user-generated content potentially influenced each other over time.

**The Industry-to-Public Pathway**

Our analysis indicated that potential exposure to alcohol brand tweets promoting alcohol delivery and isolation drinking in the early lockdown period was associated with greater odds of subsequent user-generated content on these topics. In practical terms, each standard deviation increase in potential exposure at T1 corresponded to approximately 5% higher odds of a user generating a tweet about alcohol delivery and isolation drinking at T2. While this effect may appear modest when examining average exposure, it is worth noting that individuals with the highest exposure levels (i.e., > 3 SD above the mean) are at 15%+ greater risk of tweeting about alcohol delivery and isolation drinking compared to those with minimal exposure. These findings align with our hypothesis regarding the prospective association between alcohol brand tweets and user-generated content about alcohol delivery and isolation drinking. The effect persisted after controlling for individuals' baseline tweets about these topics and both variables stability paths, suggesting that industry messaging may contribute to shaping public discourse.

These industry-to-public effects coincided with significant increases in alcohol-related health problems and deaths. White et al. (2022) reported alarming statistics: alcohol-related deaths rose by 25.6% among Americans aged 21-24 and by 37% among those aged 25-34 -demographics that closely align with Twitter's primary user base (Murthy, 2024). Given the established connections between alcohol marketing and consumption (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017; Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022), as well as between social media posts about drinking and actual alcohol use (Young, 2014; Lane et al., 2023), this study's findings raise questions about how industry messaging may have contributed to alcohol-related morbidity and mortality during this time.

The industry-to-public pathway identified here exemplifies what Lacy-Nichols et al. (2023) describe as commercial determinants of health: structural factors driven by corporate interests that shape individual health outcomes beyond personal control. By leveraging crisis conditions to promote potentially harmful consumption patterns, the alcohol industry's marketing strategies reflect broader patterns of corporate exploitation documented during the pandemic (Kim et al., 2020; Rothwell et al,. 2021; McSwane, 2022; Chetty et al., 2024; U.S. House of Representatives, Committee on Oversight and Accountability, Select Subcommittee on the Coronavirus Pandemic, 2024)

**The Public-to-Industry Pathway**

Contrary to our hypothesis, we did not find statistically significant support for user-generated content about alcohol delivery and isolation drinking being associated with subsequent potential exposure to alcohol brand tweets during the brief timeframe examined. Although these results do not support a clear bidirectional pathway, methodological factors such as strong autoregressive stability and limited temporal scope may have restricted our ability to detect subtle or gradual reciprocal influences. Thus, these findings do not conclusively rule out the possibility of bidirectional dynamics occurring over longer periods or under different conditions.

Several complementary explanations may account for this unexpected finding. First, alcohol companies may have maintained consistent marketing strategies regardless of varying user-generated content about alcohol delivery and isolation drinking, focusing on predetermined campaign schedules rather than adaptively responding. Second, the relatively short timeframe of our study may not have captured longer-term industry responses to user-generated content. Third, while we found no significant evidence of user-generated content influencing subsequent industry exposure in terms of volume, this does not rule out other forms of industry responsiveness not captured by our measures. For example, qualitative aspects of industry messaging might have shifted in response to user-generated content without changing the overall volume or distribution of content reflected in our exposure measure. Finally, the alcohol industry may not have been influenced in regard to this particular theme but may have been influenced by themes in other public tweets related to alcohol. Future research is needed which examines these additional possibilities.

**Temporal Associations**

The stability coefficients for both variables indicate that individual-level patterns of both exposure and content generation remained highly consistent throughout the study period. The particularly strong autoregressive effect for user-generated content ($\beta = 2.76$, $p < 0.001$) suggests the importance of early intervention in disruptive moments like pandemic onset, as patterns established during initial crisis phases appear to maintain considerable momentum.

Taken together, the autoregressive, cross-sectional, and cross-lagged analyses indicate that users who initially did not tweet about alcohol delivery and isolation drinking early in the pandemic, yet experienced high potential exposure to alcohol brand tweets, subsequently became significantly more likely to engage in tweeting about these topics. This behavioral shift marks an important consumer conversion - representing a theoretical transition from non-engaged to engaged customer through a novel delivery channel, while possibly adopting the high-risk practice of isolation drinking. In the current study, 160 such conversions occurred (0.24% of the eligible sample), with these converted individuals averaging 3.03 times greater potential exposure at T1 compared to non-converted users. Projecting this conversion rate onto a broader population equates to nearly 2,500 conversions per million individuals exposed. Given that social media advertisements frequently reach audiences numbering in the billions and considering that industry actors continually produce new marketing content, the potential existed for the rapid emergence and expansion of a new commercial sector - the food and beverage delivery application sector - which has proliferated significantly since the onset of the pandemic (Duthie et al., 2023).

**Public Health Implications**

Our findings have several notable implications for public health. First, the prospective association between potential exposure to alcohol brand tweets and subsequent user-generated content regarding alcohol delivery and isolation drinking supports calls for enhanced regulatory oversight of digital alcohol marketing (World Health Organization, 2021), especially during societal crises. The rapid deployment of marketing emphasizing alcohol delivery and isolation drinking coincided with documented increases in problematic consumption patterns and alcohol-related mortality (White et al., 2022), highlighting the potential public health consequences of opportunistic crisis marketing. Second, while we did not find evidence of a prospective association between user-generated content and subsequent potential exposure to alcohol brand tweets, the strong stability in both variables over time underscores the critical importance of early intervention. Public health messaging should deploy rapidly at crisis onset to establish protective behavioral norms before maladaptive patterns stabilize, as our findings suggest considerable momentum in both potential exposure and user-generated content once established. Finally, our results highlight the need for integrated consideration of both commercial determinants of health and individual behaviors in public health approaches to alcohol harm reduction. While individual-level interventions remain important, our findings demonstrate how corporate marketing strategies may systematically shape the informational environment within which individual choices occur, particularly during periods of heightened vulnerability.

While our analysis specifically examined the COVID-19 pandemic, it is essential to recognize that these findings reflect a broader pattern of marketing practices that may emerge during any public health crisis or major societal disruption. The pandemic provided a distinctive lens through which to observe industry strategies as companies swiftly adapted their messaging to exploit periods of heightened societal vulnerability, shifting consumption patterns, and evolving regulatory environments. Gaining insights into these dynamics is critical, not only for accurately interpreting the immediate impacts of alcohol marketing during COVID-19 but also for guiding the development of proactive regulatory frameworks and targeted public health interventions. Ultimately, the lessons learned from examining alcohol industry behaviors during this crisis offer a valuable foundation for strengthening public health preparedness and resilience against harmful marketing practices across diverse industries in future emergencies.

**Limitations and Future Directions**

These findings should be considered in light of several limitations. First, our methodological approach to measuring potential exposure, while conceptually grounded and theoretically informed, represents an approximation based on network characteristics and general engagement rather than direct observation of exposure. Though standardizing these variables improved their distributional properties, this indirect measurement introduces uncertainty that direct tracking would mitigate.

Second, our binary classification of user-generated content, while appropriate given the highly skewed distribution of this variable, may have reduced our ability to detect nuanced patterns among the most active content generators.

Third, our analysis focused specifically on Twitter users in the United States who engaged with COVID-19-related content, limiting generalizability to other populations, contexts, or platforms.

Fourth, our sample of alcohol brand tweets was from 11 alcohol companies with substantial social media followings (>10,000 followers) which were deemed popular by our research group of graduate and undergraduate research assistants, potentially overlooking marketing strategies employed by smaller or regional companies with fewer followers and less broad popularity.

Fifth, there was a relatively low number of tweets related to alcohol delivery and isolation drinking in the user-generated dataset; however, this is somewhat expected, as the sample specifically consisted of tweets related to COVID-19. A more general sample of tweets might yield a higher frequency of content related to these topics.

Sixth, the strong autoregressive (stability) pathways observed in our cross-lagged panel model pose analytical challenges for detecting cross-lagged (bidirectional) effects.

Specifically, high stability indicates that individual differences at baseline persist strongly over time, leaving less variance available for cross-lagged relationships to explain. This reduced variance can mask or attenuate potential bidirectional effects, thus complicating the interpretation of causality or mutual influence between variables. Consequently, caution is warranted when interpreting null findings for cross-lagged paths in contexts with high temporal stability.

Lastly, the relatively short timeframe examined in this study (approximately 2.5 months during the early COVID-19 pandemic lockdown) likely limited our ability to detect more gradual or longer-term effects, particularly those associated with industry adaptation (i.e., the public-to-industry pathway). Structural changes within industries, shifts in marketing strategies, and subsequent alterations in consumer behavior could require extended periods to emerge clearly. As a result, longer-term longitudinal research is necessary to capture these slower-developing dynamics fully and to provide a more comprehensive understanding of the longer-term bidirectional relationship between the public and the alcohol industry.

Future research should address these limitations while expanding investigation into several promising directions. Moderator analyses could identify population subgroups particularly susceptible to industry influence or contextual factors that strengthen or weaken these associations. Longer-term longitudinal studies could track how industry-user-generated content associations evolve beyond initial crisis phases; particularly as temporary regulatory changes become permanent. Additionally, comparative analyses across multiple crisis events could identify consistent patterns in alcohol industry crisis marketing tactics, informing more targeted regulatory approaches. Lastly, comparative analyses across other product types (e.g., tobacco, cannabis, highly processed foods, etc.)

could inform broader interpretations of the industry-to-public pathway identified in this study.

## Conclusion

This study provides empirical evidence of a unidirectional association between alcohol industry marketing and subsequent user-generated content about alcohol delivery and isolation drinking during the early COVID-19 pandemic. These results contribute to our understanding of how alcohol marketing on social media may influence user-generated content during societal crises, with notable implications for public health. The documented influence of strategically adapted marketing on user-generated content, coupled with the known links between marketing exposure and consumption behaviors, highlights the need for enhanced oversight of digital alcohol marketing during vulnerable periods.

As society faces ongoing and future crises, understanding these associations between industry marketing tactics and public discourse becomes increasingly critical for protecting public health. Our findings suggest that early, sustained public health messaging, coupled with appropriate regulatory guardrails around crisis marketing, may help mitigate the exploitation of vulnerability during periods of societal disruption.

# Chapter 6: Synthesis, Implications, and Future Regulations

## SYNTHESIS OF KEY FINDINGS

### Historical Perspectives Reaffirmed

The historical perspectives presented in this dissertation identified *Three Key Points* that have characterized the alcohol industry's success throughout history: (1) the alcohol industry's success is linked to advances in communication technology, (2) the alcohol industry has a long history of distributing its products to, and exploiting, vulnerable people, and (3) the media portrayals of the alcohol industry are lasting. The *Three Key Points* serve as a lens for viewing the evolution of the alcohol industry's role in society and its commercial success over centuries.

The first key point synthesizes how the alcohol industry's progress has consistently paralleled developments in communication technology. Beginning with hand-written documents like De Agri Cultura, which were some of the first resources discussing large-scale alcohol production (Cato et al., 1934), and culminating in modern digital platforms, the alcohol industry has demonstrated consistent adaptability. This pattern of technological adoption has been evident up to modern day in the era of communication technology (Miller, 2002; Hastings, 2009; Cracknell, 2012; Dutton, 2013; Moreno et al., 2014; Jernigan et al., 2017).

The second key point traces the alcohol industry's consistent pattern of targeting vulnerable populations. This exploitation has taken various forms throughout history, including providing wine rations to Roman legionaries as a form of motivation and control (Davies, 1971), marketing alcohol to war veterans struggling with psychological trauma (Pope, 1980; Shephard, 2001; Schumm, 2012; Armengol, 2014; Sturgeon, 2015;

Kamieński, 2019; Lefebvre et al., 2020; National World War II Museum, 2021), and marketing to individuals experiencing economic upheaval (Lefebvre, 2020). This pattern of exploitation has persisted through periods of societal vulnerability, whether during wartime, geopolitical shift, economic depression, or other societal disruptions.

The third key point emphasizes the lasting influence of alcohol industry media portrayals, particularly in shaping societal attitudes and behaviors. Associating drinking with soldiers and patriotism in narratives dating back to the Trojan Wars (Pope, 1980; Shephard, 2001; Schumm, 2012; Sturgeon, 2015; Kamieński, 2019; National World War II Museum, 2021), reinforcing gender norms in alcohol consumption (Parkin, 2007; Armengol, 2014; Golia, 2016; Carrotte, 2016; Auter, 2016; Lefebvre et al., 2020), and associating drinking with sophistication and success (Atkin et al., 1983, Bergkvist et al., 2016, Kellershohn, 2022) are several of many portrayals that have endured, often overlooking the potential negative consequences of these associations.

The empirical studies presented in this dissertation provide preliminary evidence that these historical perspectives were distinctly reaffirmed during the COVID-19 pandemic. Study 1 suggested that alcohol companies rapidly adapted their social media marketing strategies in response to pandemic conditions, significantly increasing messages related to alcohol delivery and isolation drinking, restaurant support, and social media promotions. If true, this strategic shift exemplifies Key Point 1, showing how the industry leveraged social media technology to maintain commercial activities despite pandemic-related restrictions. Study 1 also suggested that certain adaptations - specifically alcohol delivery and isolation drinking, and social media promotions - continued to be amplified throughout the post-pandemic period, exemplifying Key Point 3 and paralleling new legal changes that relaxed alcohol access restrictions in certain areas (Texas Legislature, 2021).

The timing of these marketing adaptations - coinciding with a period of heightened psychological distress and anxiety (Pfefferbaum et al., 2020; Panchal et al, 2021) - reaffirms Key Point 2, circumstantially suggesting that the alcohol industry may have targeted vulnerabilities brought about by the pandemic.

Similarly, Study 2 provided empirical evidence supporting all *Three Key Points*: it suggested that industry-driven messaging on social media (Key Point 1) may have targeted crisis-related vulnerabilities (Key Point 2), with industry content about alcohol delivery and isolation drinking significantly predicting subsequent user-generated content, suggesting the industry's ability to influence user-generated content (Key Point 3). While the duration of this influence remains undetermined, the transmitted messaging shows preliminary evidence of industry impact on the consumer.

The integration of historical perspectives with the empirical results from Studies 1 and 2 offers a more comprehensive understanding of alcohol marketing during the COVID-19 pandemic. The pandemic represents merely the most recent iteration of a recurring pattern wherein the alcohol industry adapts to societal disruptions, leverages advances in communication technology, targets vulnerable populations, and creates lasting impressions. This historical continuity drills in the importance of recognizing these patterns not as isolated phenomena but as part of a longstanding commercial strategy that has significant public health implications.

**Study 1 Key Findings**

Study 1 identified five themes used by alcohol companies on Twitter during the COVID-19 pandemic: Alcohol Delivery and Isolation Drinking, Restaurant Support, Social Media Promotions, Sports, and Gaming. The prevalence and/or sentiment of some of these themes varied significantly across pre-pandemic (January 2019 - January 2020), peri-pandemic (March 2020 - May 2020), and post-pandemic response periods (June 2020 - July 2021).

The findings suggest strategic shifts in marketing during the lockdown period. Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions all showed statistically significant increases during the peri-pandemic period compared to the pre-pandemic period (increases of 91.2%, 189.5%, and 150%, respectively). This pattern could suggest a deliberate reallocation of marketing resources in response to pandemic conditions, though future research is needed to further investigate this notion. In contrast, Sports and Gaming themes remained relatively stable across all periods, suggesting that they could have been part of ongoing marketing strategies rather than pandemic-specific adaptations.

Although all three pandemic-responsive themes decreased significantly from the peri-pandemic to post-pandemic period, both Alcohol Delivery/Isolation Drinking and Social Media Promotions remained significantly higher in the post-pandemic period compared to the pre-pandemic period (30.8% and 60.6% increases, respectively). This finding suggests that some marketing adaptations initially triggered by the pandemic became potentially enduring components of alcohol companies' strategies, which could suggest lasting changes in consumer behavior and industry practices.

Sentiment analysis using LIWC revealed varied communication mechanisms across the five themes. Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions employed stronger emotional appeals, with relatively high positive emotional tone (ranging from 80.16%-98.49%) and minimal negative affect. This finding is particularly important given the timing of these emotional appeals during a period of heightened psychological vulnerability, raising concerns about the potential exploitation of pandemic-related distress to promote alcohol consumption.

The Restaurant Support theme may have emerged during this time as a form of corporate exploitation of pandemic-induced sympathy for those who were unable to work or run certain businesses during this time. This theme can be considered a form of strategic corporate social responsibility (CSR) messaging. This theme exhibited the highest clout score (94.38%) and highest positive emotional tone (98.49%) across all themes, which could suggest carefully constructed content designed to build social authority and positive associations, though future research is needed to further investigate this notion. While supporting struggling restaurants during the pandemic was undoubtedly important, the alcohol industry's emphasis on restaurant support messaging represents a specific use of CSR strategies that have long been documented within the alcohol sector (Martino et al., 2021; Babor et al., 2013; Mialon et al., 2018). These strategies effectively created a dual narrative that both legitimized the industry's pandemic response through apparent social responsibility and expanded direct-to-consumer channels, a practice that Yoon et al. (2013) characterize as CSR's function in deflecting regulatory scrutiny while advancing industry market objectives.

The findings from Study 1 align with and extend the *Three Key Points* identified in the historical perspectives. The identification of pandemic-responsive themes and their strategic deployment pattern aligns with Key Point 1. The emotional appeals embedded in these themes, deployed during a period of heightened psychological vulnerability, align with Key Point 2. The lasting changes in marketing strategies observed in the post-pandemic period support Key Point 3.

**Study 2 Key Findings**

Study 2 provided empirical evidence of a unidirectional association of potential exposure to alcohol brand tweets and subsequent user-generated content related to alcohol delivery and isolation drinking during the COVID-19 pandemic lockdowns. The cross-lagged panel model revealed that potential exposure to alcohol brand tweets about "Alcohol Delivery and Isolation Drinking" in the early lockdown period (March 25-April 30, 2020) significantly predicted subsequent user-generated content about these topics in the later lockdown period (May 2020), with an odds ratio of 1.05. In practical terms, each standard deviation increase in potential exposure at Time 1 corresponded to approximately 5% higher odds of a user generating a tweet about alcohol delivery and isolation drinking at Time 2. Importantly, this association remained significant even after controlling for prior content generation patterns and both variables' stability paths.

Contrary to the study's hypothesis, user-generated content about alcohol delivery and isolation drinking in the early lockdown period did not significantly predict subsequent potential exposure to alcohol brand tweets about this theme. This asymmetrical pattern suggests that while industry messaging could have influenced user-generated content, the reverse pathway was not supported by the data. This finding also suggests that alcohol companies maintained consistent marketing strategies regardless of user-generated content, potentially focusing on predetermined campaign schedules rather than adaptively responding to user-generated content, at least in the time period under study. It is important to reiterate out that the stability pathways in the model were exceptionally strong, which makes detecting directional effects difficult, and thus the null finding from user-generated content to alcohol brand tweets should be interpreted with caution.

These findings align with what Lacy-Nichols et al. (2023) describe as commercial determinants of health: structural factors driven by corporate interests that shape individual health outcomes beyond personal control. By leveraging crisis conditions to promote potentially harmful consumption patterns, the alcohol industry's marketing strategies reflect broader patterns of corporate exploitation documented during the pandemic (Kim et al., 2020; Rothwell et al., 2021; McSwane, 2022; Chetty et al., 2024; U.S. House of Representatives, Committee on Oversight and Accountability, Select Subcommittee on the Coronavirus Pandemic, 2024).

The strong stability coefficients for both potential exposure and user-generated content indicate that despite changing associations, individual-level patterns remained highly consistent throughout the study period. The particularly strong autoregressive effect for user-generated content suggests that patterns established during initial crisis phases maintain considerable momentum, which could suggest that early intervention during crisis is critical, though future research is needed to further investigate this notion.

The findings from Study 2 align with and extend the *Three Key Points* identified in the historical perspectives. The unidirectional influence from alcohol brand tweets to user-generated content aligns with Key Point 1. The industry's strategic deployment of marketing messages during a period of heightened vulnerability aligns with Key Point 2. The rapid normalization and stability of new patterns of discourse about alcohol delivery and isolation drinking support Key Point 3.

## Implications for Theory and Knowledge

### Advancement of Theory

This dissertation provides support for the Differential Susceptibility to Media Effects Model (DSMM) (Valkenburg et al., 2013) and extends its application to understanding alcohol marketing during crisis periods. The DSMM suggests that media effects are conveyed through cognitive, emotional, and excitative response states elicited within viewers in reaction to consuming media. The findings from Study 1, which identified varying levels of cognitive, emotional, and excitative elements across different marketing themes, directly align with this theoretical framework.

The longitudinal design of Study 2 further strengthens support for the DSMM by demonstrating one example of potential exposure to industry content predicting subsequent user-generated content, suggesting that media exposure influenced attitudes and behaviors in ways consistent with the theory's predictions. The unidirectional nature of this association highlights the power of strategic media messaging, particularly during periods of heightened vulnerability.

While this dissertation supports key aspects of DSMM, it did not test how individual differences interact with media response states to further differentiate or personalize media effects. Individual differences are a crucial element to the DSMM framework, and thus identifying individual differences is a critical piece of future research in this area.

This dissertation also extends the commercial determinants of health framework by empirically documenting one example of corporate marketing strategies potentially influencing health-related behaviors through digital media channels. As defined by Lacy-Nichols et al. (2023), commercial determinants of health are structural factors driven by corporate interests that influence individual health outcomes beyond personal control. Taken together, the findings from both studies provide preliminary evidence of these determinants in action.

Additionally, this dissertation integrates historical and modern perspectives on alcohol marketing, suggesting continuity in industry strategies across centuries. The *Three Key Points* identified in the historical perspectives provide a lens for viewing these continuities. By empirically reaffirming these historical perspectives in the modern context of social media during the COVID-19 pandemic, this dissertation bridges historical perspectives with contemporary media effects theory and public health research methodology, offering a novel interpretation of alcohol marketing dynamics across time.

The theoretical implications of this dissertation extend beyond alcohol marketing to inform a broader understanding of corporate marketing practices during crisis periods. The identification of specific themes and mechanisms employed by the alcohol industry during the pandemic provides a template for analyzing similar adaptations across other industries and other societal challenges.

Furthermore, this dissertation raises important questions about whether the *Three Key Points* identified are unique to the alcohol industry or indicative of broader systemic tendencies within capitalist economies. The strategic adaptation to emergent communication technologies, exploitation of vulnerable populations, and creation of persistent media narratives align closely with critical economic theories that examine the nature of capitalism. For example, Luxemburg's (1913) analysis of capitalist accumulation suggests that the alcohol industry's aggressive marketing strategies during COVID-19 could reflect capitalism's relentless expansion into new social domains to sustain growth, often at the expense of marginalized groups. Similarly, Harvey's (2010) concept of "accumulation by dispossession" frames how capitalist enterprises systematically exploit societal vulnerabilities, crises, and inequalities to extract value. Taken together, these economic theories suggest that the alcohol industry's practices during the COVID-19 pandemic may represent a particularly illustrative case of broader capitalist dynamics, characterized by strategic opportunism, exploitation of vulnerability, and sophisticated ideological management. This theoretical extension suggests that similar patterns might be observable across other industries with sufficient longevity and market power. Industries such as tobacco, gambling, highly processed foods, and pharmaceuticals all have comparable abilities to leverage new communication technologies, target vulnerable populations, and create enduring cultural associations that normalize consumption. Society must grapple with the profound question of whether these *Three Key Points* represent industry-specific strategies or manifestations of broader structural forces within capitalist economies, particularly during periods of crisis when market constraints are temporarily suspended or transformed.

**Methodological Contributions**

The methodological approach used in Study 1 which combined BERTopic modeling, iterative Artificial Intelligence (A.I.) interpretation, and sentiment analysis allowed this dissertation to systematically identify marketing themes and quantify their emotional, cognitive, and excitative qualities. By capturing both the themes and sentiment used during COVID-19 crisis messaging, this hybrid approach contributes an innovative tool for ongoing public health surveillance. Future research could apply similar techniques on different populations (e.g., other consumer goods industries) in other contexts (e.g., elections, natural disasters) and on other platforms (e.g., Instagram, TikTok). Furthermore, the extension of these methods in Study 2, using A.I. classification combined with cross-lagged models, is also novel. This approach enabled the empirical documentation of the unidirectional association of alcohol brand tweets and subsequent user-generated content, providing a methodological approach for investigating similar patterns of influence in other contexts.

## FUTURE REGULATION

### Lessons from Past Regulatory Success

Evidence from global alcohol marketing regulation demonstrates significant public health impacts. Norway's comprehensive advertising ban, implemented in 1975, has consistently reduced alcohol consumption by an average of 7.4% annually (Rossow, 2021). Similarly, France's Loi Évin legislation, which restricted alcohol marketing in youth-targeted media and required health warnings, initially achieved an 11.36% reduction in population-level consumption over eight years (Cogordan et al, 2014).

In the U.S. we can draw valuable lessons from tobacco control, where sequential regulatory steps significantly contributed to public health gains making substantial progress on ending the tobacco epidemic. These included mandatory health warnings (Federal Cigarette Labeling and Advertising Act of 1965; FTC, 2023), broadcast advertising bans (Public Health Cigarette Smoking Act of 1969; U.S. Congress, 1970B), and prohibitions on youth-targeted marketing and sports sponsorships (Master Settlement Agreement of 1998; National Association of Attorneys General, 1998).

These global and domestic regulatory successes established both theoretical and legal precedent for implementing further restrictions on alcohol marketing, suggesting a viable pathway despite constitutional commercial speech protections (Saffer et al., 2000; World Health Organization, 2023).

### Social Media Platform Regulation

The findings from this dissertation have implications for social media platform regulation regarding alcohol marketing. The association of alcohol brand tweets and subsequent user-generated content suggests the need for more robust regulatory frameworks governing digital alcohol advertising. Based on the empirical evidence presented, several regulatory approaches should be considered.

First, content moderation proposals should be implemented to address potentially harmful alcohol marketing strategies, especially during crisis periods. Platforms should develop automated systems to flag and review alcohol-related content that exploits crisis-related vulnerabilities. These systems could incorporate frontier methodologies similar to those used in this dissertation to identify patterns in marketing content that specifically target psychological vulnerabilities and promote unhealthy behaviors. While content moderation is a controversial topic, particularly when weighed against principles of free speech, it can be approached in ways that protect both public health and constitutional rights. Legal scholars have suggested limits of commercial speech protections under the First Amendment, noting that advertising has historically been subject to greater regulation when misleading or targeting vulnerable populations (Redish, 1970; Post, 2000;). Furthermore, the U.S. Supreme Court has upheld restrictions on commercial speech that is deceptive or promotes harmful conduct (McKeon, 1981). In this context, targeted moderation of alcohol marketing can be justified under the frameworks of harm reduction and consumer protection. For example, the Protecting Americans from Dangerous

Algorithms Act (U.S. Congress, 2021) proposed mechanisms for holding platforms accountable when their algorithms amplify harmful content. Similarly, the EU's Digital Services Act (European Parliament and Council of the European Union, 2022) establishes a precedent for requiring transparency and due process in content moderation while allowing for regulation of content that poses public health risks. Ultimately, well-designed content moderation systems can serve as a vital tool for mitigating the negative societal impacts of alcohol marketing, especially during times of heightened vulnerability.

Second, enhanced age verification requirements are a critical regulatory need. While this dissertation focused on young adults and did not analyze the ages of individual Twitter users, the findings regarding emotionally-targeted marketing during periods of vulnerability have important implications for age-based protections. Given that alcohol-related mortality increased by 29.8% among Americans aged 16-20 during the pandemic (White et al., 2022), stricter measures are needed to prevent alcohol marketing exposure among underage users. Moreover, individuals aged 18-20 represent a legally underage population for alcohol consumption, yet they are active social media users who may be exposed to the types of marketing strategies identified in this research. Platforms should be required to implement robust age verification protocols consistent with alcohol consumption age restrictions, ideally at the device level, before allowing users to view alcohol-related content or follow alcohol brand accounts. These measures could include multi-factor age verification processes that go beyond simple self-reported birth dates, such as registering the device as belonging to a minor when purchasing from the manufacturer or communication service provider. Though age verification raises privacy and civil liberties concerns, there is growing legal and regulatory precedent for such protections, particularly when aimed at shielding minors from harmful content. The Children's Online

Privacy Protection Act (Federal Trade Commission, n.d.) already requires online services to obtain verifiable parental consent before collecting data from users under 13, offering a model for age-gating mechanisms. More recently, proposals such as the Kids Online Safety Act (Blumenthal et al., 2023) and state-level legislation in Texas (Texas Legislature, 2023) have called for platforms to establish clear age-verification protocols to limit youth exposure to harmful digital content, including substances, violence, and sexually explicit material. These efforts align the principle that children and adolescents are a uniquely vulnerable population deserving of heightened protections, a stance supported by international human rights frameworks such as the UN Convention on the Rights of the Child (United Nations, 1989). Implementing age verification in alcohol-related content aligns with these ethical and legal rationale. Furthermore, such systems can be developed in ways that are privacy-preserving and minimally invasive, finding a balance between user autonomy, public health, and technological feasibility. The goal is not to surveil, but to create friction in access that protects those most at risk without unduly infringing on lawful adult expression or commercial activity.

Third, algorithm and targeting transparency mandates would provide needed oversight of how alcohol marketing is disseminated across social media platforms. The findings from Study 2 regarding the association of potential exposure and subsequent user-generated content raises concerns about understanding how algorithmic systems transmit and amplify alcohol industry messages. Platforms should be required to disclose targeting parameters used by advertisers and provide transparency regarding how content is prioritized in user feeds, particularly during crisis periods (Matz, 2025).

## Alcohol Industry Regulation / Corporate Accountability

Effective regulation of alcohol marketing requires robust corporate accountability mechanisms, particularly regarding digital marketing practices. Based on the empirical findings presented in this dissertation, several accountability approaches should be considered.

First, mandatory reporting and transparency requirements should be established for alcohol companies regarding their digital marketing practices. These requirements should include detailed disclosure of marketing expenditures, targeting strategies, message themes, and engagement metrics across social media platforms. For example, the World Health Organization (2021) has recommended that alcohol producers be required to report comprehensive marketing data to better inform public health policy, including media channels, target demographics, and spending. The findings from Study 1 regarding strategic shifts in marketing themes during the pandemic affirm the importance of transparency in understanding how alcohol companies adapt their strategies during crisis periods.

Second, standards for authentic versus exploitative corporate social responsibility (CSR) should be developed. Study 1 identified Restaurant Support as a prominent theme during the pandemic lockdown period, representing a specific manifestation of CSR strategies documented within the alcohol sector (Babor et al., 2013; Yoon et al., 2013; Mialon et al., 2018; Martino et al., 2021). Regulatory frameworks should establish clear guidelines distinguishing between genuine CSR initiatives and those designed primarily to advance commercial interests while deflecting regulatory scrutiny.

Third, legal and financial liability frameworks should be developed to hold alcohol companies accountable for marketing practices that demonstrably increase harmful consumption behaviors. Given the established links between alcohol marketing and drinking behaviors (Anderson et al., 2009; Smith et al., 2009; Jernigan et al., 2017; Curtis et al., 2018; Hendriks et al., 2021; Alhabash et al., 2022), companies should face meaningful consequences for exploitative marketing practices that contribute to increased alcohol-related harm. These frameworks could include financial penalties proportional to revenue gains and requirements for companies to fund independent research on the health impacts of their marketing strategies. A legal precedent exists in the Master Settlement Agreement (National Association of Attorneys General, 1998), which held tobacco companies financially responsible for the public health costs of smoking and restricted their marketing practices.

Finally, alcohol companies should be required to implement crisis-specific marketing codes of conduct that are activated during emergencies. These codes should include commitments to avoid exploiting crisis-related vulnerabilities, to refrain from promoting alcohol as a coping mechanism, and to prioritize public health considerations in marketing strategies. Compliance with these codes should be independently monitored and publicly reported.

## LIMITATIONS, FUTURE RESEARCH AND CONSIDERATIONS

### Methodological Limitations

The research presented in this dissertation has several methodological limitations that should be considered when interpreting the findings. First, sample representativeness constraints must be acknowledged. Both studies focused exclusively on Twitter (now X) data, representing only a segment of the social media landscape. Twitter's user demographics typically skew younger, more urban, and more technologically engaged than the general population (Murthy, 2024), potentially limiting the generalizability of findings to broader populations. Additionally, Twitter has undergone notable changes since rebranding as X, including alterations in public data accessibility, censorship policies, misinformation management, and user demographics (Bisbee et al., 2024; Haman et al., 2025). Nonetheless, Twitter was highly influential during the data collection period for this dissertation (Haman et al., 2020; Rufai et al., 2020) and remains widely utilized in sociological research due to its historical data access and robust API capabilities (Murthy, 2018; Murthy, 2024).

In Study 1, the dataset was limited to tweets from 11 alcohol companies with substantial followings (>10,000 followers), potentially overlooking marketing strategies of smaller, less popular, or regional companies. Thus, findings might not generalize across the entire alcohol industry. However, these selected companies collectively had nearly 1.2 million followers, were identified as popular among young adults by undergraduate research assistants, and represent major global brands. In Study 2, analysis was limited to Twitter users in the United States engaging with COVID-19-related hashtags, restricting the generalizability to other contexts or populations. While COVID-19 content is relevant and publicly available in archived datasets, the relatively short time period examined (approximately 2.5 months during early pandemic lockdowns) may have limited detection of gradual or long-term effects, particularly industry adaptations responding to public trends. Structural industry changes, marketing strategy shifts, and resulting consumer behavior alterations may require extended observation periods to emerge clearly. Consequently, longer-term longitudinal research is necessary to fully capture these slower-developing dynamics and comprehensively understand bidirectional relationships between the public and the alcohol industry.

Beyond the methodological limitations already noted, structural limitations of the data also warrant consideration. The constrained use of specific hashtags, such as those explicitly linking to COVID-19, likely shaped the user-generated dataset in non-representative ways. This selection process may have excluded relevant conversations that did not employ these hashtags, thereby underrepresenting more casual or indirect discussions of alcohol consumption during the pandemic. Conversely, the reliance on such hashtags may have overrepresented highly engaged or promotional content, potentially skewing the sample toward industry-driven narratives. These limitations are important to acknowledge because they influence the scope of inference and raise questions about what forms of discourse were captured versus omitted. Future research should consider more diverse sampling strategies that do not rely exclusively on hashtag-based data collection to provide a more complete picture of public discourse.

Further limitations arose from strong autoregressive (stability) pathways observed in the cross-lagged panel model used in Study 2. High temporal stability indicates persistent baseline individual differences, leaving minimal variance available for cross-lagged relationships to explain. This reduced variance can obscure or attenuate potential bidirectional effects, complicating the interpretation of causality or mutual influences. Therefore, caution is warranted when interpreting null findings for cross-lagged paths in contexts exhibiting strong temporal stability.

Measurement limitations also warrant consideration. The approach to measuring potential exposure in Study 2, although conceptually grounded and theoretically informed, approximated exposure based on network characteristics and general engagement rather than direct observation. Such indirect measurement introduces uncertainty that direct tracking could mitigate. Nonetheless, both public health and computational communication research frequently rely on similar approximations.

Limitations in establishing direct causality must be recognized. Although Study 2 employed a longitudinal cross-lagged design with temporal ordering, true causal inference requires more rigorous experimental designs or extended longitudinal studies with additional control variables. Observed associations between potential exposure to alcohol brand tweets and subsequent user-generated content, while suggestive, could be influenced by unmeasured confounding or mediating variables.

Additionally, challenges in measuring actual consumption outcomes limit the direct public health implications if this research. Despite evidence suggesting social media activity effectively predicts behaviors (Young, 2014; Lane et al., 2023), this dissertation did not directly measure alcohol consumption or related harms. The relationship between social media discourse on alcohol and actual consumption patterns, especially during crisis periods, necessitates further exploration.

Lastly, alternative explanations must be considered. Several confounding factors during the pandemic period, including economic uncertainty, social isolation, and psychological distress, could independently influence observed relationships between potential exposure to alcohol marketing and user-generated content. While this dissertation identified correlations consistent with established literature on marketing's influence on drinking behaviors, the complex, multi-causal nature of increased alcohol-related mortality during the pandemic cannot be overlooked.

**Ethical Considerations**

This dissertation raises several ethical considerations. Privacy concerns in social media research are widely discussed, particularly when analyzing large datasets of public tweets. While this dissertation followed ethical guidelines for social media research, and was deemed as not human-subject research by an IRB, the public nature of Twitter content raises questions about user expectations regarding research use of their posts. Users may not anticipate that their social media activity could be analyzed to understand marketing influences or health-related behaviors.

Balancing commercial speech and public health interests represents another ethical challenge. The alcohol industry, like other commercial entities, has legal rights to marketing communication. Recommendations for restricting alcohol marketing must carefully balance these commercial speech rights against public health imperatives, particularly during crisis periods. This balancing requires nuanced ethical frameworks that recognize both the potential harms of unrestricted marketing and the importance of preserving appropriately bounded commercial expression.

Ethical considerations also extend to the use of A.I. tools in content analysis. While this dissertation employed advanced natural language processing techniques and large language models (LLMs) to analyze tweet content, these tools have inherent limitations and potential biases that could influence the identification and interpretation of themes. Transparency regarding these methodological approaches and their limitations is essential for ethical research practice.

Finally, the potential for unintended consequences from regulatory interventions should also be considered. Recommendations for restricting alcohol marketing on social media platforms could have various unintended effects, including driving marketing to less regulated channels, creating incentives for covert marketing techniques, or establishing precedents that might inappropriately restrict other forms of digital communication. Careful consideration of these potential consequences is necessary when developing regulatory approaches.

**Future Research Directions**

This dissertation showed associations between potential exposure to alcohol brand tweets and subsequent user-generated content about alcohol behaviors, and argued that - based on Key Point 3 - these drinking behaviors may persist. This is a crucial area for future research, as it suggests normative behavior changes post-crisis. While Study 1 did demonstrate that crisis-related messaging continued long after the initial crisis period, Study 2 was limited to approximately 2.5 months of time. Future research should examine whether crisis-related behavior is persistent, and if so, what implications these changes hold for public health.

Given the findings of this dissertation regarding alcohol, future research should also examine other harmful consumer product marketing strategies on social media and especially during crisis. Tobacco, cannabis, highly processed foods, and gambling products all face varying regulatory frameworks and could exhibit similar opportunistic marketing adaptations during crisis periods. Cross-product comparative research could identify common patterns of vulnerability exploitation and inform more comprehensive regulatory approaches.

Testing the effectiveness of proposed regulatory measures represents a critical next step in translating the research in this dissertation into practice. Experimental studies could evaluate various regulatory approaches to determine which most effectively mitigate the influence of alcohol marketing on consumer attitudes and behaviors, particularly during crisis periods. These studies could employ randomized designs that manipulate exposure to different regulatory environments, providing gold-standard evidence regarding intervention effectiveness.

## Dissertation Contributions

This dissertation makes contributions to public health and alcohol-related research by uniquely integrating historical perspectives with contemporary digital context, documenting ways in which alcohol industry strategies have evolved while maintaining core tactics across centuries. The historical perspectives identified *Three Key Points* that serve as a lens for viewing both historical patterns and modern marketing practices on social media platforms.

Through two complementary studies, this dissertation provided empirical evidence of industry adaptation during the COVID-19 pandemic. Study 1 identified specific marketing themes that increased during lockdown, including Alcohol Delivery and Isolation Drinking, Restaurant Support, and Social Media Promotions. Study 2 demonstrated the longitudinal association of alcohol brand tweets and subsequent user-generated content, revealing a potential pathway of a commercial determinant of health during a period of heightened vulnerability. The research extends existing literature by validating previously identified marketing themes while discovering new ones such as Social Media Promotions, Sports, and Gaming. Additionally, through exploratory sentiment analysis, the work establishes a foundation for future research that utilizes DSMM theory. Finally, its methodological contributions advance computational communication-public health hybrid approaches.

By integrating historical perspectives, modern evidence, and future regulation, the dissertation offers a more comprehensive understanding of alcohol marketing during crisis periods. This integrated approach reaffirms the industry's continued adaptation to new communication technologies, targeting of vulnerable populations, and creation of lasting media influence. This work ultimately informs evidence-based regulatory frameworks that anticipate industry adaptations during future crises.

Finally, it is important to recognize that the COVID-19 pandemic was a rare, high-impact event that generated exceptional societal conditions. While this uniqueness may limit the generalizability of the findings to more routine periods, it simultaneously offers an unparalleled lens into how the alcohol industry responds to crisis. The rapid shifts in marketing strategy, the emotional tone of brand messaging, and the alignment with public vulnerabilities reveal industry behaviors that are typically less visible outside of such disruptions. These insights, though context-specific, serve as a stress test for industry conduct and regulatory preparedness, providing valuable lessons for anticipating similar dynamics during future emergencies.

**Closing Thoughts**

In a letter of correspondence concerning biographical papers on Pablo Casals - the renowned Catalan cellist, conductor, and composer known for his political resistance against Francoist Spain - Albert Einstein (1953) stated, "Die Welt mehr bedroht ist durch die, welche das Uebel dulden oder ihm Vorschub leisten, als durch die Uebeltäter selbst." Translated from German to English:

**"The world is more threatened by those who tolerate or encourage evil than by the evildoers themselves."**

This dissertation has reviewed evidence that the alcohol industry profits from selling a product that causes death and disability. Through its marketing practices, the alcohol industry enables and encourages people - commonly vulnerable people - to consume a substance that is known to be deadly in large amounts and poisonous even in moderate amounts. If one were to measure evil by enabling widespread death and disability (a good indicator by most moral philosophies), then the alcohol industry is guilty of evil. However, by Einstein's measure, those who have the power to intervene against the alcohol industry but choose to ignore the evidence presented in this dissertation pose a bigger threat to public health than the alcohol industry itself. Following the footsteps of public health elders who scrutinized Big Tobacco while courageously imagining a tobacco-free world, this dissertation closes with putting the onus of action on the next generations of researchers, policy makers, educators, and parents to demand alcohol marketing reform.

Importantly, our approach must focus on addressing the demand side of alcohol consumption rather than simply restricting supply. 20$^{th}$ century alcohol prohibition showed that outright bans are ineffective, creating black markets and criminal enterprises while failing to address the underlying societal issues. Instead, we must pursue evidence-based interventions that reduce the appeal and perceived necessity of alcohol through education, alternative social activities, mental health support, and regulation of predatory marketing practices. The path forward requires nuanced strategies that empower individuals and communities while holding the industry accountable for its role in public health outcomes.

# Appendices

## APPENDIX 1

Python Execution for BERTopic Model

```
# Dependencies
import pandas as pd
import numpy as np
from bertopic import BERTopic
from sklearn.feature_extraction.text import CountVectorizer
from bertopic.vectorizers import ClassTfidfTransformer
from umap import UMAP
from hdbscan import HDBSCAN
from sentence_transformers import SentenceTransformer
import matplotlib.pyplot as plt
import re
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from gensim.corpora import Dictionary
from gensim.models.coherencemodel import CoherenceModel
import os
import scipy.cluster.hierarchy as sch
from sklearn.metrics.pairwise import cosine_similarity
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet
import json
from itertools import product
import shutil

nltk.download('punkt', quiet=True)
nltk.download('stopwords', quiet=True)
nltk.download('averaged_perceptron_tagger', quiet=True)
nltk.download('wordnet', quiet=True)

print("Libraries imported and NLTK data downloaded successfully.")

# Set up stopwords
with open('custom_stopwords.txt', 'r') as f:
    custom_stopwords = set(word.strip().lower() for word in f.read().split(','))

stop_words = set(stopwords.words('english')).union(custom_stopwords)

print(f"Total number of stopwords: {len(stop_words)}")

# Define text cleaning and preprocessing functions
lemmatizer = WordNetLemmatizer()

def get_wordnet_pos(word):
    tag = nltk.pos_tag([word])[0][1][0].upper()
```

```
    tag_dict = {"J": wordnet.ADJ,
                "N": wordnet.NOUN,
                "V": wordnet.VERB,
                "R": wordnet.ADV}
    return tag_dict.get(tag, wordnet.NOUN)

def clean_text(text):
    text = re.sub(r'[^a-zA-Z]', ' ', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text.lower()

def preprocess_tweet(text):
    cleaned_text = clean_text(text)
    tokens = word_tokenize(cleaned_text)
    lemmatized_tokens = [lemmatizer.lemmatize(token, get_wordnet_pos(token)) for token in tokens if
token not in stop_words]
    return ' '.join(lemmatized_tokens)

print("Text cleaning and preprocessing functions defined.")

# Define coherence computation function
def compute_coherence(topic_words, texts):
    dictionary = Dictionary([text.split() for text in texts])
    corpus = [dictionary.doc2bow(text.split()) for text in texts]
    coherence_model = CoherenceModel(topics=topic_words, texts=[text.split() for text in texts],
dictionary=dictionary, coherence='c_v')
    return coherence_model.get_coherence()

print("Coherence computation function defined.")

# Define BERTopic model creation function
def create_bertopic_model(umap_params, hdbscan_params, nr_topics=None):
    sentence_model = SentenceTransformer("all-MiniLM-L6-v2")
    umap_model = UMAP(**umap_params)
    hdbscan_model = HDBSCAN(**hdbscan_params)
    vectorizer_model = CountVectorizer(stop_words=list(stop_words))
    ctfidf_model = ClassTfidfTransformer()

    return BERTopic(
        nr_topics=nr_topics,
        embedding_model=sentence_model,
        umap_model=umap_model,
        hdbscan_model=hdbscan_model,
        vectorizer_model=vectorizer_model,
        ctfidf_model=ctfidf_model,
        verbose=True
    )

print("BERTopic model creation function defined.")
```

```
# Define function to save iteration files
def save_iteration_files(model, df, topics, n_topics, iteration, umap_params, hdbscan_params):
    iteration_dir = f"iteration_{iteration}_topics_{n_topics}"
    os.makedirs(iteration_dir, exist_ok=True)

    df['topic'] = topics
    tweets_with_topics = df[['tweet', 'topic']]
    tweets_with_topics.to_csv(os.path.join(iteration_dir, 'classified_tweets.csv'), index=False)

    model.visualize_topics().write_html(os.path.join(iteration_dir, "topic_visualization.html"))

    if len(set(topics)) > 1:
        model.visualize_hierarchy().write_html(os.path.join(iteration_dir, "topic_hierarchy.html"))
        model.visualize_heatmap().write_html(os.path.join(iteration_dir, "topic_similarity.html"))
    else:
        print(f"Skipping hierarchy and heatmap visualizations for {n_topics} topics due to insufficient
unique topics.")

    top_topics = model.get_topic_info()
    top_topics.to_csv(os.path.join(iteration_dir, 'topic_words.csv'), index=False)

    # Save parameters
    with open(os.path.join(iteration_dir, 'parameters.json'), 'w') as f:
        json.dump({
            'umap_params': {k: v if isinstance(v, (int, float, str)) else v.item() for k, v in
umap_params.items()},
            'hdbscan_params': {k: v if isinstance(v, (int, float, str)) else v.item() for k, v in
hdbscan_params.items()},
            'n_topics': n_topics
        }, f, indent=2)

print("Function to save iteration files defined.")

# Define topic model evaluation function
def evaluate_topic_model(documents, topic_range, df, umap_params, hdbscan_params, iteration):
    results = []
    for n_topics in topic_range:
        try:
            model = create_bertopic_model(umap_params, hdbscan_params, nr_topics=n_topics)
            topics, _ = model.fit_transform(documents)

            unique_topics = set(topics)
            if len(unique_topics) < 2:
                print(f"Skipping {n_topics} topics due to insufficient unique topics.")
                continue

            save_iteration_files(model, df, topics, n_topics, iteration, umap_params, hdbscan_params)

            topic_words = [model.get_topic(topic) for topic in range(n_topics)]
            topic_words = [[word for word, _ in topic] for topic in topic_words if topic]
```

```
        if len(topic_words) < 2:
            print(f"Skipping {n_topics} topics due to insufficient topic words.")
            continue

        coherence = compute_coherence(topic_words, documents)
        results.append((n_topics, coherence))
        print(f"Evaluated {n_topics} topics. Coherence: {coherence}")

      except Exception as e:
        print(f"Error evaluating {n_topics} topics: {str(e)}")
        continue

    return results

print("Topic model evaluation function defined.")

# Define parameter optimization function
def optimize_bertopic_parameters(documents, df, topic_range, iterations=50):
    umap_params = {
        'n_neighbors': [5, 10, 15, 20, 25, 30, 35],
        'n_components': [3, 4, 5, 6, 7, 8, 9, 10],
        'min_dist': [0.01, 0.05, 0.1, 0.5],
        'metric': ['cosine']
    }

    hdbscan_params = {
        'min_cluster_size': [5, 10, 15, 20, 25, 30, 35],
        'metric': ['euclidean'],
        'cluster_selection_method': ['eom']
    }

    best_coherence = -float('inf')
    best_params = None
    all_iteration_results = []

    for i in range(iterations):
        umap_config = {k: np.random.choice(v) for k, v in umap_params.items()}
        hdbscan_config = {k: np.random.choice(v) for k, v in hdbscan_params.items()}

        print(f"Iteration {i + 1}/{iterations}")
        print(f"UMAP parameters: {umap_config}")
        print(f"HDBSCAN parameters: {hdbscan_config}")

        iteration_results = evaluate_topic_model(documents, topic_range, df, umap_config,
hdbscan_config, i+1)

        if iteration_results:
            max_coherence = max(result[1] for result in iteration_results)
            if max_coherence > best_coherence:
```

```
                best_coherence = max_coherence
                best_params = (umap_config, hdbscan_config)

            iteration_result = {
                'iteration': i + 1,
                'umap_params': {k: v if isinstance(v, (int, float, str)) else v.item() for k, v in
umap_config.items()},
                'hdbscan_params': {k: v if isinstance(v, (int, float, str)) else v.item() for k, v in
hdbscan_config.items()},
                'results': [(int(n), float(c)) for n, c in iteration_results],
                'max_coherence': float(max_coherence)
            }
            all_iteration_results.append(iteration_result)

            print(f"Best coherence in this iteration: {max_coherence}")
        else:
            print("No valid results in this iteration.")

        print("------")

    if not all_iteration_results:
        print("No valid results found across all iterations.")
        return None, None

    return all_iteration_results, best_params

print("Parameter optimization function defined.")

# Main execution
if __name__ == "__main__":
    # Load and preprocess tweet data
    df = pd.read_csv('tweet_data.csv')
    df['preprocessed_tweet'] = df['tweet'].apply(preprocess_tweet)
    df[['tweet', 'preprocessed_tweet']].to_csv('preprocessed_tweets.csv', index=False)
    print("Preprocessed tweets saved to 'preprocessed_tweets.csv'")

    docs = df['preprocessed_tweet'].tolist()
    with open('bertopic_input.txt', 'w', encoding='utf-8') as f:
        for doc in docs:
            f.write(f"{doc}\n")
    print("BERTopic input saved to 'bertopic_input.txt'")

    topic_range = range(2, 26, 1)
    all_iteration_results, best_params = optimize_bertopic_parameters(docs, df, topic_range,
iterations=50)

    if all_iteration_results is not None:
        # Save all iteration results
        with open('all_iteration_results.json', 'w') as f:
            json.dump(all_iteration_results, f, indent=2)
```

```
        # Find the best overall iteration
        best_iteration = max(all_iteration_results, key=lambda x: x['max_coherence'])

        print("Best overall iteration:")
        print(f"Iteration: {best_iteration['iteration']}")
        print(f"UMAP parameters: {best_iteration['umap_params']}")
        print(f"HDBSCAN parameters: {best_iteration['hdbscan_params']}")
        print(f"Max coherence: {best_iteration['max_coherence']}")

        # Create and evaluate the best model
        best_model = create_bertopic_model(best_iteration['umap_params'],
best_iteration['hdbscan_params'])
        best_topics, _ = best_model.fit_transform(docs)

        # Save the best model results
        save_iteration_files(best_model, df, best_topics, "best", "final", best_iteration['umap_params'],
best_iteration['hdbscan_params'])

        print("Optimization completed. Results saved in 'all_iteration_results.json'")

        # Plot results for the best model
        best_results = best_iteration['results']

        plt.figure(figsize=(10, 6))
        plt.plot(*zip(*best_results))
        plt.xlabel('Number of Topics')
        plt.ylabel('Coherence Score')
        plt.title('Topic Coherence Scores (Best Model)')
        plt.savefig('best_model_coherence_scores.png')
        plt.close()

        optimal_topics = max(best_results, key=lambda x: x[1])[0]
        print(f"Optimal number of topics for the best model: {optimal_topics}")
    else:
        print("No valid models could be created. Please check your data and parameters.")

    print("Script execution completed.")
```

### APPENDIX 2

Python Execution for Topic Naming

```
# Dependencies
### This script was written under previous OpenAI API versions. Before Running: pip install
openai==0.27.0; after running: pip install --upgrade openai
import openai
import pandas as pd
from tqdm import tqdm
import time

# Set up OpenAI API key
openai.api_key = "********"  # Replace "api_key" with your actual OpenAI API key

# Load the dataset
df = pd.read_csv('topics_for_gpt.csv') # Where topics_for_gpt.csv is a dataset with two columns:
numeric topic number (value from BERTopic model), words (array of representative words from
BERTopic model); rows iterated 5000 times

# Function to call GPT-4o and get a topic name with error handling
def get_topic_name(words, max_retries=5):
    prompt = (f"The following words represent a topic derived from tweets generated by alcohol
companies "
              f"during the COVID-19 pandemic: {words}. "
              f"Please provide a concise, meaningful topic name that clearly summarizes these words. ")

    retries = 0
    while retries < max_retries:
        try:
            response = openai.ChatCompletion.create(
                model="gpt-4o",
                messages=[
                    {"role": "system", "content": "You are an expert at naming topics."},
                    {"role": "user", "content": prompt}
                ],
                max_tokens=15,
                temperature=0.5,
                n=1,
                stop=None
            )
            topic_name = response['choices'][0]['message']['content'].strip()
            return topic_name

        except openai.error.APIError as e:
            print(f"APIError: {e}. Retrying ({retries + 1}/{max_retries})...")
            retries += 1
            time.sleep(2 ** retries)  # Exponential backoff

        except openai.error.RateLimitError as e:
            print(f"RateLimitError: {e}. Retrying ({retries + 1}/{max_retries})...")
```

```
            retries += 1
            time.sleep(2 ** retries)  # Exponential backoff

        except openai.error.Timeout as e:
            print(f"TimeoutError: {e}. Retrying ({retries + 1}/{max_retries})...")
            retries += 1
            time.sleep(2 ** retries)  # Exponential backoff

        except Exception as e:
            print(f"Unexpected error: {e}. Retrying ({retries + 1}/{max_retries})...")
            retries += 1
            time.sleep(2 ** retries)  # Exponential backoff

    # If all retries fail, return a placeholder or handle it accordingly
    print(f"Failed to get a response after {max_retries} retries.")
    return "Error: Unable to generate topic name"

# Loop through each row and generate the topic name with a progress bar
tqdm.pandas()

df['GPT'] = df['Words'].progress_apply(lambda words: get_topic_name(words))

# Save the updated dataframe back to CSV
df.to_csv('topics_for_gpt_output.csv', index=False)

print("Completed!")
```

## APPENDIX 3

Python Execution for Tweet Classification using GPT-4o

```
# Dependencies
### This script was written under previous OpenAI API versions. Before Running: pip install
openai==0.27.0; after running: pip install --upgrade openai
import openai
import pandas as pd
from tqdm import tqdm  # Import tqdm for the progress bar

# Set up OpenAI API key
openai.api_key = "********"  # Replace "api_key" with your actual OpenAI API key

# Load the dataset
df = pd.read_csv('tweet_data.csv')
# Where tweet_data.csv is a dataset with two columns:
# tweet (verbatim text from study), GPT (empty, to be filled by API Call)

# Function to call GPT-4o and classify text
def get_topic_name(words):
    prompt = (f"Classify the following text into one of the following 6 themes: 1. Alcohol Delivery and
Isolation Drinking, 2. Restaurant Support, 3. Social Media Promotions, 4. Sports, 5. Gaming, 6. The text
does not fit into any of these themes."
              f"Text: {words}."
              f"Respond with verbatim classification using only the specified 6 options.")

    response = openai.ChatCompletion.create(
        model="gpt-4o",
        messages=[
            {"role": "system", "content": "You are an expert at classifying text into themes."},
            {"role": "user", "content": prompt}
        ],
        max_tokens=30,
        temperature=0.5,
        n=1,
        stop=None
    )

    topic_name = response['choices'][0]['message']['content'].strip()
    return topic_name

# Loop through each row and generate the topic name with a progress bar
tqdm.pandas()

df['GPT'] = df['tweet'].progress_apply(lambda words: get_topic_name(words))

# Save the updated dataframe back to CSV
df.to_csv('tweet_data_output.csv', index=False)

print("Completed!")
```

## APPENDIX 4

**Full LIWC Output (csv)**

Corpus,WC,Analytic,Clout,Authentic,Tone,WPS,BigWords,Dic,Linguistic,function,pronoun,ppron,i,we,
you,shehe,they,ipron,det,article,number,prep,auxverb,adverb,conj,negate,verb,adj,quantity,Drives,affiliat
ion,achieve,power,Cognition,allnone,cogproc,insight,cause,discrep,tentat,certitude,differ,memory,Affect
,tone_pos,tone_neg,emotion,emo_pos,emo_neg,emo_anx,emo_anger,emo_sad,swear,Social,socbehav,pr
osocial,polite,conflict,moral,comm,socrefs,family,friend,female,male,Culture,politic,ethnicity,tech,Lifest
yle,leisure,home,work,money,relig,Physical,health,illness,wellness,mental,substances,sexual,food,death,
need,want,acquire,lack,fulfill,fatigue,reward,risk,curiosity,allure,Perception,attention,motion,space,visua
l,auditory,feeling,time,focuspast,focuspresent,focusfuture,Conversation,netspeak,assent,nonflu,filler,All
Punc,Period,Comma,QMark,Exclam,Apostro,OtherP,Emoji

Alcohol Delivery and Isolation
Drinking,3598,85.63,76.92,36.78,80.16,15.51,17.07,76.21,53.7,38.8,9.09,6.28,1.5,1.28,2.92,0.25,0.22,2.
81,11.2,5.31,4.17,12.98,3.64,4,3.47,0.61,10.56,6.73,3.81,4.5,2.56,1.08,1.03,7.31,1.28,6.03,1.45,0.89,0.8
3,1.17,0.47,1.72,0.14,4.86,4.36,0.44,1.36,1.11,0.22,0.06,0,0.03,0.03,9.34,3.09,0.86,0.64,0,0.06,1.08,6.23
,0.19,0.75,0.31,0.69,1.17,0,0.03,1.14,3.53,0.72,1.28,1.47,0.11,0,5.84,0.11,0.06,0.06,0,2.5,0,4.89,0.06,0.1
4,0.42,1.75,0,0.11,0.08,0.03,0.31,0.08,9.42,9.64,0.25,1.14,5.75,1.11,0.08,1.08,5.78,0.69,3.06,1.28,1.03,0
.86,0.22,0.03,0,27.85,6.92,2.06,0.44,1,1.53,15.9,0

Restaurant Support
,1661,89.68,94.38,9.14,98.49,16.61,24.14,82.24,54.43,41.3,9.81,7.41,0.42,3.37,2.05,0.36,0.42,2.41,11.2,
4.7,4.46,15.77,3.85,2.47,4.21,0,11.92,5.66,3.13,11.68,7.77,0.78,3.25,6.98,1.2,5.66,0.9,0.9,1.63,1.38,0.12
,1.75,0,7.83,7.41,0.36,0.96,0.9,0,0,0,0,0,17.04,8.01,6.38,0.72,0,0.12,1.26,8.97,0.12,0.12,0.42,0,1.93,0.06
,0.3,1.57,11.02,5.6,0.06,2.35,3.01,0,6.68,0.36,0.36,0,0,0.12,0,5.78,0,0.66,0.24,0.96,0,0,0,0.12,0.18,0,7.2
8,7.71,0.3,1.26,5.6,0.6,0,0.06,2.83,1.57,2.59,0.9,0.36,0.3,0.06,0,0,27.33,4.76,3.07,1.38,1.87,0.72,15.53,0

Social Media
Promotions,3962,88.89,93.8,32.37,88.52,15.36,19.06,76.55,55.25,39.63,9.04,6.66,0.56,2.52,2.4,0.58,0.3
8,2.37,10.93,5.3,3.58,14.59,3.28,3.23,4.64,0.15,12.47,4.16,3.03,6.64,4.72,1.62,0.48,5.2,0.43,4.75,0.98,0.
28,0.93,1.34,0.45,0.96,0.2,5.22,4.95,0.2,1.54,1.41,0.05,0,0,0,0,12.75,3.79,0.71,0.1,0.03,0.13,1.74,8.63,0.
15,1.49,0.25,1.74,1.44,0.05,0.15,1.24,2.93,0.88,0.88,0.91,0.25,0.03,3.99,0.03,0,0,0,0.83,0,1.59,0.08,0.25
,0.23,0.66,0.03,0.05,0.03,0.86,0.23,0.15,9.06,12.04,0.61,2.73,6.51,1.44,0.81,0.53,5.38,0.5,2.7,1.49,1.97,
1.97,0,0.05,0,24.31,4.11,2.02,0.76,2.52,0.83,14.06,0

Sports,2514,92.32,87.37,26.37,67.22,14.37,19.05,73.51,48.41,38.07,8,6.05,1.43,1.51,2.43,0.32,0.2,1.95,
11.69,5.93,5.85,13.84,3.86,2.23,2.98,0.44,9.63,4.42,3.78,6.32,3.94,2.31,0.91,6.05,0.6,5.41,0.84,0.48,1.3
1,1.15,0.72,1.35,0.08,3.46,3.22,0.24,0.56,0.56,0,0,0,0,0,11.61,3.78,1.23,0.12,0.04,0.2,1.35,8.55,0.44,1.5
1,0.04,2.11,0.64,0.04,0,0.6,5.53,3.06,1.47,0.84,0.24,0,1.27,0.24,0.16,0,0.04,0.44,0,0.56,0,0.16,0.2,0.8,0.
04,0.08,0,0.2,0.12,0.16,7.4,10.26,0.48,1.67,6.28,1.95,0.36,0.16,4.69,0.68,2.67,1.15,0.44,0.44,0,0,0,33.33
,6.09,2.19,2.27,2.11,0.72,19.97,0
Gaming,489,98.37,71.9,19.05,51.75,12.87,26.99,69.73,37.01,29.86,3.68,2.45,0,0.61,1.23,0,0.2,1.23,7.36
,4.29,10.02,16.56,2.45,0.82,1.64,0.41,7.16,2.25,1.02,3.89,1.02,2.04,1.43,3.07,0.41,2.66,0.2,1.02,0.41,0.4
1,0,0.41,0.41,3.68,2.86,0.82,1.23,1.23,0,0,0,0,0,8.38,3.89,1.23,0,0.41,0.61,0.82,4.5,0,1.43,0,1.64,3.48,0,
0,3.48,6.54,3.68,0,1.02,1.84,0,1.23,0.41,0,0.2,0,0,0,0.2,0.41,0,0,1.02,0,0,0,0.2,0.2,0,7.57,11.04,1.64,2.04
,5.52,2.25,1.02,0,5.52,0.61,2.04,1.84,0.2,0.2,0,0,0,39.26,4.29,3.68,1.02,3.07,0.41,26.79,0

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